

Politecnico di Torino

Politecnico di Torino

Master's Degree in Computer Engineering

A.Y. 2025/2026

Graduation Session December 2026

GroundOne: A Specialist Decision Model for Groundedness Verification in Local Retrieval-Augmented Generation

Fine-tuning compact language models as calibrated single-forward-pass verifiers on Apple Silicon

Supervisor: Prof. Paolo Garza

Candidate: Francesco Pio De Dominicis

Summary

Retrieval-augmented generation (RAG) is the standard way to give language models access to private documents, and running the whole pipeline on the user's own device is now practical on consumer hardware, which helps with privacy, offline use and cost. Retrieval does not, however, eliminate hallucination: generated answers still contain sentences that the retrieved context does not support or that contradict it. Detecting them is a prerequisite for trustworthy local assistants, and since the check must run on every answer, it has to fit on the same constrained machine as the generator.

Locality is the backbone of this work, and it has a precise technical consequence. On a single device, reading text is cheap and writing it is expensive: the model reads its input in a single forward pass, that is, one pass through all its layers, limited by compute, but generates its output one token at a time, each token limited by the memory bandwidth needed to read all the weights. On the laptop used here, writing a token costs fourteen to sixteen times more than reading one, and a review of current local hardware, from phones and laptops to dedicated machines such as Apple's Mac Studio and NVIDIA's DGX Spark, shows the same asymmetry everywhere, with new hardware accelerating reading much more than writing. Small open-weight models now trail the closed frontier by only months, so capable local pipelines are realistic; the question is how to verify them at a cost the device can bear.

Verification currently follows two paradigms. **Encoder-based checkers**, such as AlignScore, MiniCheck and HHEM, are small and fast, but return one fixed score per pair of document and sentence. **Generative judges**, by contrast, are flexible but write their verdicts token by token, so their cost grows with what they write; small judges often break the output format, and the confidence they express is rarely usable as a probability. A third family has recently emerged: **decision models**, such as the proprietary Jev and the open SemIf and Rizzo-Flow, answer closed questions in a single forward pass by reading the probability of each answer option from the logits. The evidence for them is mostly reported by their authors, and the open implementations are general-purpose. Whether a compact decision model specialised in one task can match generative judges and dedicated encoders,

where its advantage comes from, and whether its probabilities are usable, all on consumer hardware, had not been studied in a controlled way.

This thesis designs, trains and evaluates **GroundOne**, a decision model specialised in one decision: for each sentence of a RAG answer, whether the retrieved document supports it, contradicts it or does not mention it. A marker follows every sentence of the answer, and the verdicts for all K sentences are read from the logits of three answer letters in one forward pass, without generating any text. Qwen2.5 models with 0.5B and 3B parameters are fine-tuned with LoRA on the human-labelled answers of RAGTruth, with a class-weighted loss restricted to the three letters; a temperature and a bias fitted on development data calibrate the probabilities, and a threshold chosen on the same data turns them into consistent labels and strict or lenient flags. Everything, from building the dataset to training, evaluation and serving, runs on one Apple M4 laptop with 16 GB of memory, with MLX and no external service, and every baseline is run on the same machine. The evaluation uses three test sets built and frozen before training (the RAGTruth test split, LLM-AggreFact and VitaminC) and compares GroundOne with cues that never read the document, the untrained models, generative judges of the same size with and without probabilities, two general-purpose decision models and specialised encoders, reporting cluster-bootstrap intervals.

On the RAGTruth test set GroundOne reaches a three-way balanced accuracy of 0.769 with 0.5B parameters and 0.832 with 3B, against 0.37 and 0.44 for the untrained models; its AUROC (0.905 and 0.940) exceeds that of the encoders (about 0.82), and it clearly outperforms both general-purpose decision models: Rizzo-Flow, eight times larger than the 0.5B, and Liquid AI's d1-3B, of the same size as the 3B (0.657), which in turn is much stronger on the out-of-domain VitaminC test. On LLM-AggreFact the 3B is on par with MiniCheck-RoBERTa (74.8 against 74.0), above the published zero-shot 7B model and below the closed decision model Jev as measured independently (78.6). When an answer contains fourteen or more sentences ($K \geq 14$), GroundOne is about two times faster than a terse judge and six times faster than a verbose judge of the same size and, unlike the judges, always produces a valid output. It also uses less energy (5.5 joules per answer with the 0.5B, 33 with the 3B), but the saving is smaller than the saving in time, because a long prefill draws more power than decoding. Combining vendor specifications with the measured length of the judges' outputs, a 3B judge must spend at least 1.8 seconds only on writing its verdicts for a long answer on the thesis laptop, and 0.12 seconds even on the fastest consumer graphics card: a cost that GroundOne does not pay on any hardware. Asked for the same output, generative judges provide probabilities that barely separate correct from incorrect decisions, while calibration brings GroundOne's expected calibration error on the test set, the average gap between the stated confidence and the observed frequency of correct answers, to 0.019. How the decision is read also matters: with the same trained

model, comparing only the three admissible letters is more accurate than letting the model generate its verdict freely, with a gain in balanced accuracy of 0.25 for the 0.5B and 0.07 for the 3B. Some aspects remain to be improved, and they point to natural directions for future work: performance drops on domains far from the training data, such as the fact checking of VitaminC; the subtlest hallucinations, hidden in sentences that otherwise follow the source faithfully, are detected only in part; and training is not yet fully stable across random seeds, which simple changes to the optimiser could mitigate.

Acknowledgements

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Tristique senectus et netus et malesuada fames ac turpis. Pretium nibh ipsum consequat nisl vel pretium lectus quam. Urna molestie at elementum eu facilisis sed odio morbi quis. Sed egestas egestas fringilla phasellus faucibus scelerisque eleifend. At in tellus integer feugiat. Mauris rhoncus aenean vel elit scelerisque mauris pellentesque pulvinar. Commodo sed egestas egestas fringilla. Nunc lobortis mattis aliquam faucibus purus in massa. Facilisis magna etiam tempor orci eu lobortis elementum nibh. Elementum curabitur vitae nunc sed velit dignissim. Neque volutpat ac tincidunt vitae. Massa id neque aliquam vestibulum morbi blandit cursus risus at. Porta non pulvinar neque laoreet suspendisse interdum consectetur. Turpis in eu mi bibendum. Ut tristique et egestas quis ipsum suspendisse. Integer quis auctor elit sed vulputate mi sit amet. Viverra nam libero justo laoreet sit amet cursus.

Table of Contents

List of Figures x

1 Introduction 1
- 1.1 Motivation . . . 1
  - 1.1.1 Retrieval-augmented generation on the user's device . . . 1
  - 1.1.2 Hallucination and the need for verification . . . 2
  - 1.1.3 The cost of checking an answer on a laptop . . . 2
- 1.2 Decision models . . . 3
- 1.3 Approach and research questions . . . 5
- 1.4 Scope . . . 6
- 1.5 Contributions . . . 6
- 1.6 Main results . . . 7
- 1.7 Thesis structure . . . 8

2 Local Inference of Language Models 9
- 2.1 From hosted to local models . . . 9
- 2.2 Open-weight and small language models . . . 12
  - 2.2.1 Open weights are not open source . . . 12
  - 2.2.2 The gap with closed models is closing . . . 13
  - 2.2.3 Small language models . . . 16
- 2.3 The cost of inference on one device . . . 16
  - 2.3.1 Prefill and decode . . . 17
  - 2.3.2 The roofline model . . . 18
  - 2.3.3 A lower bound on generation . . . 18
- 2.4 Hardware for local inference . . . 19
  - 2.4.1 Classes of hardware . . . 20
  - 2.4.2 The asymmetry holds on every class . . . 24
- 2.5 Runtimes and quantisation . . . 25
  - 2.5.1 Runtimes . . . 25
  - 2.5.2 Specialised inference engines . . . 25
  - 2.5.3 Quantisation . . . 28

2.5.4 Faster generation without fewer tokens . . . 29
2.6 Energy . . . 29
2.7 Consequences for verification on the device . . . 30

3 Background and Related Work 31
3.1 Decision models . . . 31
3.2 Retrieval-augmented generation and hallucination . . . 34
3.3 Groundedness verification . . . 34
3.4 Calibration and selective prediction . . . 35
3.5 Parameter-efficient fine-tuning . . . 36
3.6 Research gap . . . 38

4 GroundOne: Method and Implementation 39
4.1 Task and decision . . . 39
4.2 All verdicts in a single forward pass . . . 40
4.3 Training objective . . . 43
4.4 Calibration . . . 45
4.5 Software . . . 49

5 Data 52
5.1 Task and labels . . . 52
5.2 Source datasets . . . 53
5.3 Construction . . . 54
5.4 The resulting dataset . . . 55
5.5 Reliability of the test sets . . . 56
5.6 Exploratory analysis . . . 58
5.7 Leakage control . . . 60

6 Experimental Setup 61
6.1 Models . . . 61
6.2 Baselines . . . 62
6.3 Metrics . . . 64
6.4 Statistical protocol . . . 64
6.5 Latency, energy and hardware . . . 65

7 Results 67
7.1 Training . . . 67
7.2 Variability across seeds . . . 67
7.3 Accuracy on the RAG test set (T1) . . . 69
7.4 Comparison with the baselines . . . 70
7.5 Transfer to a different domain (T3) . . . 72
7.6 Readout versus generation . . . 73

7.7 Calibration and selective prediction . . . 73
7.8 Probabilities at equal output . . . 75
7.9 Latency as a function of K . . . 77
7.10 Energy per verification . . . 78
7.11 Locality: offline run, quantisation and co-residence . . . 79
7.12 Error analysis . . . 80

8 Discussion, Limitations and Threats to Validity 81
8.1 When a specialist decision model pays off . . . 81
8.2 Limitations . . . 84
8.3 Threats to validity . . . 85

9 Conclusion and Future Work 87
9.1 Summary of findings . . . 87
9.2 Future work . . . 88

A Output of every system on one answer 89
A.1 Verdicts sentence by sentence . . . 90
A.2 Raw outputs . . . 91

Bibliography 93

List of Figures

1.1 Two ways of verifying the K sentences of an answer. Parallelograms: inputs and outputs; blue rectangles: components of the model; yellow rectangle: processing outside the model; diamonds: decisions. The decision model reads the document and the answer once and, for every sentence, chooses among three admissible answers, also returning their probability. The generative LLM passes through the model again for every token it writes (red loop), and the verdicts must then be extracted from the text, which may not follow the required format. 4

2.1 Capability of the best closed and the best open-weight model over time on Epoch AI's aggregate capabilities index (dots: individual models; lines: best so far). Plotted from Epoch AI's published data table [34]. 14

2.2 Artificial Analysis Intelligence Index (version 4.3.2, ten evaluations) on 20 October 2026, redrawn from the published chart [35]. Closed models in orange, open-weight models in blue; hatched, the only model that Artificial Analysis classifies as small. In brackets, the reasoning setting at which each model was evaluated. The open-weight status is taken from the data of the same leaderboard. It is a snapshot: scores and ranking change with every new release. 15

2.3 Time per token when the model reads its input (prefill) and when it writes its output (decode), measured on the thesis machine (Apple M4, 16 GB, MLX 8-bit, one request at a time). Prefill: median over contexts of 500, 1,000 and 2,000 tokens. 17

2.4 Roofline model of the thesis machine [43]. Sloped roof: Apple's memory bandwidth (120 GB/s). Flat roof: throughput reached in prefill, a lower bound of the unpublished peak. Intensities computed for one feed-forward layer of Qwen2.5 with 8-bit weights and 16-bit activations; attention is ignored. 19

2.5 Memory needed for the weights of a model as a function of its size and precision (parameters times bits per weight, including half a bit of per-group scales for the quantised formats). Dashed lines: common memory sizes of local hardware. The key-value cache, activations and other running models come on top. 20

2.6 The two architectures of local inference hardware. Blue rectangles: compute units; green rectangles: memory; orange rectangle: the model weights; line width: indicative bandwidth. Capacity and bandwidth values from Table 2.2. In a discrete system a model that does not fit in VRAM must go through the much slower PCIe link; in a unified-memory system the model can occupy almost all of the machine's memory, but every generated token reads it at a lower speed than a high-end graphics card. 21

2.7 Some of the local inference systems compared in this section (images not to scale). Sources: (a), (c) Apple; (b) LoMit, Wikimedia Commons, CC0; (d), (e) NVIDIA. The vendors' images are promotional material, reproduced for the purpose of scientific documentation. . . 22

2.8 Local inference hardware on the two axes that decide what can run: memory capacity (which models fit) and memory bandwidth (how fast tokens are generated). Vendors' specifications: markers at the largest memory configuration, shaded bars over the range of memory sizes offered; the thesis machine is shown at its actual 16 GB. 24

2.9 Lower bounds on the cost of generating text on each class of local hardware, computed as weight bytes divided by the vendor's bandwidth (one request at a time; key-value cache ignored). (a) Time per generated token with the two GroundOne base models in 8 bits; diamonds: decode time measured on the thesis machine. (b) Minimum time that a 3B generative judge spends only on writing the verdicts for an answer of fourteen or more sentences (median generated tokens of the latency protocol). A decision model generates no token, so both costs are zero for it on every machine. 26

2.10 Streaming experts from the SSD in an engine such as Colibri. Memory holds the part of the model used by every token (attention, router, shared experts) and a cache of recently used experts; every expert is stored on the SSD. For each token the router chooses a few experts: those already cached are used at once, the missing ones are read from disk, at a bandwidth of a few gigabytes per second. For example, GLM-5.2 keeps about 10 GB resident and leaves about 429 GB of weights on disk [36]. The diagram is not to scale. 27

3.1 Three ways of verifying the K sentences of an answer. A specialised encoder evaluates each (document, sentence) pair in a separate forward pass and returns only a probability of support. A generative judge reads everything, but writes its verdict token by token (red loop), so its cost grows with the text produced and the output has to be interpreted by a parser. A specialist decision model reads document and answer in a single prefill and derives three probabilities from each marker, without generating any token. 36

3.2 Sketch of a risk–coverage curve (illustrative shape, not a result). The decisions are sorted from the most to the least confident; for each share decided (coverage) the error rate (risk) of the accepted decisions is reported. At coverage 1 the risk equals the overall error; giving up the least confident decisions lowers the error. The area under the curve (AURC) summarises the curve in one number: the smaller it is, the better the confidence separates correct decisions from wrong ones. 37

3.3 Low-rank adaptation applied to a linear layer. The weight matrix W stays frozen (8-bit in GroundOne); only the two small matrices A and B are trained, and their product, scaled by α, is added to the output of the layer. With rank r the adapter has $r(d+k)$ parameters instead of the dk of W. 38

4.1 The marker prompt of GroundOne (design b). The sentences of the answer are numbered and every scored sentence is followed by the marker “ `Verdict:`”; a sentence without checkable content (here a heading) remains as context, without a marker. The model processes the whole sequence in a single forward pass, and the output head, restricted to the letters A, B and C, is applied only at the marker positions. Because of causal attention, the verdict on sentence i depends on the document and on sentences 1 to i. 41

4.2 Estimated cost of verifying an answer of $S = 800$ tokens as a function of K, computed with only the constants measured on the M4: the per-token time of prefill and decoding and the mean number of tokens each judge generates per verdict on the latency subset (mean K 8.7). It is an analytical model, not a measurement: the measured latencies are in Section 7.9. The cost of the decision model grows by only three prefill tokens per decision, that of the judges by a few decoding tokens per decision. 42

4.3 The two solutions for obtaining K verdicts from a single reading. (b) The marker readout, adopted as the default, inserts a marker after every sentence and reads all the verdicts in the same pass. (a) The fallback solution reads the shared prefix once and appends a short question for each sentence, evaluated in a batch; it requires, however, K copies of the KV cache of the prefix, and it is the natural format for an untrained model. 42

4.4 Where GroundOne applies LoRA. Each transformer block of Qwen2.5 contains seven linear projections: four in the attention (W_q, W_k, W_v, W_o) and three in the MLP block (W_{gate}, W_{up}, W_{down}). Their weights remain frozen at 8 bits, and each is paired with a rank-16 adapter (in blue) that corrects its output following the scheme of Figure 3.3. The block is repeated 24 times in the 0.5B and 36 in the 3B; the embedding and the output head have no adapter. 44

4.5 How the AUROC arises (a sketch with synthetic scores, not a result). (a) Distribution of P(supported) for supported and unsupported sentences: the two distributions overlap, and every threshold accepts part of each. (b) Moving the threshold from 1 to 0 yields the ROC curve; the black dot corresponds to the threshold of panel (a). The area under the curve is the AUROC: the more separated the two distributions, the closer the curve gets to the top-left corner and the area to 1. The diagonal corresponds to random scores. 49

4.6 From logits to returned labels. The raw decision, that is, the argmax of the logits, serves the classification metrics of Chapter 7. The calibrated probabilities, obtained with temperature and bias fitted on the dev set, feed instead the single rule of the returned output: supported if $P(\text{supported}) \geq \tau$, otherwise the more probable of contradicted and not mentioned. The strict and lenient flags derive from this. 50

4.7 Architecture of the software. At the top, the experimental pipeline, from the public datasets to the frozen tests, run under the memory guard; at the bottom, the verification service, in which the command line and the HTTP server (native and compatible endpoints) share the same Engine, which serialises the requests and runs every MLX call on a single worker thread. The adapter and the calibration file produced by the pipeline are the only artefacts the service reads. . . 51

5.1 Class balance of the labelled sentences in each split. RAGTruth (train, development and T1) is about 88–91 % supported; T3 is sampled to 1,000 pairs per class. 56

5.2 Distribution of K, the number of scored sentences per answer, in training and in T1. Dashed lines mark the bands used to report results as a function of K. 57
5.3 Share of contradicted or not-mentioned sentences by the model that wrote the answer, in training and in T1. 58
5.4 Word overlap between a sentence and its document, by class, in training, T1 and T3 (mean in brackets). 59
5.5 Share of contradicted or not-mentioned sentences by position in the answer. 60

7.1 Three-way balanced accuracy on the development set during training of the 0.5B model, overall (left) and per task (right). The dotted line marks chance. The 3B shows the same shape on its 200-answer development subset. 68
7.2 Development three-way balanced accuracy of the 0.5B during training with the same recipe and three seeds. Two runs converge to the same level; the third collapsed to a constant predictor and was stopped. . 68
7.3 T1: AUROC of P(supported) for every system (left), and three-way balanced accuracy for the systems with a three-way output (right). Rizzo-Flow is evaluated on 300 answers and the untrained 3B on 600; all other systems, including d1-3B, on the full test set. 71
7.4 Control B2 on the full T1: with the same weights and the same forward pass, the verdict is read from the three answer letters (GroundOne) or taken as the token that greedy generation would emit (format failures counted as errors). 74
7.5 Reliability of P(supported) of the 0.5B before (raw) and after calibration on the development set, for the development set, T1 and T3. Points are bins with at least 20 sentences. 74
7.6 Risk–coverage curves of the binary decision of the 0.5B with calibrated probabilities: error rate as the least confident sentences are deferred. 75
7.7 Equal-output comparison on the 200 latency answers: generative judges of the same size asked for GroundOne's output, with probabilities read from the logits of the generated letter or written in JSON; all systems calibrated on development data with the same map. Sentences with a valid verdict only. 76
7.8 Median time per answer as a function of K on the same 200 T1 answers (in-process MLX, one answer at a time, M4 with 16 GB), for GroundOne, the decision model with one question per sentence, and the terse and verbose generative judges. Logarithmic scale. . . 77

7.9 On the latency subset: three-way balanced accuracy (format failures counted as errors) and share of sentences whose verdict could not be parsed. 78
7.10 Energy per verified answer on 100 latency answers (powermetrics, CPU, GPU and Neural Engine, net of idle power), with the mean net power during the run. 78

Chapter 1

Introduction

1.1 Motivation

1.1.1 Retrieval-augmented generation on the user's device

Retrieval-augmented generation (RAG) has become the standard way to let a language model answer questions about documents it was never trained on. A retriever selects the passages that are relevant to the question, and the model writes its answer with those passages in its context [1]. The same recipe serves a company searching its internal reports, a lawyer reading a contract and a student questioning their own notes, and in each case the value of the answer depends on its staying faithful to the documents.

Until recently such a pipeline required a hosted model, reached over the network. This is no longer the case. Compact open-weight models and efficient runtimes now run on consumer hardware [2, 3], and platform vendors ship models of a few billion parameters directly on their devices [4]. The gap with the best closed models is small and shrinking: models that fit on a single consumer graphics card reach the performance of the frontier of the time within six to twelve months [5], and the lead of the best closed model over the best open-weight one on a popular leaderboard fell from 8 % to under 2 % in a single year [6]. The whole pipeline, from retrieval to answer, can therefore stay on the user's own machine. This matters for four reasons. When the documents are private or regulated, they never leave the device; privacy becomes a technical property of the system rather than a promise of a provider, which the user cannot verify. A recent dispute over credit between OpenAI and two mathematicians, who had written unpublished drafts inside one of the company's products, showed how real the question is: the provider itself stated that it could not rule out that data derived from that usage had helped improve its models [7]. When there is no network, the assistant keeps working. When the behaviour of the system must stay stable, the weights change only if the user changes them,

instead of following a provider's updates. Finally, the cost becomes that of the hardware and its energy, instead of a price per token. Local models also expose their logits, which hosted interfaces often hide or limit, and this property turns out to be essential for the approach studied here.

Locality is a constraint as well as an advantage. A laptop has limited memory, limited bandwidth and a battery, and every component of the assistant must fit within them at the same time. This thesis treats locality as a requirement rather than as an afterthought: every step of the work, from building the dataset to training, evaluation and serving, runs on one consumer laptop with 16 GB of memory, and no step depends on an external service. Because locality is the backbone of the work, Chapter 2 is devoted entirely to it: why it matters, which models and hardware make it possible, and what it costs.

1.1.2 Hallucination and the need for verification

Retrieval does not remove hallucination. Even with the right passages in context, generated answers still contain sentences that the passages do not support, or that contradict them [8, 9]. An example from the data used in this thesis shows how subtle the problem can be. Asked to describe a restaurant from a structured record, a model writes six fluent sentences. Four of them are correct. One states that the restaurant is open seven days a week, whereas the record shows that it is closed on Mondays: the sentence contradicts the source. Another states that the restaurant has a rating of 4.5 stars on Google, whereas the record gives the rating without naming any source: the sentence adds a detail that the source does not mention (Appendix A).

Two observations follow from this example. First, the error sits in a single sentence of an otherwise correct answer, so a verdict on the whole answer is of little help: the user needs to know which sentence to distrust. Second, the two erroneous sentences are wrong in different ways. A contradiction is a factual error with respect to the source, whereas an unsupported addition may be harmless background or an invention, and an application may want to treat the two differently. A useful verifier therefore decides, for every sentence, whether the document supports it, contradicts it or does not mention it. Since no user can be expected to check every sentence against its source, the verifier must run on every answer, automatically, and on the same machine that generated it.

1.1.3 The cost of checking an answer on a laptop

A verifier that runs next to the generator competes with it for the same memory, time and battery. The most common solution, asking a language model to judge the answer, is expensive precisely where a laptop is weakest. A judge writes its

verdicts token by token, and on local hardware each generated token costs far more than each token it reads. Reading the prompt is limited by computation, which a laptop GPU handles well, whereas writing the output is limited by memory bandwidth, because every new token must read all the weights [10]. On the laptop used in this thesis, reading one token takes about 0.37 milliseconds for a model of 0.5B parameters, while writing one takes about 5.9, sixteen times more, and the asymmetry holds on every class of local hardware, from phones to dedicated AI machines (Chapter 2). A judge that explains its reasoning before each verdict multiplies this cost by the length of its explanation, and the cost grows again with the number of sentences in the answer. Small judges also break the required output format, and the confidence they express is rarely usable as a probability.

Encoder-based verifiers avoid generation altogether. Models such as AlignScore, MiniCheck and HHEM read a pair of document and sentence and return a score of support [11, 12, 13]. They are small and fast, but they give one fixed binary score per pair: they do not distinguish a contradiction from an omission, they read each sentence without the rest of the answer, and adapting them to a new decision requires a new model.

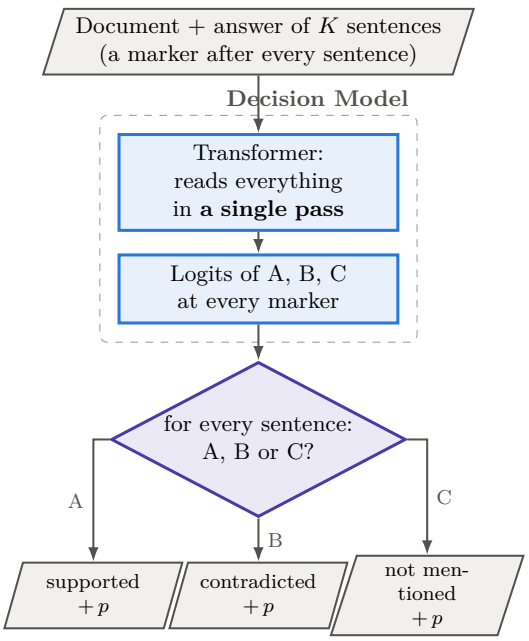
1.2 Decision models

A different way to use a language model for decisions has recently attracted attention: *decision models*. The category was introduced under the name of *System One models* [14], which evokes the fast, intuitive mode of thinking as opposed to deliberate reasoning; this thesis adopts the term decision model, later taken up by the ecosystem. Instead of generating an answer, such a model reads the question and the context once and returns a probability for each admissible answer, for example by reading the logits of the option letters at the end of the prompt (e.g., A: 90 %, B: 5 %, C: 5 %) [15, 16]. Three properties follow directly. No text is generated, so the decision costs only the reading of the input. The output is always one of the admissible answers, so it is always valid. Many questions about the same context can share a single forward pass[1] over it, so each additional question costs little.

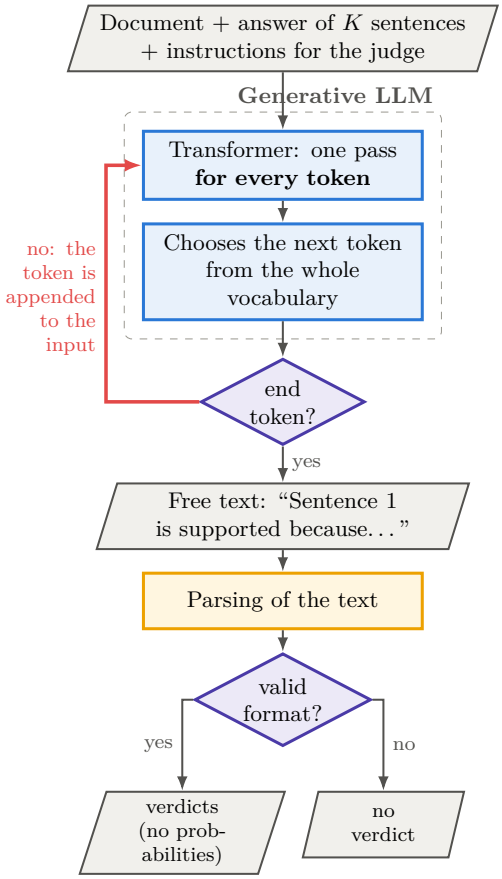
The idea fits verification well. An answer of K sentences poses K closed questions about the same document, and each of them has three admissible answers. A decision model can answer all of them while reading the document only once, and

[1] A *forward pass* is one complete pass of the input through all the layers of the network: for a language model, the computation that turns the input tokens into the logits of the next token. A generative model runs one for every token it writes; a decision model runs a single one to read the whole input.

the probability it returns for each sentence can be calibrated and compared with a threshold chosen by the application. A traditional LLM, instead, produces its answer autoregressively: after reading the document it must generate the verdicts one token at a time, with a new pass through the model for every token written, and only at the end can the text it produced be interpreted as a decision (Figure 1.1).



(a) Decision model: one reading and, for every sentence, a choice among three admissible answers.



(b) Traditional LLM: one pass through the model for every token, until the text is complete.

Figure 1.1: Two ways of verifying the K sentences of an answer. Parallelograms: inputs and outputs; blue rectangles: components of the model; yellow rectangle: processing outside the model; diamonds: decisions. The decision model reads the document and the answer once and, for every sentence, chooses among three admissible answers, also returning their probability. The generative LLM passes through the model again for every token it writes (red loop), and the verdicts must then be extracted from the text, which may not follow the required format.

The field is young and moves quickly. After the first hosted model, open implementations followed [15, 16], and on 9 October 2026 the Hugging Face hub officially introduced a tag for decision models, which the following day already grouped about 1,500 models, more than half of them published in the previous weeks [17]. Established organisations released their own, such as Liquid AI's d1-3B and Cloudflare's Clef [18, 19]. The evidence, however, lags behind the releases. Most of it comes from the authors of each system, independent evaluations are recent and often less favourable [20, 21], and a common leaderboard finds that general-purpose decision models with fewer than one billion parameters barely exceed the trivial baseline on hallucination detection in retrieval-augmented answers [22]. All the open implementations, moreover, are general-purpose systems that answer any typed question.

Three problems therefore remain open. The first is whether, for a single task, a small specialised decision model is worth using. The second is how such a model compares with the alternatives: a general-purpose decision model, a generative judge and a dedicated verifier. The third is where any advantage comes from: from the specialisation, from the way the decision is read or, more simply, from the size of the model.

1.3 Approach and research questions

This thesis designs, trains and evaluates GroundOne, a compact decision model specialised in one decision: for every sentence of a RAG answer, whether the retrieved document supports it, contradicts it or does not mention it. The answer is written with a short marker after each sentence, and the verdicts of all sentences are read in one forward pass, from the logits of three answer letters at the markers, without generating any text. Qwen2.5 models with 0.5B and 3B parameters are fine-tuned with LoRA on human-labelled answers from RAGTruth [23, 24, 8], and their probabilities are calibrated on development data. Three test sets were built and frozen before any model was trained, so that no design choice could be tuned on them. The three open problems become five research questions: the first two restate them in a testable form, and the other three add what matters for real use on the device, namely the reliability of the probabilities, the cost and the value of a three-way decision.

RQ1, specialisation. Can a decision model of 0.5B or 3B parameters, fine-tuned on the task, decide whether each sentence is faithful to the document? And does it do so as well as larger models used zero-shot[2] and as the encoders specialised in this kind of verification?

[2]A model is used *zero-shot* when it tackles a task without any task-specific training and

RQ2, readout versus generation. How much does the decision-model mechanism, that is, the readout[3] of the decision from the logits, contribute at the same model size? The comparison is twofold: against generative judges that write their verdicts and against the same model left free to generate its answer. And how does the difference change as the number of sentences to verify grows?

RQ3, calibration. Are the probabilities returned by the model reliable as a measure of confidence? In particular, can they be used to set a decision threshold and to defer uncertain cases to the user? And are they better than those that can be obtained from a generative judge?

RQ4, cost on the device. Does the verifier fit within the memory, time and energy limits of a 16 GB laptop? And how much does it cost compared with a generative judge running on the same machine?

RQ5, beyond a binary score. What does a user of the verifier gain from a three-way decision for every sentence, compared with the plain score of a binary verifier?

1.4 Scope

The thesis studies the verification step in isolation. The retrieved document is given, so the quality of retrieval is not evaluated, and an answer is judged only against the document it was generated from, not against world knowledge. GroundOne flags sentences but does not correct them, and it returns verdicts and probabilities rather than written explanations. The data are in English. Documents longer than the context used in training are split into parts, as described in Chapter 6. Finally, the experiments target one class of hardware, a consumer laptop with unified memory, and every system, including every baseline, runs on the same machine.

1.5 Contributions

The thesis makes four contributions.

---

without examples in the prompt: it receives only an instruction that describes the task.

[3]The *readout* is the reading of the decision directly from the logits of the model: instead of letting it write an answer, the scores it assigns to the admissible answers alone (here the letters A, B and C) are compared and the most probable one is chosen.

1. **GroundOne**, an open specialist decision model for sentence-level groundedness verification that runs entirely on a laptop. Thanks to a marker placed after every sentence, it reads all the verdicts of an answer in one forward pass. It is released as an installable package with a command-line interface and a local HTTP API, which also includes an interface compatible with the System One interface of Jev, already used by other clients.

2. **A sentence-level dataset derived from public annotations.** No label was annotated anew: the human annotations of RAGTruth, which mark spans of text, were converted into a three-way label for every sentence, with rules fixed in advance. The three test sets come from public benchmarks, were checked automatically, for instance on sentence segmentation, and were frozen before any model was trained, with leakage controlled at the level of documents.

3. **A controlled decomposition of the gains.** The comparison isolates the contribution of specialisation, of the restricted readout and of model size. GroundOne is compared with cues that never verify a sentence, untrained models, generative judges with and without probabilities, two general-purpose decision models (Rizzo-Flow, with 4B parameters, and Liquid AI's d1-3B, of the same size as the 3B) and specialised encoders, all on the same hardware and with intervals obtained by cluster bootstrap.

4. **A study of calibration, time and energy**: how the probabilities can be calibrated and turned into consistent labels, how much they help in deferring uncertain decisions, and how the time and energy of a verification grow with the length of the answer. The study is set in an analysis of local inference that combines vendor specifications with own measurements to bound, for each class of consumer hardware, what a verifier that generates text must pay and a decision model does not.

1.6 Main results

The results show that specialisation works in the domain it was aimed at. On the RAGTruth test set GroundOne reaches a three-way balanced accuracy of 0.769 with 0.5B parameters and 0.832 with 3B, against 0.37 and 0.44 for the same models before training. It ranks supported sentences better than the specialised encoders, and it clearly outperforms both general-purpose decision models: Rizzo-Flow, eight times larger than the 0.5B, and d1-3B, of the same size as the 3B. On LLM-AggreFact, a benchmark drawn from different sources, the 3B is on par with the best small encoder.

The decision-model mechanism pays off in speed and in accuracy. When an answer contains fourteen or more sentences, GroundOne verifies it about two times faster than a terse judge and six times faster than a verbose judge of the same size, with an output that is always valid and with less energy. How the decision is read also matters: with the same trained model, choosing the most probable of the three admissible letters is more accurate than letting the model generate its verdict freely, with a gain in balanced accuracy of 0.25 for the 0.5B and 0.07 for the 3B. After calibration, the probabilities of GroundOne are close to the observed frequencies, whereas the probabilities of generative judges asked for the same output barely separate correct from incorrect decisions.

Some aspects remain to be improved, and they point to natural directions for future work. Performance drops on domains far from the training data, where the general-purpose decision model d1-3B is stronger; the subtlest hallucinations, hidden in sentences that otherwise follow the source faithfully, are detected only in part; and training is not yet fully stable across random seeds[4].

1.7 Thesis structure

Chapter 2 is devoted to local inference: the reasons for keeping models on the device, open-weight and small models, the cost model of inference on one device, the hardware landscape, runtimes and quantisation, energy, and the consequence for verification. Chapter 3 reviews decision models and their ecosystem, hallucination in retrieval-augmented generation, verification methods and calibration. Chapter 4 describes GroundOne: the decision, the single-pass readout, the training objective, the calibration and the served output. Chapter 5 defines the labels and describes how the training and test sets were built, checked and frozen. Chapter 6 presents the trained models, the baselines, the metrics and the measurement protocol. Chapter 7 reports the results, Chapter 8 discusses what they mean and where they stop, and Chapter 9 concludes. Appendix A shows the output of every system on the example answer described in Section 1.1.2.

[4]The *seed* is the number that initialises the random number generator used during training, for instance to initialise the added weights and to shuffle the order of the data. Repeating the training with different seeds produces slightly different models: if the results change a lot from one seed to another, the procedure is not robust.

Chapter 2

Local Inference of Language Models

This thesis builds a verifier that runs entirely on the user's own machine, and this choice shapes every other one: the size of the models, the way decisions are read from them, the measurements and the comparisons. This chapter explains why local inference matters, what makes it possible today and what it costs. It starts from the reasons for keeping a model on the device and from the open-weight models that make this possible. It then derives a simple cost model of inference on one device, which shows why reading a token costs far less than writing one, and surveys the hardware on which local models run, from the laptop used in this work to dedicated desktop machines. It continues with the software and the quantisation that bring models within reach of that hardware, and with energy. It closes by drawing the consequence for verification, which motivates the design of GroundOne in the following chapters.

2.1 From hosted to local models

Most people use language models through hosted services. The user's input travels over the network to a provider, a model in a datacenter processes it, and the answer travels back. The model, the hardware and the policies that govern the data all belong to the provider. Running the model on the user's own device changes all of this at once. Five properties follow, and each of them matters for a retrieval-augmented assistant that works on private documents.

Privacy as a technical guarantee. With a hosted model, every document and every question is sent to a third party, and what happens to them afterwards, how long they are kept, who can access them and whether they are used to improve

future models, is governed by contract and policy. The user can read the policy but cannot verify that it is respected. With a local model the documents never leave the machine, so the question does not arise: privacy becomes a property of the system rather than a promise. Surveys of edge intelligence, that is, of running models on peripheral devices close to the user, state the point directly: on-device inference eliminates privacy leakage and the need for an Internet connection [25].

European law makes the difference concrete. Under the General Data Protection Regulation (GDPR), the processing of personal data explicitly includes its "disclosure by transmission" [26, Art. 4(2)]. Sending a document that contains personal data to a hosted model therefore makes the provider a processor[1], bound by a contract with the specific guarantees of Article 28. If, in addition, the provider operates outside the European Union, the transfer of the data must have a legal basis: an adequacy decision[2], appropriate safeguards or one of the derogations provided by Articles 44–49 [26]. A verifier that runs on the user's device involves no processor and no transfer. It does not remove every obligation, since the organisation that uses it must still respect the principles of Article 5 and secure the device, but it removes the obligations that depend on trusting someone else. Many confidential documents, such as unpublished research, contracts or source code, contain no personal data at all; for them the regulation does not apply, and the argument becomes one of confidentiality.

A recent episode illustrates what is at stake. In September 2026 OpenAI announced that an internal multi-agent system had produced a proof of blow-up (the formation of a singularity in finite time) for the forced Navier–Stokes equations, one of the Millennium Prize problems, formalised in the Lean proof assistant [7]. Two mathematicians, Tristan Buckmaster and Levent Alpöge, had released related results for the Euler equations at about the same time, and disputed the credit. Buckmaster had written his drafts in OpenAI's own coding assistant, and asked the company "whether the model had been trained on, or had access to, our sessions in Codex, into which we had been putting all our drafts" [27]. OpenAI denied that its researchers or agents had seen the work before its release, but added: "While unlikely, we cannot rule out that de-identified data derived from their usage of our products helped improve our models" [7, 28]. Nothing in the public record shows that the drafts were misused, and the dispute is mainly about credit. What the episode does show is that, once unpublished work is inside a hosted model, neither the user nor, by its own statement, the provider can exclude that it contributed to the provider's models. With a local model this question cannot be asked.

[1] In the GDPR, the *controller* decides why and how personal data are processed; the *processor* processes them on behalf of the controller, for instance by providing a computing service.

[2] A decision by which the European Commission recognises that a third country ensures a level of data protection essentially equivalent to that of the European Union.

Providers are aware of the problem. Apple describes on-device processing as the cornerstone of its approach to privacy, and for requests that exceed the device it built Private Cloud Compute, a dedicated cloud with custom servers, no privileged access for staff, no persistent storage of requests and a public log of the software images that run on it [29]. The design is a serious attempt to make hosted inference verifiable, and its very complexity is an argument for locality. Privacy in the cloud requires custom silicon, a hardened operating system and cryptographic attestation, that is, a signed proof of which software is running on the servers. Even so, the system rests on trust in the hardware, in the supply chain and in the attestation itself, which only security researchers, not users, can inspect [29].

Availability. A local model works without a network and without depending on the availability of a provider. For an assistant used on a train, in a hospital ward with restricted connectivity or in an air-gapped environment, that is, one physically isolated from any network, this is a requirement rather than a convenience.

Stability and reproducibility. The weights of a local model change only when the user changes them. A hosted model can be updated, re-tuned or withdrawn at any time, often silently, so that the same request produces a different answer from one month to the next. For a verifier whose thresholds are calibrated on development data (Chapter 4), a silent change of the underlying model would invalidate the calibration without warning.

Cost. A hosted model is paid per token, and a verifier that checks every answer multiplies the number of tokens processed. A local model is paid once, with the hardware, and then costs only its energy.

Access to the model's internals. A local model exposes its logits, the scores from which every probability is computed. Hosted interfaces often hide them or return them only for a few tokens. The approach studied in this thesis reads the decision directly from the logits of a few answer letters and calibrates them, so it requires this access; it is, in this sense, possible only with a local model.

The price of locality. These advantages have a cost. A laptop holds only small models, and it must hold them next to the operating system, the user's applications and every other component of the assistant. It has less memory bandwidth and less computation than a datacenter accelerator, and it runs on a battery. Part of the literature concludes that pure on-device deployment is too limited and proposes to split the work with an edge server, that is, a server close to the user, or with the cloud [25, 30]: a small local model handles easy requests and forwards hard ones,

or drafts an answer that a larger remote model checks. Apple's combination of an on-device model with Private Cloud Compute is the most visible example [4, 29]. These hybrid designs reduce the cost of locality but sacrifice part of its benefits, since whatever is forwarded leaves the device. This thesis takes the strict position: the whole verification stays on the device, and the rest of the chapter examines what this requires.

Table 2.1 summarises the comparison.

Table 2.1: Hosted and local inference compared on the properties that matter for a verifier of private documents.

Property	Hosted model	Local model
Privacy	Contractual: documents are sent to a processor; the user cannot verify how they are used.	Technical: documents never leave the device.
Regulation (GDPR)	Processor contract (Art. 28); third-country transfers (Arts. 44–49).	No processor and no transfer; Art. 5 principles and device security still apply.
Availability	Requires network and provider uptime.	Works offline.
Stability	The provider may update or withdraw the model.	Weights change only when the user changes them.
Cost	Price per token.	Hardware once, then energy.
Logits	Often hidden or limited.	Fully accessible: readout and calibration possible.
Capability	Frontier models.	Small models, limited by memory, bandwidth and battery.

2.2 Open-weight and small language models

Local inference requires a model whose weights can be downloaded. This section describes what such models are, how far they trail the best closed models, and which of them are small enough for a personal device.

2.2.1 Open weights are not open source

A model is called *open-weight* when its trained parameters are published and can be run and modified by anyone. This is not the same as open source. The Open Source Initiative defines an open-source AI system through the four freedoms of the

free-software tradition, to use, study, modify and share the system, and requires that the form preferred for modifying it be available: the parameters, the complete code used to train and run it, and enough information about the training data for a skilled person to build a substantially equivalent system [31]. The data themselves need not be released, but weights alone do not qualify. Most models that are commonly called open publish their weights and an inference recipe, but neither the training code nor a detailed description of the data. Being open is therefore a matter of degree rather than a binary property. A recent index scores models on five aspects: the availability of the weights, the disclosure of the pre-training and of the post-training data (the two phases of training, first on generic text and then on instructions and preferences), the methodology and the licence [32]. No model reaches the maximum score; fully documented research models such as OLMo score high, whereas some popular open-weight models publish little beyond the weights.

Licences add a further distinction. The two models fine-tuned in this thesis come from the same family but carry different licences: Qwen2.5-0.5B is released under the permissive Apache 2.0 licence, whereas Qwen2.5-3B is released under a research licence that restricts commercial use [23]. In the terminology above both are open-weight models, and the thesis uses this term for them. The distinction also matters for regulation. The European AI Act relaxes some transparency obligations for general-purpose models released under free and open-source licences, but not for models that pose systemic risk. A model is presumed to pose systemic risk when its training required more than 10^{25} floating-point operations (FLOP[3]) [33], far more than those used for the models of this thesis.

2.2.2 The gap with closed models is closing

Local inference would be of limited interest if open-weight models were far behind closed ones. They are not. On the Chatbot Arena leaderboard, where users compare the answers of two anonymous models, the lead of the best closed model over the best open-weight model fell from 8.04 % in January 2024 to 1.70 % in February 2025 [6]. On Epoch AI's capabilities index, which aggregates many benchmarks, the best open-weight models trailed the closed frontier by about four months on average between January and May 2026 (Figure 2.1) [34].

A more recent snapshot confirms the picture and also shows its limit. On the Artificial Analysis Intelligence Index, which aggregates ten evaluations, on 20 October 2026 the best closed model, Claude Opus 5.5, scores 58 points, while the best open-weight model, MiMo-V2.6-Pro, scores 46 (Figure 2.2) [35]. Several

[3]A FLOP is a single arithmetic operation on floating-point numbers, such as an addition or a multiplication. Computing speed is measured in FLOP per second: one TFLOP/s corresponds to 10^{12} operations per second.

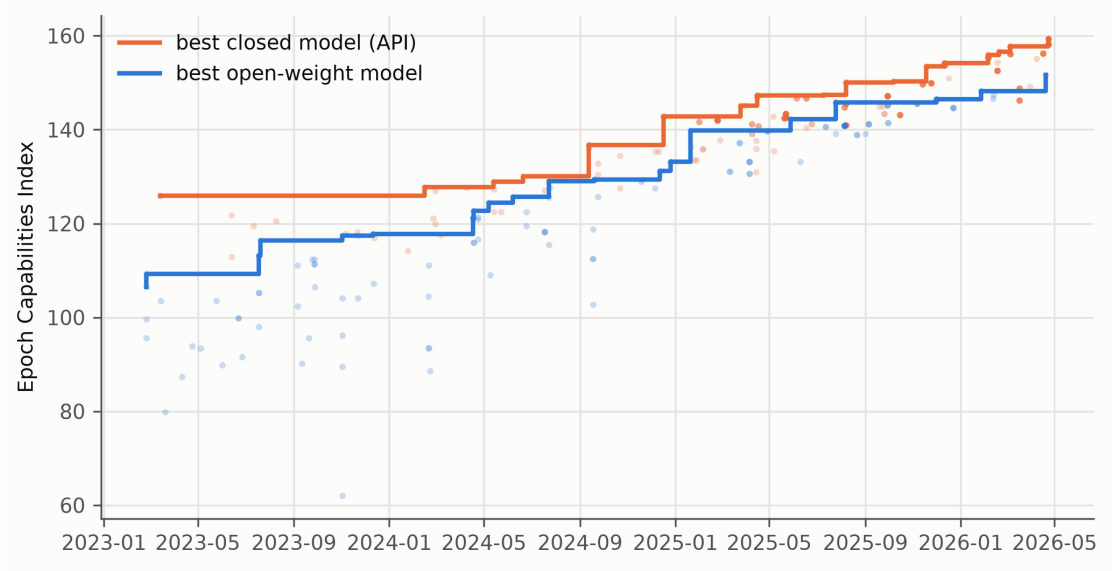


Figure 2.1: Capability of the best closed and the best open-weight model over time on Epoch AI's aggregate capabilities index (dots: individual models; lines: best so far). Plotted from Epoch AI's published data table [34].

open-weight models, such as DeepSeek V4.1 Flash, GLM-5.3 and Kimi K3, match or exceed mid-range closed models such as Claude Haiku 5.5 and Gemini 3.8 Flash. All these open-weight models, however, belong to the class of large models, whereas the only model in the chart that Artificial Analysis classifies as small, Qwen3.8 27B, stops at 34 points.

This does not mean that large models are excluded from local hardware. The specialised inference engines described in Section 2.5.2, such as ds4 and Colibri, run large open-weight models, for example DeepSeek V4 Flash and GLM-5.2, even when their weights do not fit in memory: they keep the always-active part resident and read the rest from the SSD. The limit then shifts from capacity to speed. According to the authors of Colibri, GLM-5.2 generates 1.83 tokens per second on a Strix Halo machine with 128 GB and less than one token per second on a laptop [36]. For a personal device, therefore, the question is not only which model can be run, but how fast it can generate text; and generation is precisely the phase that a decision model avoids.

For interactive use, what matters most is the best model that fits entirely in the memory of consumer hardware, and can therefore generate without depending on the bandwidth of the SSD. Epoch AI tracks this quantity directly. Considering models small enough to run on a single consumer graphics card, at most about 28 billion parameters for a card with 24 GB and about 40 billion for one with 32 GB, it finds that across four benchmarks they reach the performance of the

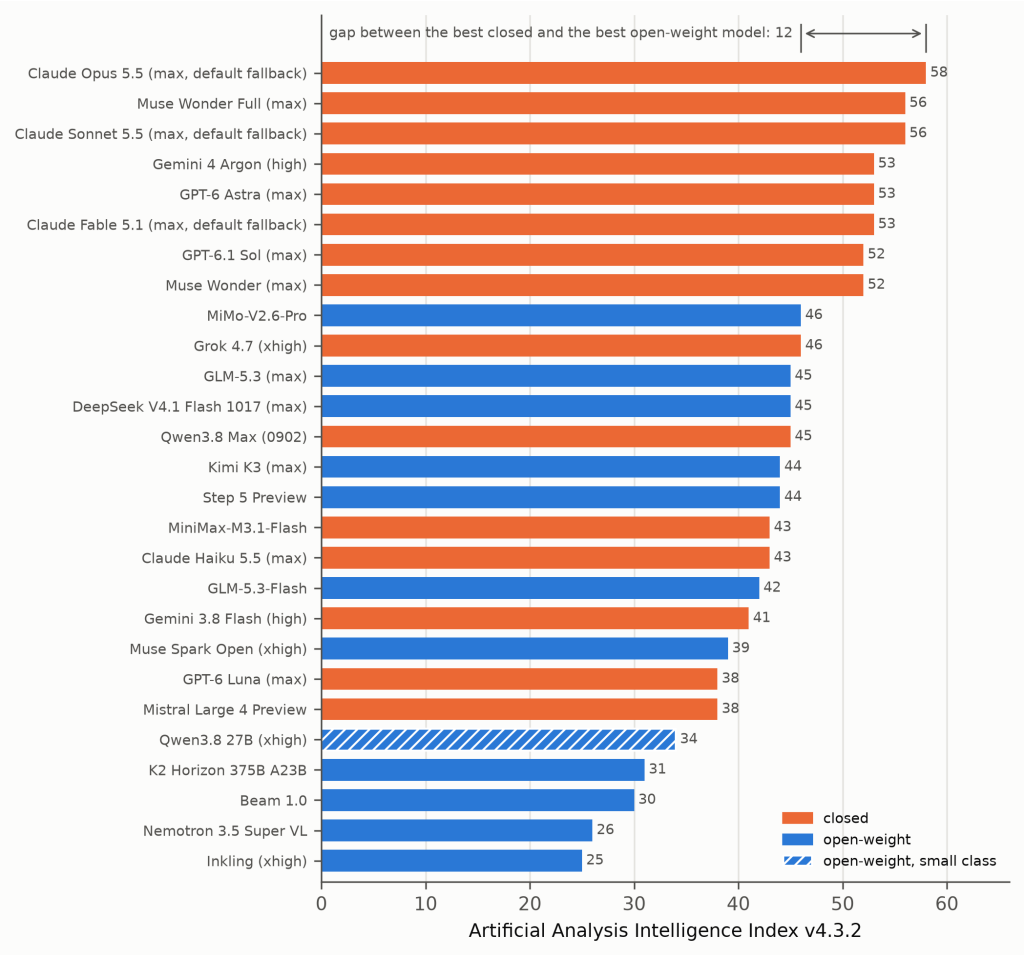


Figure 2.2: Artificial Analysis Intelligence Index (version 4.3.2, ten evaluations) on 20 October 2026, redrawn from the published chart [35]. Closed models in orange, open-weight models in blue; hatched, the only model that Artificial Analysis classifies as small. In brackets, the reasoning setting at which each model was evaluated. The open-weight status is taken from the data of the same leaderboard. It is a snapshot: scores and ranking change with every new release.

frontier model of the time after six to twelve months [5]. Three forces drive the catch-up: open-weight developers scale their training compute about as fast as closed ones, distillation, that is, training a small model to imitate the answers of a large one, transfers much of its capability, and each new generation of graphics cards raises the size of the models that fit [37]. The same trend appears from the other side: the smallest model scoring above 60 % on the MMLU benchmark, a set of multiple-choice questions on 57 subjects, had 540 billion parameters in 2022 and 3.8 billion in 2024, a reduction of 142 times [6].

2.2.3 Small language models

At the low end of the scale are *small language models*. Definitions vary; a survey that measures several dozen of them on devices places the range at 100 million to 5 billion parameters, with a decoder-only architecture, that is, the architecture of common LLMs that generate text from left to right [38]. Small models are trained on far more data than would be worthwhile if the only aim were to minimise the cost of training. The reason is that, for them, the cost that matters is inference, repeated millions of times after training: it is therefore worth spending more once to obtain a small but capable model. A model of half a billion parameters can thus see trillions of tokens [38]. Measured on edge devices with 4-bit weights, models between 0.1 and 3 billion parameters need between 0.3 and 2.5 GB of memory, and at long context most of it is taken by the key-value cache, which stores the context already read (Section 2.3.1), and by the computation buffers [38]. Research on models below one billion parameters argues that, on phones, an application should not use more than about a tenth of the device's memory, and shows that a carefully designed 350-million-parameter model matches a general 7-billion-parameter model on a narrow task such as calling an application interface [39].

Platform vendors have adopted the same idea. Apple's foundation model for its devices has about three billion parameters, is quantised to under four bits per weight and is specialised for each feature with small low-rank adapters, additional modules with few parameters trained on the single task (Section 3.5) [4]; Google's Gemma 3n family is designed to run on phones and laptops with a reduced memory footprint, and is marketed explicitly for privacy and offline use [40]. GroundOne follows the same pattern: a small open-weight model, quantised, specialised by an adapter for one task.

2.3 The cost of inference on one device

The previous sections explain why a model should run locally and which models can. This section explains how much it costs to run one, and why the cost depends so strongly on what the model is asked to do.

2.3.1 Prefill and decode

A decoder-only language model processes a request in two phases. In the *prefill* phase it reads the whole prompt at once. To do so it uses the attention mechanism, through which each token gathers information from the tokens that precede it: every layer computes, for each input token, two vectors (a *key* and a *value*) that later tokens will consult, and stores them in the *key-value cache*. In the *decode* phase it produces the output one token at a time; each new token consults the cache through attention and extends it by one position [10]. Prefill processes many tokens in parallel; decoding processes one. Serving systems exploit the difference in several ways, for instance by managing the cache in pages to fit more concurrent requests [41] or by running the two phases on different machines suited to each [42].

The consequence is visible on the laptop used in this thesis (Figure 2.3). With the 0.5B model, reading one token of the prompt takes less than 0.4 milliseconds, while writing one token of output takes 5.9 milliseconds, sixteen times more. With the 3B model the figures are 2.2 and 31.8 milliseconds, fourteen times more. A model that writes a short paragraph thus spends more time on it than on reading a document of several thousand tokens.

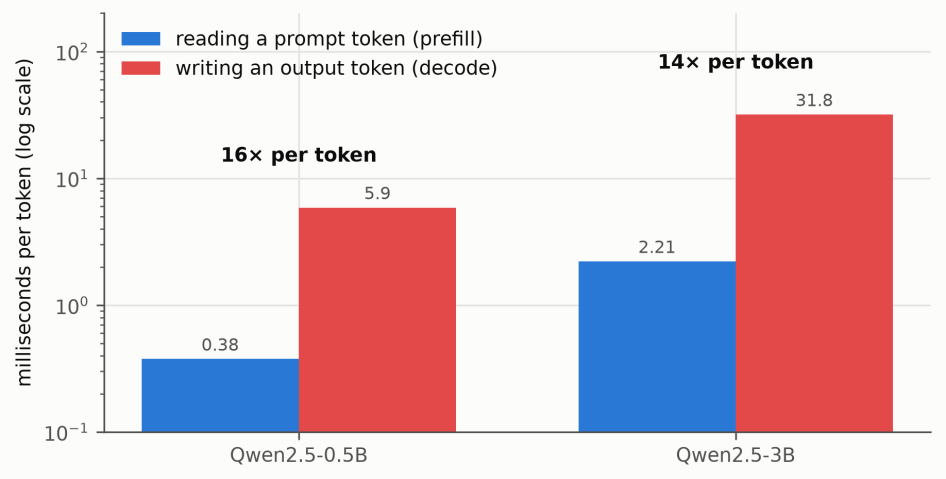


Figure 2.3: Time per token when the model reads its input (prefill) and when it writes its output (decode), measured on the thesis machine (Apple M4, 16 GB, MLX 8-bit, one request at a time). Prefill: median over contexts of 500, 1,000 and 2,000 tokens.

2.3.2 The roofline model

The reason for this asymmetry is captured by the roofline model, a simple model of performance introduced for multicore processors [43]. A computation is characterised by its *arithmetic intensity*, the number of floating-point operations it performs for each byte it reads from memory. A processor has two limits: a peak rate of computation and a peak memory bandwidth. The attainable rate is the minimum of the peak computation and the product of bandwidth and intensity. A computation with low intensity is limited by the bandwidth, and one with high intensity by the computation; the intensity at which the two limits meet is called the ridge point [43].

Applied to a transformer, the model separates the two phases cleanly [44]. Each weight matrix of a linear layer must be read from memory once per forward pass. During prefill the same matrix multiplies the hidden states of hundreds or thousands of tokens, so each byte of weights is used for hundreds of operations: the intensity is high and the computation is the limit. During decoding, the matrix multiplies a single vector, so each weight is used for about two operations: the intensity is close to one and the bandwidth is the limit. For a 7-billion-parameter model in 16-bit precision, the linear layers reach intensities above one thousand operations per byte in prefill and about one in decode [44]. In datacenters this imbalance is softened: the decoding of many concurrent requests is grouped into batches, so that each weight read from memory serves to generate several tokens and the intensity rises [10]. A personal device serves a single user, one request at a time, and cannot rely on this remedy: this is where the asymmetry between the two phases weighs most.

Figure 2.4 draws the roofline of the thesis machine. Apple publishes its memory bandwidth, 120 GB/s, but no peak rate of computation [45]; the flat roof in the figure is therefore the rate actually reached during prefill, about 2.8 TFLOP/s, which is a lower bound of the peak. The ridge point lies at about 23 operations per byte. Generating a token sits at about two operations per byte, an order of magnitude below the ridge, whereas reading a document of hundreds of tokens sits an order of magnitude above it. The two phases of the same model, on the same machine, live in different regimes.

2.3.3 A lower bound on generation

Because decoding reads every weight once per generated token, its time per token cannot be lower than the size of the weights divided by the memory bandwidth, whatever the hardware. For the 3B model in 8-bit precision, whose weights occupy about 3.3 GB, this floor is 27 milliseconds per token on a machine with 120 GB/s; the measured 31.8 milliseconds are only about one sixth above it. The bound depends only on two numbers, the size of the model and the bandwidth of the

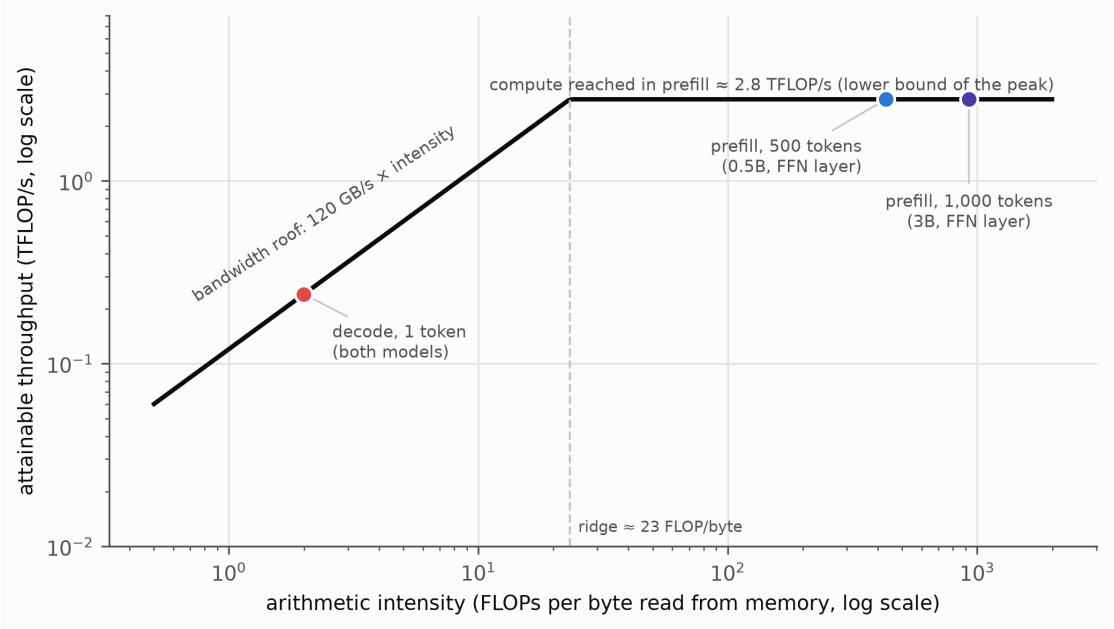


Figure 2.4: Roofline model of the thesis machine [43]. Sloped roof: Apple's memory bandwidth (120 GB/s). Flat roof: throughput reached in prefill, a lower bound of the unpublished peak. Intensities computed for one feed-forward layer of Qwen2.5 with 8-bit weights and 16-bit activations; attention is ignored.

machine, and it applies to every model that generates text. It is the key to comparing hardware in Section 2.4 and to the consequence drawn in Section 2.7.

Memory capacity sets a second, harder limit: the weights must fit. A model needs about two bytes per parameter in the 16-bit format used for training, one in 8 bits and half in 4 bits, plus a few small scale factors for each group of weights, needed to reconstruct their values (Figure 2.5). A 7-billion-parameter model in 16 bits needs 14 GB for its weights alone, beyond what most personal devices can spare [46]; in 4 bits it needs about 4 GB. The key-value cache, the activations, the operating system and every other model that runs at the same time come on top. A laptop with 16 GB can therefore hold a generator of 7–8 billion parameters in 4 bits next to a small verifier, but not much more.

2.4 Hardware for local inference

The cost model reduces the choice of hardware for local inference to three quantities. *Memory capacity* decides which models fit. *Memory bandwidth* decides how fast they can generate. *Computation* decides how fast they can read. Power and price complete the picture. This section surveys the hardware available in October 2026

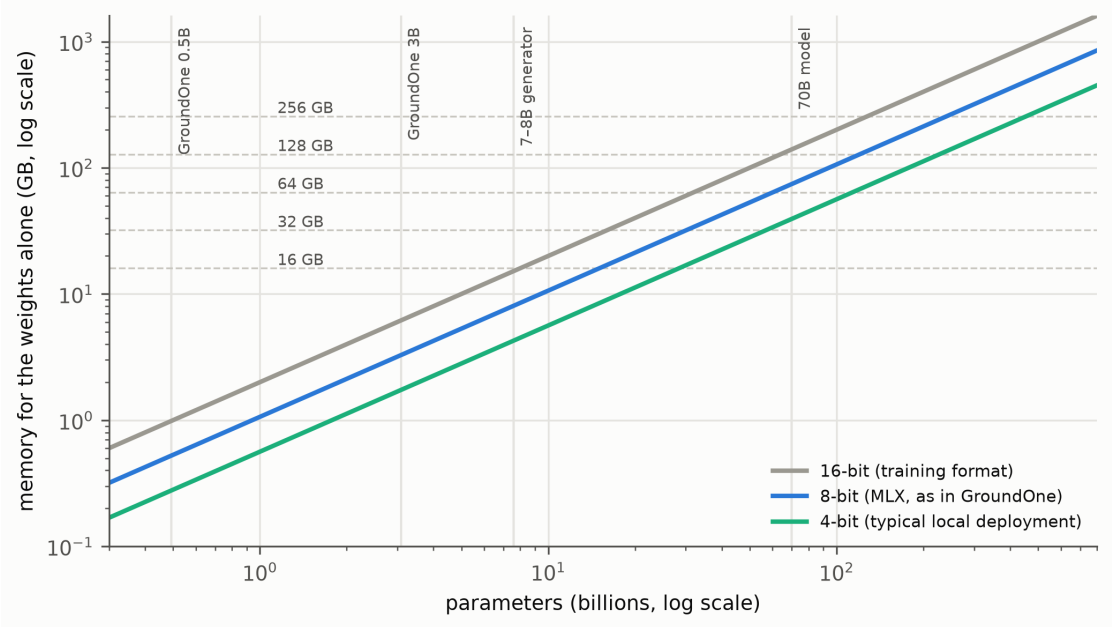


Figure 2.5: Memory needed for the weights of a model as a function of its size and precision (parameters times bits per weight, including half a bit of per-group scales for the quantised formats). Dashed lines: common memory sizes of local hardware. The key-value cache, activations and other running models come on top.

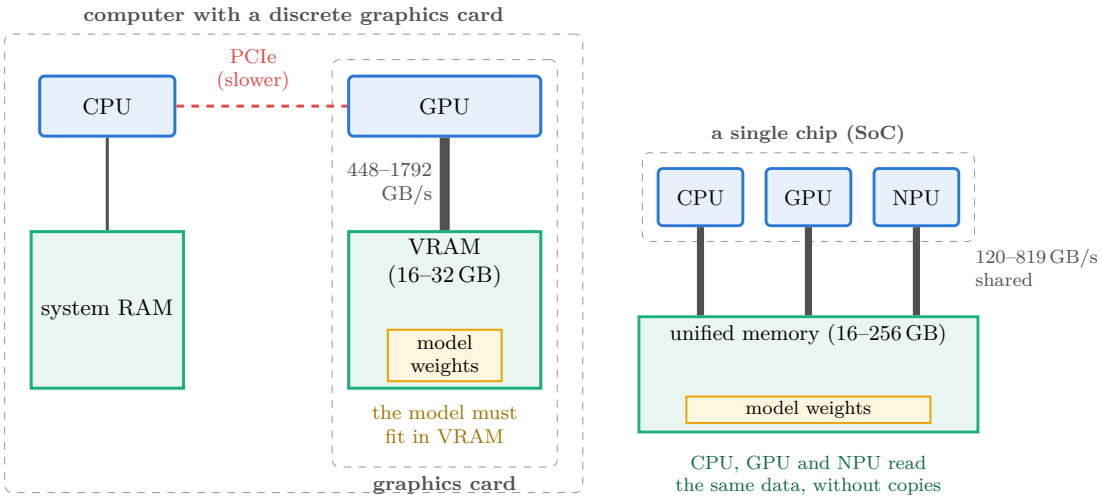
along these axes, using the vendors' published specifications.

2.4.1 Classes of hardware

Two architectures dominate. In a *discrete* system, the graphics card has its own memory, fast but small, and the model must fit in it. In a *unified-memory* system, the processor and the graphics unit share one large pool of memory, so a model can use most of the machine's memory, at a lower bandwidth than a high-end graphics card. Apple's processors, NVIDIA's DGX Spark and AMD's Ryzen AI Max (Strix Halo) all follow the second design. Figure 2.6 compares the two architectures.

Table 2.2 and Figure 2.8 place the main options on the two memory axes. Figure 2.7 shows some of the systems discussed, to give a concrete idea of what they look like.

Laptops. Laptops with 16 to 32 GB and 120 to 150 GB/s, such as the MacBook Pro used in this thesis, are the most common personal computers on which local models run. They hold small models comfortably and models of 7–8 billion parameters in 4 bits. The thesis machine sits at the bottom left of Figure 2.8: if a



(a) Discrete system: the GPU reads the weights from its own VRAM, very fast but limited to 16–32 GB; the rest of the system is reachable only through PCIe.

(b) Unified memory: CPU, GPU and NPU on the same chip share one pool of memory, larger but with lower bandwidth.

Figure 2.6: The two architectures of local inference hardware. Blue rectangles: compute units; green rectangles: memory; orange rectangle: the model weights; line width: indicative bandwidth. Capacity and bandwidth values from Table 2.2. In a discrete system a model that does not fit in VRAM must go through the much slower PCIe link; in a unified-memory system the model can occupy almost all of the machine's memory, but every generated token reads it at a lower speed than a high-end graphics card.

verifier runs well there, it runs on everything above it.

Desktop machines with unified memory. A second class, within or close to a budget of about three thousand dollars, offers much more memory at moderate bandwidth. AMD's Strix Halo processor, sold for instance in the Framework Desktop, lets the graphics unit use up to 96 GB of its 128 GB of memory; the 128 GB configuration was priced at 1,999 dollars at launch in August 2025, with a processor power of 55 W [51]. Apple's Mac mini with the M5 Pro reaches 64 GB at 307 GB/s [48]. Such machines can hold models of tens of billions of parameters or mixture-of-experts models, which are composed of many sub-networks (the experts) of which each token activates only a part, so that decoding reads only the weights of the active experts, at the price of a slower generation than on a graphics card.



(a) 14-inch MacBook Pro with M4, the thesis machine.



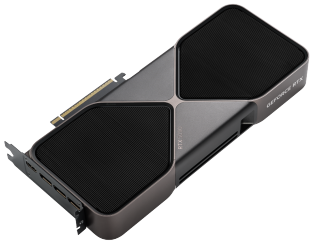
(b) Mac mini (2024, M4).



(c) Mac Studio.



(d) NVIDIA DGX Spark.



(e) GeForce RTX 5090 Founders Edition graphics card.

Figure 2.7: Some of the local inference systems compared in this section (images not to scale). Sources: (a), (c) Apple; (b) LoMit, Wikimedia Commons, CC0; (d), (e) NVIDIA. The vendors' images are promotional material, reproduced for the purpose of scientific documentation.

Dedicated AI machines. Above this budget, the Mac Studio with the M3 Ultra combines up to 256 GB with 819 GB/s [49], whereas NVIDIA's DGX Spark is a small desktop box, with a 240 W power supply, that contains 128 GB of unified memory with 273 GB/s of bandwidth and a Blackwell graphics unit rated at one petaflop (10^{15} operations per second) in 4-bit arithmetic, a figure that exploits sparsity, that is, the skipping of zero weights [50]. The two machines make opposite choices, and a direct comparison shows the cost model at work: on an 8-billion-parameter model with an 8,000-token prompt, the DGX Spark completes the prefill 3.8 times faster than the Mac Studio, thanks to its higher computation, while the Mac Studio generates 3.4 times faster, thanks to its three times higher bandwidth [53]. The same study combines the two machines, running the prefill on the Spark and the decoding on the Mac, which is the disaggregation of the two phases [42] brought to a desk.

Table 2.2: Hardware for local inference: vendors' published specifications (accessed 2026-10-11). Memory: range of the configurations offered; bandwidth: for those configurations; n/s: not stated by the source.

System	Memory (GB)	Bandwidth (GB/s)	Memory type	Source
MacBook Pro, M4 (thesis machine: 16 GB)	16–32	120	unified	[45]
MacBook Pro / Mac mini, M5	16–32	153	unified	[47, 48]
Mac mini, M5 Pro	24–64	307	unified	[48]
MacBook Pro, M5 Max	36–128	460–614	unified	[47]
Mac Studio, M4 Max	36–128	410–546	unified	[49]
Mac Studio, M3 Ultra	96–256	819	unified	[49]
NVIDIA DGX Spark (GB10)	128	273	unified	[50]
Framework Desktop, AMD Strix Halo	32–128 (96 to GPU)	n/s	unified	[51]
NVIDIA RTX 5090	32	1,792	discrete	[52]
NVIDIA RTX 5080	16	960	discrete	[52]
NVIDIA RTX 5060 Ti	16	448	discrete	[52]

Graphics cards. Discrete cards offer by far the highest bandwidth, up to 1,792 GB/s for the RTX 5090, but at most 32 GB of memory, at a power of several hundred watts [52]. They are the fastest option for any model that fits, and unusable for any model that does not.

Phones and mobile processors. At the other end, phones have 6 to 16 GB of memory and a bandwidth of a few tens of gigabytes per second shared by all processors [30, 54]. Their neural processing units (NPUs) are fast at 8-bit matrix multiplication and therefore at prefill: on a phone, a 7-billion-parameter model in 4 bits reads its prompt at about 770 tokens per second on the NPU, against 37 on the graphics unit and 8 on the processor cores [54]. Research systems exploit this by running the prefill on the NPU and the decoding elsewhere [55].

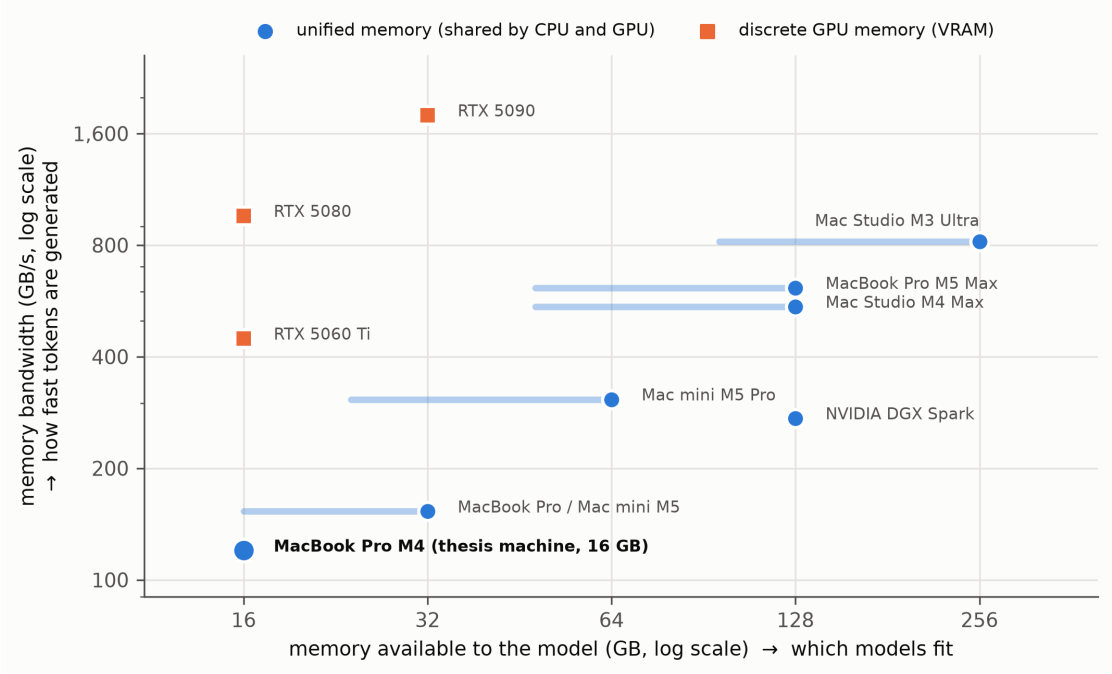


Figure 2.8: Local inference hardware on the two axes that decide what can run: memory capacity (which models fit) and memory bandwidth (how fast tokens are generated). Vendors' specifications: markers at the largest memory configuration, shaded bars over the range of memory sizes offered; the thesis machine is shown at its actual 16 GB.

2.4.2 The asymmetry holds on every class

The two phases respond to different properties of the hardware, and recent hardware generations make the point explicit. Apple reports that its M5 processor, which adds dedicated matrix units to each graphics core, reduces the time to first token, that is, the wait before the model starts answering, which is dominated by prefill, by 3.3 to 4.1 times with respect to the M4, while generation becomes only 1.19 to 1.27 times faster, in line with a memory bandwidth that grew from 120 to 153 GB/s [56]. Apple itself attributes the first figure to computation and the second to bandwidth. On edge devices, a study that compares 18 pairs of configurations finds that slowdowns accumulate in decoding, while the time to evaluate the prompt changes by at most 16 % [57]. On phones, for tasks whose output is short, prefill takes most of the latency [55]. Hardware is thus evolving to read faster, much more than to write faster.

Figure 2.9a applies the lower bound of Section 2.3.3 to the hardware of Table 2.2. On the fastest card, the 3B model could in principle generate a token in 1.8

milliseconds; on the thesis machine the floor is 27 milliseconds, and the measured time is close to it. Better hardware lowers the floor, but no hardware removes it.

2.5 Runtimes and quantisation

2.5.1 Runtimes

The software for local inference has matured quickly. llama.cpp provides inference in C and C++ without external dependencies, with quantisation and support for most consumer accelerators [2]. It also defined the GGUF format, which gathers in a single file the quantised model and everything needed to use it, such as the tokenizer and the conversation format. Tools such as Ollama and LM Studio build on it and reduce downloading and starting a model to a single command. MLX is a computing framework designed by Apple for the unified memory of its processors, with an interface close to NumPy [3]. Its language-model package supports quantisation and LoRA fine-tuning directly on the device [58], and it is the runtime used throughout this thesis.

The runtime matters as much as the chip. A comparison of five runtimes on the same Apple machine reports decoding rates that differ by more than an order of magnitude, with MLX and MLC-LLM at the top and PyTorch's generic Metal back end at the bottom [59]; the study's numbers are not fully consistent with each other and should be read as indicative, but the spread is large enough to survive the uncertainty. On phones, an evaluation of several runtimes and devices finds large differences in speed, energy and thermal behaviour between runtimes and between devices [60].

2.5.2 Specialised inference engines

Next to the general runtimes, a second family of software appeared in 2026: inference engines written for a few models, which give up generality to push further the memory limits described in Section 2.3. DwarfStar (ds4), developed by Salvatore Sanfilippo, is a native engine optimised for a few large open-weight models, starting from DeepSeek V4 Flash, and aimed at Macs with at least 96 GB of memory, the DGX Spark and Strix Halo machines [61]. Two of its choices concern this thesis directly. The first is a key-value cache saved on disk: a conversation that has already been read can be resumed without repeating the prefill, which for models of this size is the most expensive part of a long request. The second is reading the weights directly from the SSD, which makes it possible to run models larger than the available memory, at the price of speed.

Colibri takes this second idea to the extreme [36]. It exploits the fact that a mixture-of-experts model uses only a small part of its weights for each token: it

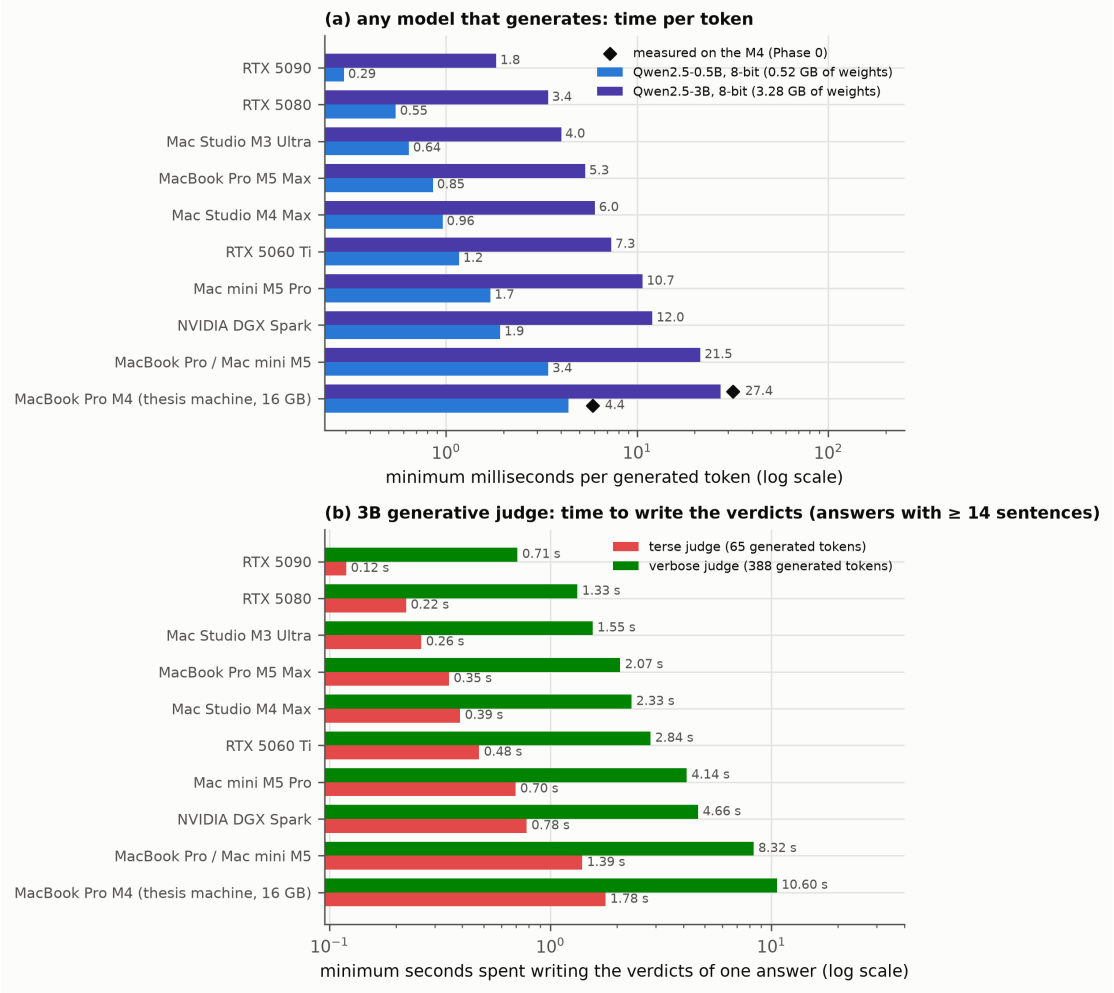


Figure 2.9: Lower bounds on the cost of generating text on each class of local hardware, computed as weight bytes divided by the vendor's bandwidth (one request at a time; key-value cache ignored). (a) Time per generated token with the two GroundOne base models in 8 bits; diamonds: decode time measured on the thesis machine. (b) Minimum time that a 3B generative judge spends only on writing the verdicts for an answer of fourteen or more sentences (median generated tokens of the latency protocol). A decision model generates no token, so both costs are zero for it on every machine.

keeps the always-active part in memory and reads the experts from disk when the router requests them. In this way GLM-5.2, a model of 744 billion parameters, runs on a laptop with 25 GB of memory. The price is the one predicted by the cost model: the bandwidth of an SSD is a few gigabytes per second, far below that of memory, and according to the authors generation drops to 0.05–0.1 tokens per second when cold on that laptop and to 1.83 tokens per second on a 128 GB Strix Halo machine [36]. It is the idea of LLM in a flash [46] applied to expert models: memory capacity stops being a hard limit and turns into a bandwidth limit, an even stricter one. Figure 2.10 illustrates the mechanism.

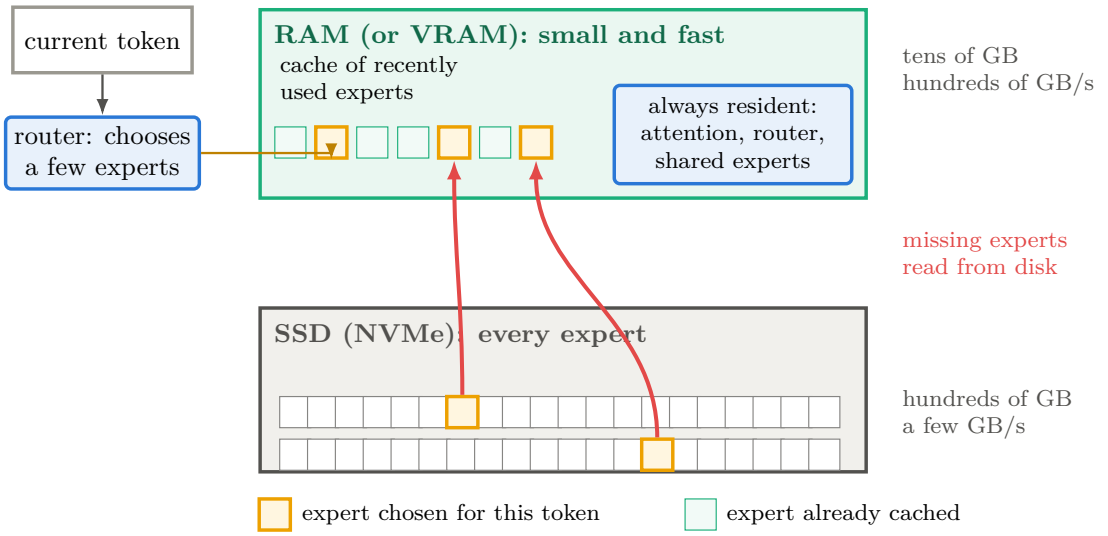


Figure 2.10: Streaming experts from the SSD in an engine such as Colibri. Memory holds the part of the model used by every token (attention, router, shared experts) and a cache of recently used experts; every expert is stored on the SSD. For each token the router chooses a few experts: those already cached are used at once, the missing ones are read from disk, at a bandwidth of a few gigabytes per second. For example, GLM-5.2 keeps about 10 GB resident and leaves about 429 GB of weights on disk [36]. The diagram is not to scale.

Colibri matters to this thesis for a second reason. It exposes the same System One interface as Jev, through which any model it runs answers closed questions by reading the probability of each option instead of generating the answer, and it includes three decision models natively: Laya, GLiNER2.5-Decide and Clef [36]. The authors report that, with Qwen3.6, four questions about the same document are answered 5.7 times faster than by generating the same answers on the same CPU. This is a figure reported by the project and not verified in this thesis, obtained with a different model and hardware, but it points in the same direction as the

results of Chapter 7. That an engine designed to run huge models on modest hardware adopts the reading of decisions as a first-class mode confirms that the decision-model idea is spreading precisely where generation costs the most.

2.5.3 Quantisation

Quantisation stores the weights with fewer bits than the 16 used in training. On local hardware it serves two purposes at once: the model fits in less memory, and decoding becomes faster, because decoding time is dominated by reading the weights. Table 2.3 summarises the methods that shaped the field.

Table 2.3: Weight quantisation methods for language models.

Method	Idea and main result	Source
LLM.int8()	8-bit weights and activations; the few dimensions with very large anomalous values (outliers) are kept in 16 bits. No loss up to 175B parameters; saves memory but does not speed up small models	[62]
GPTQ	3–4-bit weights, quantised layer by layer by estimating how much each weight affects the error. 4-bit OPT-175B loses 0.03 perplexity[a]; small models lose more; 3.2–4.5× faster decoding from bandwidth	[63]
AWQ	4-bit weights; the 1 % of channels that matter most, found from activation magnitudes, are protected by scaling. Robust to the calibration data; designed for on-device use	[64]
GGUF k-quants	2–8-bit block formats of llama.cpp, the de facto standard for consumer distribution	[2]
MLX affine	4- or 8-bit weights with a scale and a bias per group of 64; supports LoRA training on the quantised model (used in this thesis)	[58]
QLoRA	4-bit normal-float base model with 16-bit adapters, for fine-tuning on limited memory	[65]

[a] *Perplexity* measures how well a model predicts a text: the lower it is, the less the model is "surprised" by the words it encounters.

Two results from this literature matter for this thesis. The first is that quantisation helps the bandwidth-bound phase. GPTQ attributes its speed-ups entirely to the matrix-vector products of decoding and explicitly obtains no reduction in computation [63]; AWQ illustrates the same point with a roofline: on a desktop graphics card, a 200-token prefill takes 10 milliseconds while generating 20 tokens takes 310 [64]. Reducing the bits of the weights therefore accelerates a model that generates far more than one that only reads. The second is that small models are more fragile. GPTQ loses much more accuracy at 4 bits on a 1.3-billion-parameter

model than on a 175-billion one, and models of three billion parameters or fewer become unusable at 3 bits on phones [63, 60]. LLM.int8() shows that a handful of outlier features control a large share of the attention probability mass [62]. For a model whose output is a probability that must be calibrated, rather than a text that must merely be fluent, these observations argue for a conservative precision: GroundOne uses 8-bit weights. A measurement made in this thesis confirms that, at this precision, quantisation costs no accuracy: applying the same adapter of the 0.5B model to the original 16-bit weights leaves the three-way balanced accuracy on T1 practically unchanged (0.768 against 0.769), the probabilities change by 0.011 on average and the predicted class agrees on 98 % of the sentences (Section 7.11).

2.5.4 Faster generation without fewer tokens

A different family of methods accelerates generation without changing the model. In speculative decoding, a small draft model proposes several tokens, and the large model checks all of them in a single forward pass, accepting the longest prefix that it would have produced itself; the output is identical to that of the large model, and the speed-up reaches two to three times [66]. The method is useful for any system that must generate, but it does not change the nature of the cost: every accepted token still requires a forward pass of the large model, and the total amount of computation grows [66]. It makes writing cheaper; it does not make it free.

2.6 Energy

On a battery-powered device, energy is a property of the method, not only of the hardware. Measurements on datacenter hardware show that the energy of a request grows with the number of generated tokens and depends strongly on model size and task [67, 68]. On phones, a 3-billion-parameter model in 4 bits spends between 0.58 and 0.76 joules per generated token, with sustained power around 14 W and surface temperatures close to 48 °C [60]; a rule of thumb from work on small models puts the cost at about 0.1 joules per generated token for each billion parameters [39]. Moving the prefill to a phone's NPU reduces its energy by 1.9 to 4.3 times with respect to the graphics unit [55].

The distinction between the two phases also applies to power. Prefill keeps the arithmetic units busy and draws more power; decoding waits for memory and draws less, but for much longer [42]. A method that saves time therefore saves energy too, but by a smaller factor, and Chapter 7 measures exactly this effect: GroundOne spends 5.5 joules per long answer with the 0.5B model and 33 with the 3B, less than any generative judge of the same size, but with an advantage smaller than in time (Section 7.10).

2.7 Consequences for verification on the device

The chapter can now be summarised in one observation: on a local device, which serves one request at a time, reading a token costs far less than writing one, fourteen to sixteen times less on the thesis machine (Sections 2.3.1–2.3.3), and recent hardware generations accelerate reading much more than writing (Section 2.4.2). This has a direct consequence for a verifier that runs next to the generator.

A generative judge verifies an answer by reading the document and the answer and then writing its verdicts. It pays a prefill, like any method that reads the document, and then one decoding step for every token it writes. In the latency protocol of this thesis (Section 7.9), a terse 3B judge writes a median of 65 tokens for an answer of fourteen or more sentences, and a verbose one, which explains each verdict, writes 388. The lower bound of Section 2.3.3 turns these counts into a minimum time that no implementation can beat (Figure 2.9b): on the thesis machine, at least 1.8 seconds for the terse judge and 10.6 for the verbose one, on top of the prefill; even on the fastest graphics card, 0.12 and 0.71 seconds.

A decision model removes this term altogether. It reads the document and the answer once and takes every decision from the logits computed during that single reading, so its cost is the prefill alone: the compute-bound phase, the one that local hardware performs well and that new hardware accelerates most. Its advantage over a judge is therefore not a property of one machine but of the cost structure of local inference, and it is largest exactly where inference is local. Locality, which at first sight is only a constraint on the size of the model, becomes the reason why the design matters.

Three requirements for the verifier follow from this chapter, and they guide the method of Chapter 4. It must generate no tokens, so that its cost is a single prefill. It must use a model small enough to share a 16 GB machine with a generator and a retriever, quantised conservatively because its output is a probability. And it must read its decisions from logits, which only a local model exposes, so that they can be calibrated.

Chapter 3

Background and Related Work

This chapter reviews the work on which GroundOne builds and the work against which it is compared. The cost of inference on a local device, from which the design follows, was already treated in Chapter 2. The chapter starts from decision models, that is, from the idea of reading the decisions of a language model instead of having it generate them, and from their rapidly growing ecosystem. It then turns to the problem of hallucination in retrieval-augmented generation (RAG) and to the methods that verify answers against their sources. It closes with calibration, with parameter-efficient fine-tuning and with the gap that this thesis sets out to fill.

3.1 Decision models

The term *System One model* was introduced by TypeSafe with Jev, a model offered as a hosted service. Jev answers closed questions about a given state, that is, about the context supplied in the request, and returns typed outputs, such as a boolean, a choice among options or a score, without generating text [14]. The name evokes the fast, intuitive thinking described by Kahneman, as opposed to deliberate reasoning. The vendor reports latencies ranging from tens to hundreds of milliseconds and large savings with respect to generative judges. These figures, however, remain vendor claims, and the architecture is not disclosed.

The open implementations that followed differ in how they produce the probabilities, and three families can be identified. The most common one presents the question as a multiple choice and keeps the head of the language model, that is, the final layer that assigns a score to every token of the vocabulary: the decision is read from the logits of the option letters, at the position where the model would write

the answer. SemIf applies this readout to a 4B-parameter model kept frozen, that is, without modifying its weights, and reports that 21 criteria about one document are answered in about one second; the best generative baseline able to produce valid JSON, by contrast, takes more than five seconds [15]. Rizzo-Flow, unlike SemIf, fine-tunes a 4B model with LoRA on typed decisions and exposes the same interface as Jev. Its authors report that fine-tuning improves both its accuracy and its calibration [16]. This is the general-purpose decision model used as a baseline in this thesis.

The second family encodes the state and each candidate answer separately, and compares their embeddings, that is, the numerical vectors that represent each text. Contrastive Language Models, which belong to this family, are very fast when the candidates are numerous and reusable. In the comparison reported by the authors themselves, however, they agree with Jev's decisions much less than a model in which state and candidate interact [69]. The third family uses small encoders[1] with a dedicated decision head. Laya, for example, is built on a ModernBERT encoder of about 400M parameters and assigns a score to each option at the corresponding mask token. The questions of a request are evaluated in a single call, grouped into a batch, but each of them is encoded together with its own copy of the context [70, 71].

The evidence supporting these systems is uneven. Most of it comes from the authors themselves, often on a single machine and on a single benchmark. Independent evaluations are recent and less favourable to small open models. An evaluation of Jev on 37 datasets finds well calibrated the probabilities that the model assigns to choices among options. Binary probabilities, instead, rank examples well but are poorly positioned with respect to a fixed threshold of 0.5. The same study reports for Jev a mean balanced accuracy[2] of 78.6 on LLM-AggreFact [20]. An independent reproduction of Laya observes a calibration different from the declared one and a strong dependence on a calibration temperature estimated on the target data [21]. A benchmark built on contrastive pairs, that is, on nearly identical decisions that differ in a single decisive detail, places small open models far below Jev [72]. Finally, it has been shown that, when the label of an option and its definition conflict, typed decision models follow the label rather than the

[1] An *encoder* is a transformer in which every token attends to all the others, both those that precede it and those that follow it. It produces a representation of the text instead of generating text, as BERT and the models derived from it do. The *mask token* is a special token that, during pre-training, takes the place of the words the model has to reconstruct.

[2] *Balanced accuracy* is the mean, computed over the classes, of the recall, that is, of the fraction of examples of each class that are classified correctly. Unlike plain accuracy, it does not reward a model that always predicts the most frequent class: a constant predictor obtains 0.5 in the binary case (50 when expressed as a percentage, as here) and about 0.33 with three classes.

definition [71].

The ecosystem, meanwhile, is growing rapidly. On 9 October 2026 the Hugging Face hub officially introduced a tag dedicated to these *decision models* [17]. The following day the tag already grouped about 1,500 models, published by about 600 authors and of very uneven quality. This thesis uses the term *decision model*, adopted by the ecosystem; *System One* remains the name with which TypeSafe introduced the category, and in what follows it appears only as the name of the Jev-compatible interface.

A few weeks after the release of Jev, established organisations also published their own open decision models. Cloudflare's Clef adds a dedicated decision head to a decoder and answers all the questions with a single forward pass over the state: this is the architecture that had only been hypothesised for Jev [19]. Liquid AI's d1-3B reads the decision from a restricted set of answer letters, with a 3B model whose state is shared by all the questions [18]. Perplexity, finally, fine-tunes a 27B model with a head on the letters [73]. A community leaderboard, the Decision Index, evaluates these models on a common battery of public benchmarks, including RAGTruth evaluated at the level of the whole answer [22]. The results draw a sharp line. General-purpose decision models with fewer than one billion parameters barely beat the trivial baseline on RAGTruth, and only models with tens of billions of parameters reliably detect hallucinated answers.

The studies on Jev add three observations. Its probabilities are calibrated when aggregated over many tasks, but not on a single task. Thresholds and temperatures have to be estimated on the target workload [74, 75]. Finally, choosing the threshold through a lower confidence bound makes it safer to put into production a judge that returns only decisions [75]. Runtimes have adapted as well: llama.cpp now natively supports the System One interface [2]. None of these models, however, is specialised in groundedness, that is, in the adherence of the answer to the document. Nor does any of them report results of three-class, sentence-level verification on the benchmarks used in this thesis.

The letter readout has also been studied outside the decision-model literature, in multiple-choice question answering. Here two ways of obtaining the answer of an instruction-tuned model can be compared: reading which letter has the highest probability as the first token, or letting the model write out the answer in full. The two answers often do not coincide [76]. The readout, moreover, favours particular option identifiers, such as certain letters, and this bias can be estimated and removed [77]. Both results concern models used zero-shot. They suggest that a letter readout, if it is to serve as a decision, needs to be trained, calibrated or both. Moreover, no open decision model has been specialised in groundedness verification and then compared, under controlled conditions and on the same hardware, with generative judges, with specialised verifiers and with general-purpose decision models, one of them of the same size.

3.2 Retrieval-augmented generation and hallucination

Retrieval-augmented generation (RAG) gives a language model access to documents on which it was not trained: a retriever selects the passages relevant to the question, and the model writes the answer with these passages in its context [1]. It is the standard approach for building assistants over private documents, and it can run entirely on a local machine. It does not, however, eliminate hallucinations. Surveys of the literature distinguish two types of error. Errors of *factuality* are measured against world knowledge, errors of *faithfulness* against the input or the context given to the model [9]. A RAG answer, in particular, can be unfaithful to the very passages it received, by contradicting them or by adding information that they do not contain.

RAGTruth documents how frequent this is. The dataset collects about 18,000 answers, written by six language models on three RAG-style tasks, and annotates in them, with character-level precision, the spans of text that conflict with the context or are not supported by it [8]. RAGTruth's distinction between conflicting content and baseless content corresponds to the classes *contradicted* and *not mentioned* adopted in this thesis.

3.3 Groundedness verification

Verifying an answer against its source is a problem closely related to natural language inference (NLI). In NLI the task is to establish whether a premise implies a hypothesis (entailment), contradicts it or is neutral with respect to it. Large NLI datasets provide training data: MultiNLI [78] and ANLI [79], the latter built adversarially, that is, by collecting examples written on purpose to fool the models of the time. VitaminC [80] adds contrastive pairs, in which a minimal revision of a Wikipedia sentence flips the verdict: the model is thus forced to look at the precise fact rather than at word overlap.

Applying these models to long generated texts requires decomposing them. SummaC splits both the document and the summary into sentences and aggregates the inference scores computed sentence by sentence [81]. FActScore, instead, decomposes long generations into atomic facts and verifies them one by one [82]. Sentence-level verification, adopted by GroundOne, lies between a single score for the whole answer and the decomposition into atomic facts. Sentences are the natural units of an answer, the user can act on a flagged sentence, and the number of decisions grows with the length of the answer.

In practice two families of verifiers prevail. The first is that of specialised encoders, small and fast. AlignScore trains a single alignment function on a

broad mixture of tasks [11]. MiniCheck trains small verifiers on synthetic data, built so as to require reasoning over several sentences, and matches much larger models on grounding benchmarks [12]. HHEM is a hallucination evaluation model of the same kind, released as an open model [13]. These models evaluate each document–sentence pair independently and return a single probability of support.

The second family uses a language model as a judge, following the LLM-as-a-judge scheme[3] [83]. Evaluation frameworks for RAG such as RAGAS [84] and ARES [85] rely on judges of this kind. Lynx fine-tunes a judge dedicated to hallucination detection [86], while Self-RAG trains the generator itself to emit critique tokens, that is, special tokens with which the model evaluates the retrieved passages and its own output [87]. Judges are flexible, but their cost grows with the amount of text they write. Small models used as judges, moreover, often violate the required output format. Guaranteeing an answer that the parser can interpret then requires grammar-constrained decoding, which at each step admits only the tokens compatible with the required format [88]. The reference benchmark for both families is LLM-AggreFact, which gathers eleven manually annotated grounding datasets and maps them to a common binary protocol [12, 89]. Figure 3.1 compares the two families with the specialist decision model studied in this thesis.

3.4 Calibration and selective prediction

A probability is calibrated when, among the predictions made with a given confidence, the fraction of correct ones matches that confidence: in a calibrated model, for example, about 80 % of the predictions made with confidence 0.8 turn out to be correct. Modern neural networks are often poorly calibrated, and a simple and effective correction is temperature scaling: the logits are divided by a single constant, the temperature[4], estimated on held-out data [90]. The usual measure of calibration is the Expected Calibration Error (ECE), which compares confidence and frequency of correct predictions within confidence bins [91] (Section 4.4). For language models queried through a prompt, a per-class bias may also be needed, to correct a systematic preference for some answers [92].

Language models know, within limits, what they know: on closed questions, the probabilities of their tokens are often informative about the correctness of the answer [93]. When the probabilities are not accessible, the model can be asked to

[3] In the *LLM-as-a-judge* scheme a language model receives, through a prompt, the text to be evaluated and the evaluation criteria, and writes its own judgement, sometimes preceded by a rationale.

[4] With a temperature $T > 1$ the probabilities become less extreme, with $T < 1$ more extreme. Since all the logits are divided by the same constant, the most probable class does not change.

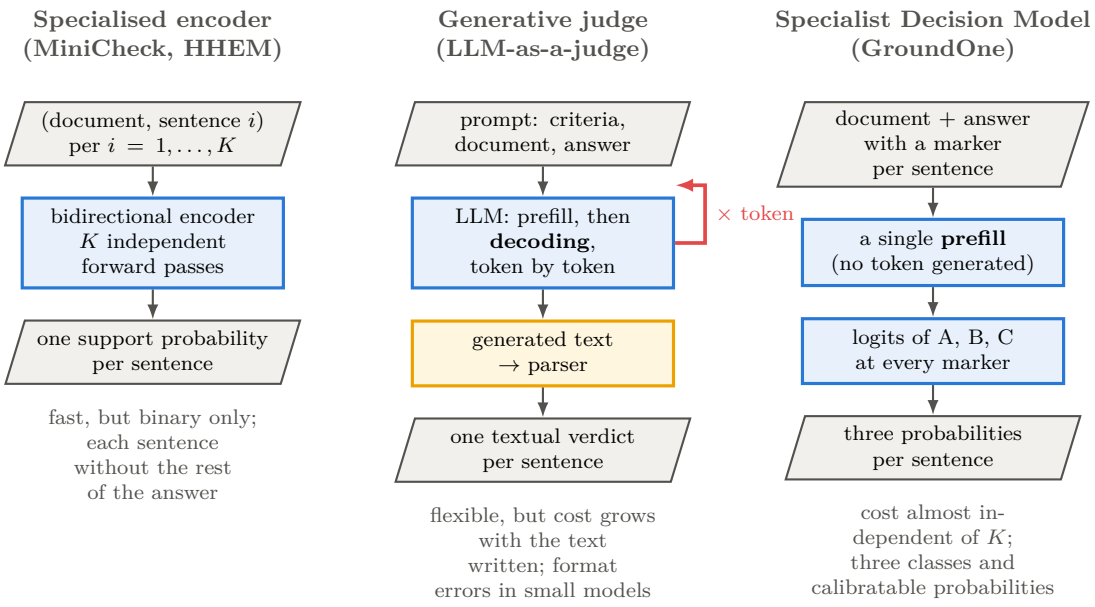


Figure 3.1: Three ways of verifying the K sentences of an answer. A specialised encoder evaluates each (document, sentence) pair in a separate forward pass and returns only a probability of support. A generative judge reads everything, but writes its verdict token by token (red loop), so its cost grows with the text produced and the output has to be interpreted by a parser. A specialist decision model reads document and answer in a single prefill and derives three probabilities from each marker, without generating any token.

express its confidence in words. In large models fine-tuned with human feedback, this verbalised confidence can be better calibrated than probabilities estimated by sampling [94]. Systematic evaluations, however, find it coarse and overconfident, with values concentrated on a few round numbers [95].

A calibrated probability becomes useful through selective prediction: the system acts on the decisions it is confident about and defers the others. The risk–coverage curve describes this trade-off. For each share of accepted decisions (the coverage) it reports the error rate on them (the risk), and thus shows how the error rate falls as more decisions are deferred [96]. Figure 3.2 illustrates its typical shape.

3.5 Parameter-efficient fine-tuning

Specialising a pre-trained model does not require updating all of its weights. Parameter-efficient fine-tuning (PEFT) methods train only a small share of the parameters and leave the rest of the model unchanged. Low-rank adaptation

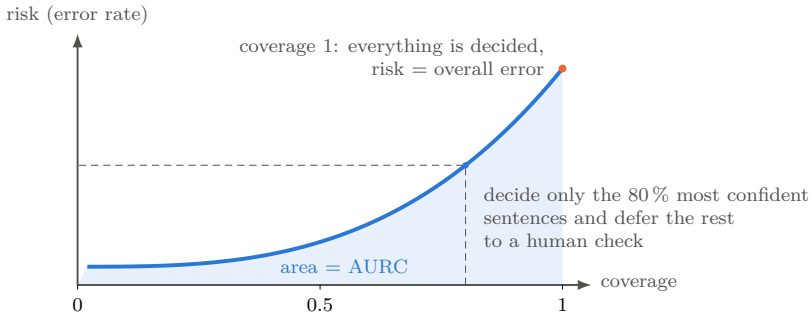


Figure 3.2: Sketch of a risk–coverage curve (illustrative shape, not a result). The decisions are sorted from the most to the least confident; for each share decided (coverage) the error rate (risk) of the accepted decisions is reported. At coverage 1 the risk equals the overall error; giving up the least confident decisions lowers the error. The area under the curve (AURC) summarises the curve in one number: the smaller it is, the better the confidence separates correct decisions from wrong ones.

(LoRA) freezes the model and, for some weight matrices, trains only a low-rank[5] update. At the end of training, the update can be merged into the weights or kept as a small separate adapter [24]. Applying LoRA to all linear layers matters more than using a high rank, and the technique combines with quantised base weights, so that even large models fit in little memory [65]. The Qwen2.5 family offers open models starting from 0.5B parameters with a shared tokenizer, and therefore makes it possible to compare different sizes with the same recipe [23]. Figure 3.3 sketches a LoRA adapter applied to a linear layer.

Fine-tuning on small datasets, however, is notoriously unstable. Results vary with the random seed, which determines both the initialisation of the new parameters and the order of the training data [97]. Some training runs fail altogether, and this behaviour is linked to optimisation details such as the absence of bias correction in Adam[6] [98]. Gradient clipping, which bounds the norm of the gradient,

[5]The *rank* of a matrix is the maximum number of its linearly independent rows (or columns). LoRA constrains the correction learned for a $d \times k$ weight matrix to have a small rank r. A matrix of this kind can be written as the product of two narrow matrices, one $d \times r$ and one $r \times k$, so it suffices to store and train these two: $r(d + k)$ parameters instead of the dk of the full matrix. In GroundOne 3B, for example, a 2048×2048 matrix has about 4.2 million weights, while its rank-16 correction requires about 65 thousand, 1.6 %.

[6]Adam is an optimiser that updates each weight using two moving averages, one of the gradient and one of its square, called moment estimates. Since these averages start from zero, in the first steps they underestimate the true values; *bias correction* rescales them to compensate. AdamW is the variant of Adam that applies weight decay separately from the gradient.

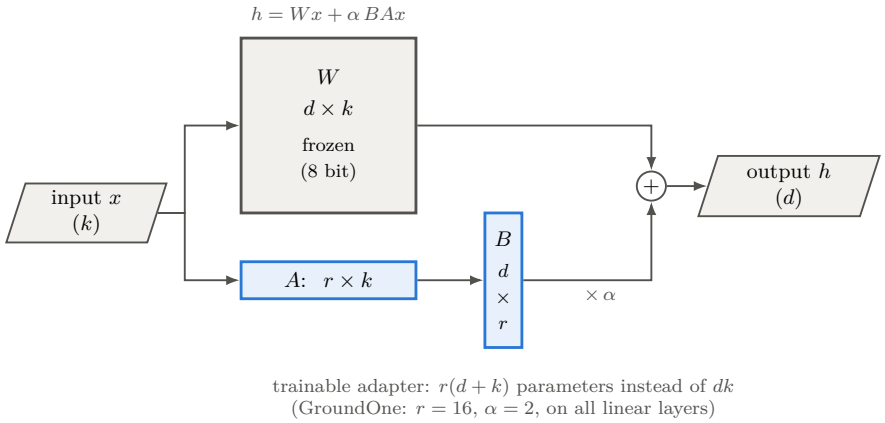


Figure 3.3: Low-rank adaptation applied to a linear layer. The weight matrix W stays frozen (8-bit in GroundOne); only the two small matrices A and B are trained, and their product, scaled by α, is added to the output of the layer. With rank r the adapter has $r(d+k)$ parameters instead of the dk of W.

is finally the standard defence against occasional exploding updates [99].

3.6 Research gap

The components needed for a local, fast and calibrated verifier already exist, but separately: compact open models and runtimes that run them on a laptop, the decision-model idea of reading decisions instead of generating them, specialised verifiers and post-hoc calibration. What is missing is an independent, controlled study that brings them together on a single task. In particular, no work has measured whether a compact decision model, specialised in groundedness verification, can match generative judges and dedicated encoders on this task. Nor is it known where its advantage would come from: from specialisation, from the readout or from model size. Finally, it remains to be established whether its probabilities are good enough to set thresholds and defer decisions, and how its time and energy compare with those of a judge on the same consumer hardware. The rest of the thesis addresses these questions.

Chapter 4

GroundOne: Method and Implementation

GroundOne is a compact language model used as a decision model. It reads the document and the answer once and, without generating text, returns a verdict for each scored sentence of the answer, deriving it from the logits of a few answer letters. The chapter first describes the decision the model makes and the prompt that gathers all the decisions about one answer into a single forward pass. It then presents the training objective, the calibration of the output probabilities and the rule that turns the probabilities into the returned labels, and it closes with the software that implements all of this.

4.1 Task and decision

For every sentence of an answer, GroundOne chooses among three classes, defined in Section 5.1: with respect to the document, the sentence is *supported*, *contradicted* or *not mentioned*. The *document* is the whole context that the generator received. It may comprise several retrieved passages, possibly from different sources, concatenated in the prompt: this is the case of question answering in RAGTruth (Section 5.2). A sentence is therefore supported when at least one passage backs it, including by combining information scattered across different passages.

Each class corresponds to a letter (A, B and C), and the model returns a probability for each of them. This is the only question the model receives: everything else is derived from these three probabilities. The binary groundedness score is the probability that the sentence is supported. The strict policy flags every unsupported sentence, while the lenient policy flags only contradicted sentences. A single forward pass is therefore enough for all these uses.

Asking the model a separate question for each policy would add nothing: once

the class of a sentence is known, how each policy judges it is already known too. Separate questions would only multiply the number of decisions and, in training, would risk teaching the model mutually inconsistent answers about the same sentence.

4.2 All verdicts in a single forward pass

The prompt contains a short instruction, the document and the answer. The sentences of the answer are numbered, and every sentence associated with a decision is followed by a marker, that is, a short fixed string that indicates where the verdict is read: the string " `Verdict:`". Sentences without checkable content (Section 5.3) remain in the text as context, but without a marker. The model processes the whole sequence once. At the last token of each marker the logits of the letters A, B and C are read and normalised with a softmax[1] restricted to these three letters. In a decoder each position sees only the preceding ones, so the verdict on sentence i depends on the document and on sentences $1, \ldots, i$. This is exactly what verification needs, because references, such as pronouns, point backwards. What follows the last marker affects no verdict, so the sequence ends there. Figure 4.1 shows the structure of the prompt and the points where the readout takes place.

Two details make the readout exact and efficient. The first concerns exactness: the prompt is built by tokenising its parts separately and then concatenating them, so that the position of every marker is known exactly and stays identical in training and at inference. The second concerns efficiency: the output projection onto the vocabulary is applied only at the marker positions. The transformer body processes the whole sequence, but the large vocabulary matrix is multiplied only by the few hidden states that are actually read, that is, by the vectors the model computes at the marker positions.

The marker readout also determines the cost of a verification. Let S denote the total length of document and answer, K the number of decisions and m the length of a marker in tokens. The decision model requires a single prefill of $S + Km$ tokens. A generative judge, in contrast, must also produce its verdicts during decoding. If each verdict requires n tokens and each decoded token takes a time t_{dec}, the cost of the judge is roughly $\text{prefill}(S) + K\,n\,t_{\text{dec}}$. A marker is only three tokens long, whereas on the reference hardware decoding is much slower than prefill for the same number of tokens; the gap between the two costs therefore widens as K grows.

[1] The *softmax* turns a vector of scores into a probability distribution: each score z_i becomes $e^{z_i} / \sum_j e^{z_j}$. Restricting it to three letters means computing it only over their three logits, ignoring the rest of the vocabulary.

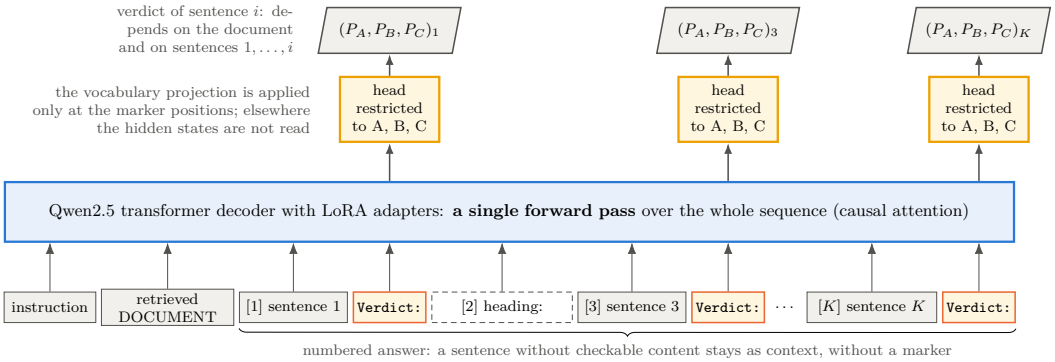


Figure 4.1: The marker prompt of GroundOne (design b). The sentences of the answer are numbered and every scored sentence is followed by the marker " `Verdict:`"; a sentence without checkable content (here a heading) remains as context, without a marker. The model processes the whole sequence in a single forward pass, and the output head, restricted to the letters A, B and C, is applied only at the marker positions. Because of causal attention, the verdict on sentence i depends on the document and on sentences 1 to i.

On the M4 used in this thesis, with the 0.5B model prefill costs about 0.37 ms per token and decoding 5.9 ms per generated token; with the 3B the two values are 2.2 and 31.8 ms. One decoded token thus costs as much as fourteen to sixteen tokens in prefill. The two phases also stress the hardware differently. Prefill processes many tokens together and is limited by compute, whereas decoding produces one token at a time and is limited by the bandwidth needed to read the weights [10, 42]. The difference resurfaces in the energy measurements of Chapter 7. Figure 4.2 translates the cost model into seconds per answer, using only these measured constants.

There is also a second solution, which reads the shared prefix once and then appends a short question for each sentence, evaluating the questions in a batch. Unlike the marker solution, which needs only a single forward pass, this one requires a KV cache replicated K times. It is kept as a fallback and is the natural format for an untrained model, which understands an explicit question but has no reason to interpret a marker as a request for a verdict. For this reason it is the format used by the untrained baseline (Section 6.2). Figure 4.3 compares the two solutions.

A classifier, not a faster generator. The readout restricts the decision to three letters, and the restriction is not a detail. After the marker, the fine-tuned model would not write the letter that is read. Its most probable next token is in fact the same letter preceded by a space, which for the tokenizer is a different token. At the markers, the three readout letters together collect only about 4–5 % of the probability mass. A loss restricted to these letters trains their relative

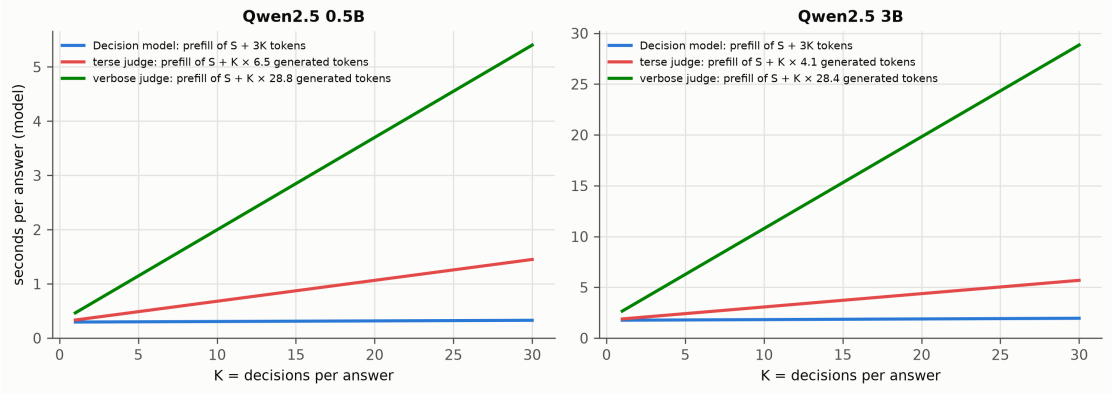


Figure 4.2: Estimated cost of verifying an answer of $S = 800$ tokens as a function of K, computed with only the constants measured on the M4: the per-token time of prefill and decoding and the mean number of tokens each judge generates per verdict on the latency subset (mean K 8.7). It is an analytical model, not a measurement: the measured latencies are in Section 7.9. The cost of the decision model grows by only three prefill tokens per decision, that of the judges by a few decoding tokens per decision.

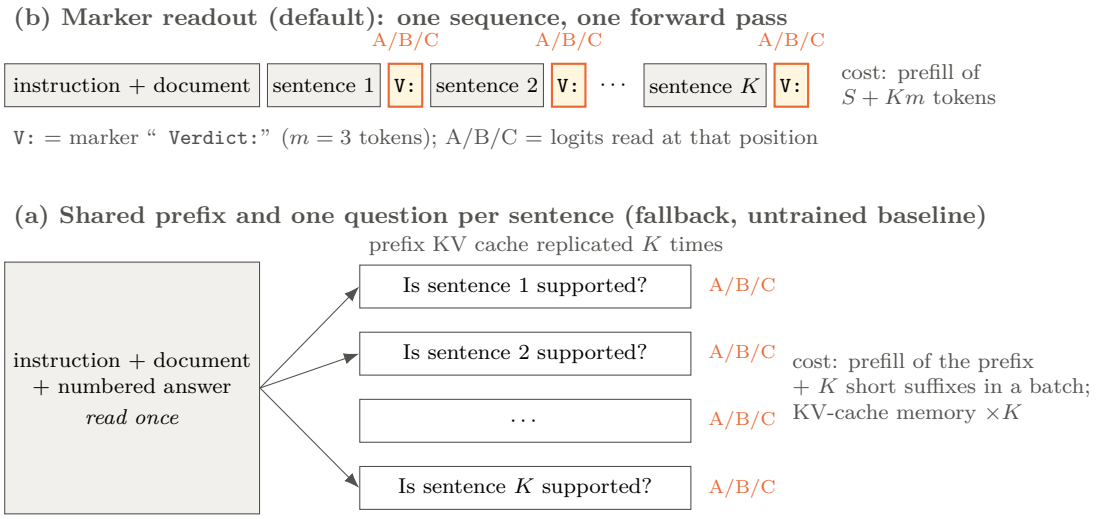


Figure 4.3: The two solutions for obtaining K verdicts from a single reading. (b) The marker readout, adopted as the default, inserts a marker after every sentence and reads all the verdicts in the same pass. (a) The fallback solution reads the shared prefix once and appends a short question for each sentence, evaluated in a batch; it requires, however, K copies of the KV cache of the prefix, and it is the natural format for an untrained model.

logits, not their absolute probability, and the readout therefore behaves as a three-class classifier grafted onto the language model. This is what defines a decision model: the answer never goes through decoding, but is chosen among the admissible options. The phenomenon also recalls two results from the literature: the probability of the first answer token and the answer the model actually writes may not coincide [76], and the choice among letters is subject to selection bias [77]. A specialised, calibrated readout has to absorb both effects. A control experiment quantifies its difference from generation: it keeps the same weights and the same forward pass, but instead of the restricted readout it takes the token that generation would emit (Section 7.6).

4.3 Training objective

GroundOne is obtained from Qwen2.5-0.5B-Instruct and Qwen2.5-3B-Instruct [23] through low-rank adaptation (LoRA) [24], described in Section 3.5 (Figure 3.3). The adapters have rank 16, scaling factor 2 and dropout 0.05[2]. They are applied to all the linear layers of all the transformer blocks, that is, to the seven projections of each block (Figure 4.4), because covering every layer matters more than using a high rank [65]. In total, 8.8 million parameters are trained in the 0.5B and 29.9 million in the 3B, about 1.8 % and 1.0 % of the respective models. The weights of the base model remain frozen at 8-bit precision. Each training example is an answer with all its scored sentences, and the loss is computed at each of its markers.

The loss is a cross-entropy[3] computed over the three letters only. The prompt is never supervised, so the training signal comes entirely from the verdicts. About 88 % of the sentences in the training set are supported, so the classes are weighted so that each contributes equally to the loss. With the distribution of the training set this gives weights of about 0.38 for supported sentences, 7.5 for contradicted ones and 4.4 for not-mentioned ones. Without these weights, a model can reach a low loss by answering "supported" almost always. A useful consequence follows: a model that had learnt only the class frequencies would remain at a loss of $\ln 3 \approx 1.10$, so a loss below this value indicates that the model actually exploits the text.

The optimiser is AdamW, used without weight decay, that is, without the penalty that shrinks the magnitude of the weights at every step. As in the default

[2]The *scaling factor* multiplies the update computed by the adapter before it is added to the output of the frozen layer. *Dropout* randomly zeroes, during training, a fraction of the adapter's inputs (here 5 %), so as to reduce the risk that the model memorises the training set.

[3]The *cross-entropy* of a prediction is the negative logarithm of the probability the model assigns to the correct class: it is zero when this probability is one and grows without bound when the model is confidently wrong.

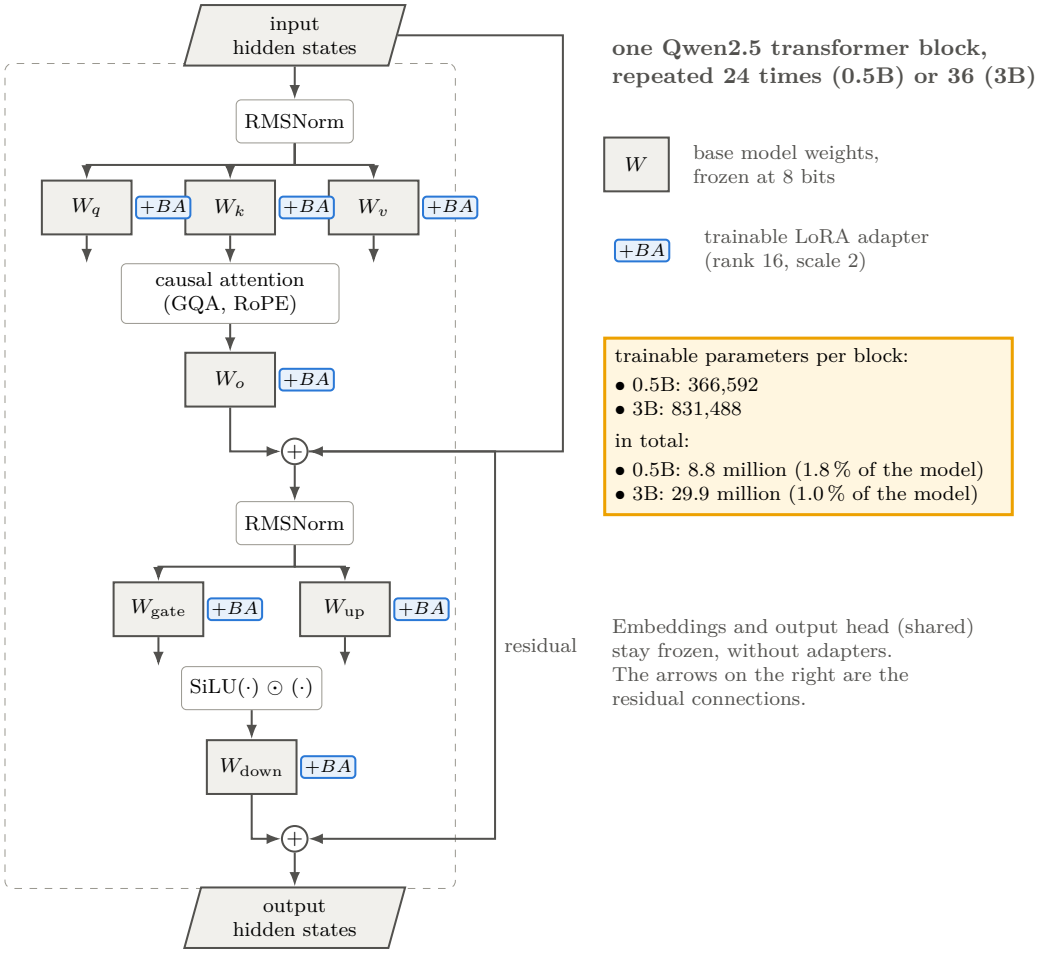


Figure 4.4: Where GroundOne applies LoRA. Each transformer block of Qwen2.5 contains seven linear projections: four in the attention (W_q, W_k, W_v, W_o) and three in the MLP block (W_{gate}, W_{up}, W_{down}). Their weights remain frozen at 8 bits, and each is paired with a rank-16 adapter (in blue) that corrects its output following the scheme of Figure 3.3. The block is repeated 24 times in the 0.5B and 36 in the 3B; the embedding and the output head have no adapter.

configuration of MLX, the moment estimates receive no bias correction, and no gradient clipping is applied. Training lasts one epoch, that is, a single pass over the answers of the training set. The learning rate rises to 10^{-4} with a linear warm-up over the first 3 % of the steps and then decreases to zero with a cosine schedule[4]. Answers of similar length are grouped into micro-batches, each within a token budget that also includes padding, that is, the filler tokens that equalise the lengths. The gradients of several micro-batches are then accumulated, so that each optimiser step sees about 16 answers, or about 110 decisions. In this way every step contains some contradicted sentences, despite the rarity of the class. Gradient checkpointing, which recomputes some activations during the gradient computation instead of storing them, keeps memory usage within the 16 GB of the laptop. During training the model is evaluated periodically on the dev set, and the checkpoint, that is, the copy of the weights, with the highest three-class balanced accuracy, averaged over the three tasks of RAGTruth (question answering, summarisation and data-to-text), is kept. Section 6.1 reports the settings of each run.

4.4 Calibration

The class weights train the model as if the three classes were equally frequent, so its raw probabilities do not reflect the real class frequencies. A post-hoc transformation is therefore fitted on the dev set and applied to the logits after training, without changing the weights. The three logits are divided by a temperature T and shifted by a per-class bias, $\mathrm{softmax}(z/T + b)$, with the bias of the supported class fixed at zero. The parameters T and b are chosen by minimising the negative log-likelihood (NLL) of the dev-set labels[5], that is, the mean cross-entropy [90]. The bias plays the role of the contextual calibration proposed for language models queried through prompts [92]; unlike that correction, however, here it is fitted on labelled data. Temperature alone changes the probabilities but neither the most probable class nor the ranking of the sentences. The bias, instead, can change the most probable class and, with three classes, can also slightly reorder $P(\text{supported})$ from one

[4]During the *warm-up* the learning rate grows gradually to the chosen value, so as to avoid abrupt updates at the start of training. The *cosine schedule* then reduces it following half a period of a cosine: slowly at first, faster in the middle and slowly again towards the end.

[5]The *negative log-likelihood* of a set of n predictions is the mean, with the sign changed, of the logarithm of the probability assigned to the correct class: $\mathrm{NLL} = -\frac{1}{n}\sum_{i=1}^{n} \log p_i(y_i)$, where y_i is the correct class of prediction i. It coincides with the mean cross-entropy. It is zero when the model always assigns probability one to the right class and grows rapidly when it assigns a low probability to the right class, that is, when it is confidently wrong: this is why it penalises both errors and overconfidence.

sentence to another. For this reason the classification metrics and the AUROC[6] are computed on the raw output. The calibrated probabilities are used instead wherever a probability has to correspond to a frequency: in the calibration metrics, in the thresholds and in selective prediction (Section 7.7).

Calibration during or after training. GroundOne calibrates its probabilities after training. Jev, the decision model that introduced the category, takes the opposite route: according to the vendor, its probabilities become reliable already during training, thanks to a reinforcement learning phase with rewards designed to reward calibration (*Reinforcement Learning with Calibrated Decisions*, RLCD) [14]. The recipe is not public, and a direct comparison was not possible: Jev is a remote, closed service, and evaluating it on the test sets of this thesis would have required sending the data to an external service. The comparison therefore concerns the approaches, on the basis of the independent evaluations (Table 4.1).

These evaluations show that calibration obtained during training holds in aggregate, but not on the individual workload. Over 31 benchmarks the ECE of Jev is 0.047 when computed on all the data together, but 0.168 as the median of the individual benchmarks [74]. Its binary probabilities rank the examples well, but they are slightly underconfident, so the fixed threshold of 0.5 ends up poorly placed: the mean probability of an affirmative answer is 0.465, against an observed frequency of 0.518 [20]. For this reason the studies recommend fitting thresholds and temperatures on the target workload [75]. Even the openly released RLCD recipe offers no measurable advantage: in a controlled comparison it remains 2.5 to 3 points below plain cross-entropy, with a variability across seeds up to four times larger, and after temperature scaling no training objective turns out to be better calibrated than the others [100].

GroundOne starts from here. A specialist works on a single domain, so it can fit its own calibration on exactly the distribution in which it will be used, with a dev set and a few parameters. On T1 the binary ECE falls from 0.30 to 0.019 for the 0.5B and from 0.18 to 0.019 for the 3B. The price is the same one that the studies find for Jev, seen from the other side: the calibration fitted on RAGTruth does not transfer to VitaminC (Section 7.7). The advantage is that the procedure is open, reproducible and bound to the model that produced it, and that the decision

[6]The *AUROC* (area under the ROC curve) measures how well a score separates two classes, here supported from unsupported sentences. Given a threshold, the sentences with a higher score are accepted as supported: the share of supported sentences accepted is the true positive rate, the share of unsupported sentences accepted by mistake is the false positive rate. The ROC curve plots these two quantities for all possible thresholds, and the AUROC is the area under it. It equals the probability that a randomly chosen supported sentence receives a higher score than a randomly chosen unsupported sentence: it is 0.5 for a random score and 1 for perfect separation.

threshold is part of the calibration instead of being left to the user.

Table 4.1: Two approaches to the calibration of a decision model. The values for Jev come from independent evaluations on benchmarks different from those of this thesis; those for GroundOne are measured on the T1 test set.

	Jev	GroundOne
When	during training: SFT followed by RL with calibration rewards (RLCD)	after training: temperature and per-class bias fitted on the dev set
Scope	one model for all tasks	a single domain, the deployment domain
Recipe	not public (vendor's statement)	open, bound to the model fingerprint
Threshold	left to the user; to be fitted for each workload	fitted on the dev set and part of the output rule
Evidence	ECE 0.047 in aggregate, 0.168 per individual benchmark; underconfident binary probabilities	binary ECE on T1 from 0.30 to 0.019 (0.5B), from 0.18 to 0.019 (3B); does not transfer to VitaminC

Calibration is measured mainly with the Expected Calibration Error (ECE) [91, 90], which measures how far, on average, the confidence stated by a model departs from the frequency with which its predictions turn out to be correct. The n predictions are divided, according to their confidence, into M intervals of equal width, with $M = 15$ in this thesis. For each interval B_m the mean confidence of its predictions, $\mathrm{conf}(B_m)$, is compared with the fraction of them that turned out to be correct, $\mathrm{acc}(B_m)$. The ECE is the mean of the gaps, weighted by the size of the intervals:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \Big|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\Big|. \tag{4.1}$$

A perfectly calibrated model has an ECE of zero; an ECE of 0.05, for example, indicates that the stated confidence departs on average by five percentage points from the observed frequency. In the binary variant the confidence is $P(\text{supported})$ and the observed frequency is the share of supported sentences in the interval. In the three-class top-label variant, instead, the confidence is the maximum probability and the observed frequency is the share of correct predictions. The ECE does not compare the distribution of the predicted classes with that of the test set. Rather, it checks whether, among the sentences to which the model assigns a given confidence, that confidence matches reality. The measure has two limits,

however: it depends on the choice of intervals, and a constant predictor that always returns the base rate obtains an ECE close to zero while discriminating nothing. For these reasons the ECE is always reported together with the Brier score[7], the negative log-likelihood and reliability diagrams, the plots that show, for each interval, the observed frequency as a function of the mean confidence. The ability of the probabilities to separate correct from wrong decisions is measured separately instead, with two measures. The first is the AUROC, described below. The second is the AURC (area under the risk–coverage curve), that is, the area under the risk–coverage curve of Section 3.4 (Figure 3.2): it measures the mean error rate obtained by accepting the decisions from the most to the least confident, and it is smaller the more the errors are concentrated among the least confident decisions, which are those a system would defer.

Calibration and AUROC answer two different questions, which is why the thesis always reports both. The ECE asks whether a probability of 0.8 really corresponds to a sentence that is supported eight times out of ten. The AUROC asks instead whether supported sentences receive higher scores than unsupported ones, whatever the scale of the scores (Figure 4.5). The two properties are independent. A constant predictor is perfectly calibrated with respect to the base rate, but has an AUROC of 0.5; conversely, a model can rank the sentences almost perfectly, with an AUROC close to 1, and still assign overly extreme probabilities. The AUROC, moreover, depends neither on any threshold nor on the class frequencies, and it does not change if the scores are transformed monotonically, as temperature does. For this reason it is the main measure by which GroundOne is compared with verifiers that return a binary score, such as the specialised encoders, while the choice of the threshold and its consistency with the probabilities shown to the user are treated separately.

From probabilities to returned labels. A user of the verifier sees, for each sentence, a label and a probability, which must be consistent with each other. This thesis evaluates the raw decision, that is, the most probable class according to the uncalibrated output. The raw decision, however, can contradict the calibrated probability shown next to it: a sentence can be labelled "not mentioned" while having a calibrated $P(\text{supported})$ slightly above one half. The returned output therefore derives every element from the calibrated probabilities with a single rule. A sentence is labelled supported if and only if $P(\text{supported}) \geq \tau$; otherwise it receives the more probable of the two remaining classes. The strict policy flags every unsupported sentence, the lenient one only contradicted sentences. The

[7]The *Brier score* is the mean squared error between the predicted probability and the observed outcome, coded as 1 if the event occurs and 0 otherwise: the lower it is, the better the probabilities.

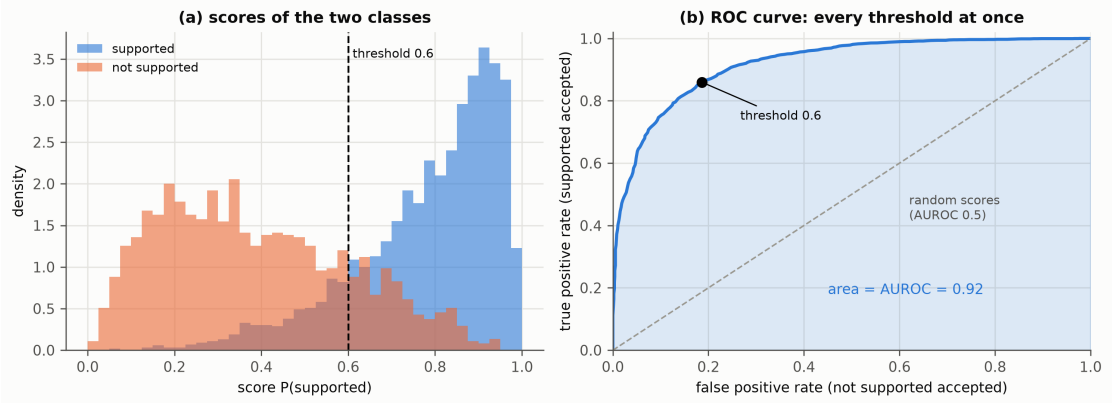


Figure 4.5: How the AUROC arises (a sketch with synthetic scores, not a result). (a) Distribution of $P(\text{supported})$ for supported and unsupported sentences: the two distributions overlap, and every threshold accepts part of each. (b) Moving the threshold from 1 to 0 yields the ROC curve; the black dot corresponds to the threshold of panel (a). The area under the curve is the AUROC: the more separated the two distributions, the closer the curve gets to the top-left corner and the area to 1. The diagonal corresponds to random scores.

threshold τ is fitted on the dev set so as to maximise the three-class balanced accuracy. The user can change it in every request, thereby choosing a different operating point without retraining the model. Figure 4.6 summarises the path from the raw logits to the returned labels.

4.5 Software

GroundOne is implemented as an installable Python package based on MLX [3, 58], the array framework designed for Apple silicon. All the training and inference phases described so far run with MLX on the same laptop. The package comprises the dataset builder with its checks, the training loop, calibration, evaluation on the frozen test sets, the baselines and a memory-protection mechanism. The latter runs long processes one at a time and stops a process before the system runs out of memory.

The verifier is exposed through a command-line interface and an HTTP server. The server offers a native endpoint and an endpoint compatible with the System One interface of Jev, already used by other decision models, so that existing clients can adopt GroundOne by changing only the address. The native endpoint receives a document and an answer. It segments the answer with the same rule used to build the data and runs the single forward pass of Section 4.2. For each sentence

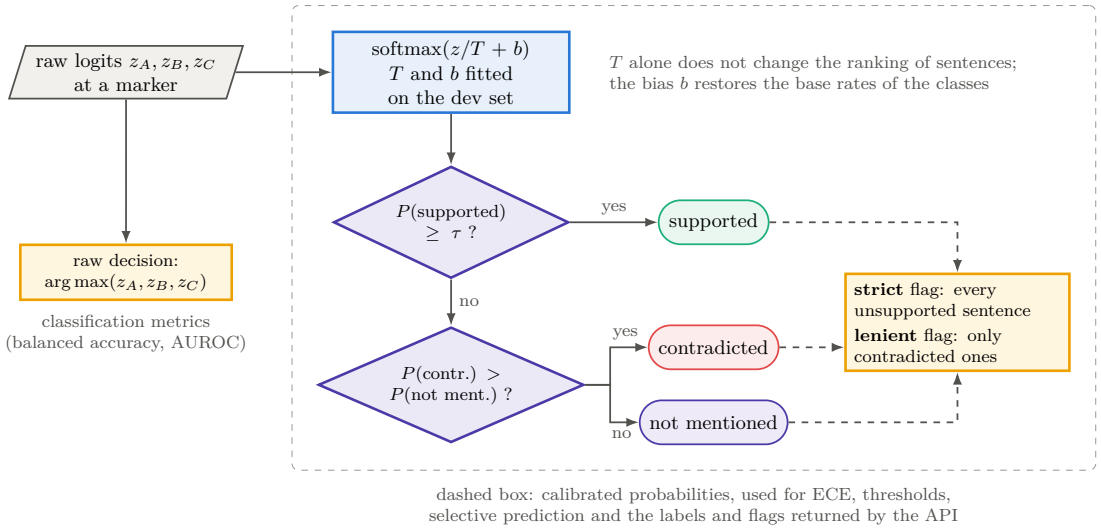


Figure 4.6: From logits to returned labels. The raw decision, that is, the argmax of the logits, serves the classification metrics of Chapter 7. The calibrated probabilities, obtained with temperature and bias fitted on the dev set, feed instead the single rule of the returned output: supported if $P(\text{supported}) \geq \tau$, otherwise the more probable of contradicted and not mentioned. The strict and lenient flags derive from this.

it reports whether it was scored, the three calibrated probabilities, the returned label and the flags of the strict and lenient policies, together with the decision rule and the threshold that produced them. It also reports the number of input tokens, the number of generated tokens, always zero, and the time taken. The compatible endpoint, instead, answers generic closed questions about a shared context, with the one-question-per-sentence format. All the model computation takes place on a single long-lived worker thread, because the Metal context of the GPU, that is, the state through which the program uses it via Apple's graphics API, must not move from one thread to another. Finally, nothing leaves the machine: models, data and requests stay local. Figure 4.7 summarises the architecture of the package.

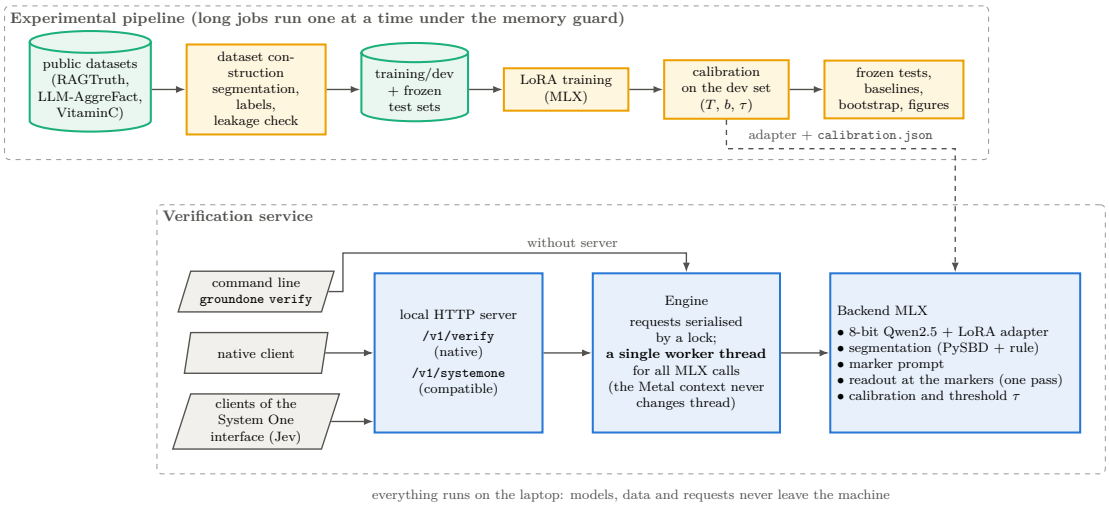


Figure 4.7: Architecture of the software. At the top, the experimental pipeline, from the public datasets to the frozen tests, run under the memory guard; at the bottom, the verification service, in which the command line and the HTTP server (native and compatible endpoints) share the same Engine, which serialises the requests and runs every MLX call on a single worker thread. The adapter and the calibration file produced by the pipeline are the only artefacts the service reads.

Chapter 5

Data

GroundOne learns a single decision, and the data must teach exactly that decision and measure it reliably. This chapter first defines the decision and its labels, then describes the three public datasets that provide them and the way they are turned into training, development and test sets. The last sections report how the reliability of the test sets was checked without new annotation, what an exploratory analysis of the result reveals, and how leakage between training and test data was ruled out.

5.1 Task and labels

The input is a retrieved document and an answer written by a language model, split into sentences. For every sentence the model decides whether, *with respect to the document*, the sentence is *supported*, *contradicted* or *not mentioned*. A sentence is supported when everything it states is said or directly implied by the document. It is contradicted when some part of it conflicts with the document, and not mentioned when it adds information that the document neither contains nor contradicts. A sentence that contains both a conflicting and an unverifiable part is labelled as contradicted, because a contradiction is the more serious error. Only the document counts: a statement that happens to be true in the world but is absent from the document is still not mentioned.

The number of decisions for one answer is the number of its scored sentences, denoted K. This quantity is central to the thesis. A decision model reads the document and the answer once, so each additional decision is almost free, while a generative judge has to produce text for every sentence. The comparison with generative judges is therefore made as a function of K, and the data must contain long answers as well as short ones.

The model outputs three probabilities per sentence, and every other quantity is derived from them rather than learned separately. The binary groundedness

score, used for comparison with published binary benchmarks, is $P(\text{supported})$. A *strict* policy, under which anything not stated in the document is an error, flags a sentence when $P(\text{supported})$ falls below a threshold τ. A *lenient* policy, under which only contradictions are errors, flags it only when it is judged contradicted. The user can thus choose the notion of error, and the threshold, at inference time without retraining the model; Section 4.4 gives the exact rule used by the served verifier.

5.2 Source datasets

The dataset is deliberately built from as few sources as possible: one for training and three for testing (Table 5.1). Fewer sources mean a single definition of the labels, a single text format and fewer places where the construction can go wrong.

Table 5.1: Source datasets and their role in GroundOne.

Dataset	Licence	Role	Labels
RAGTruth [8]	MIT	train, dev, test T1	character spans of four hallucination types on LLM answers
LLM-AggreFact [12, 89]	CC BY-ND 4.0	test T2 only	binary (supported / unsupported) per claim
VitaminC [80]	CC BY-SA 3.0	test T3 only	supports / refutes / not enough information

RAGTruth. RAGTruth [8] contains 17,790 answers that six language models (GPT-3.5, GPT-4, Llama-2 7B, 13B and 70B, and Mistral-7B) wrote in three retrieval-augmented tasks. In question answering the context is a set of passages retrieved for an MS MARCO question [101]. In summarisation it is a news article from CNN/DailyMail [102, 103] or from recent news, and in data-to-text generation it is a Yelp business record in JSON [104]. Two trained annotators marked the character spans of each answer that the context does not support, with a third review in case of disagreement, and assigned each span one of four types: evident or subtle *conflict*, and evident or subtle *baseless information*. The paper reports an agreement of 91.8 % at the level of answers and 78.8 % at the level of spans. It is the only public dataset that combines natural multi-sentence answers from several models with a distinction between contradicted and unsupported content, which is

why it is the only training source. Its official split separates 450 test sources from the remaining training sources.

LLM-AggreFact. LLM-AggreFact [12, 89] aggregates eleven human-labelled fact-checking datasets of (document, claim) pairs. They cover summarisation (AggreFact-CNN and XSum [105], TofuEval [106]), retrieval-augmented and long-form question answering (RAGTruth [8], ClaimVerify [107], ExpertQA [108], LFQA [109]) and fact checking (WiCE [110], REVEAL [111], FactCheck-GPT [112]). Its public leaderboard makes results comparable with published systems. The licence states that the data must not be used to train models [113], so in this thesis it is used only for evaluation: nothing is trained, calibrated or selected on any of its splits.

VitaminC. VitaminC [80, 114] is a fact-verification dataset built from Wikipedia revisions that change a fact. For each revision, trained annotators wrote claims that one version of the sentence supports and the other refutes; when no contradiction could be written, the pair was labelled as having not enough information. Because the two versions of the evidence are almost identical, a model must look at the exact fact rather than at word overlap. On a re-annotated sample of 2,000 pairs the authors report a Fleiss' κ of 0.71. VitaminC plays a specific role in this thesis. RAGTruth publishes no agreement for the hallucination *type*, so it cannot alone show that the boundary between contradicted and not mentioned is learned. VitaminC provides that boundary with labels of known reliability, in a different domain.

Several other candidates were examined and not adopted. Natural language inference corpora such as ANLI [79] have short premises far from retrieval-augmented contexts, and VitaminC covers the same three-way distinction with factual text. Synthetic or binary-only training data, such as the data released with MiniCheck [12], does not carry the three-way labels. Datasets whose labels were generated by a language model, or which reuse RAGTruth test answers, were excluded to keep the test sets independent and the labels human.

5.3 Construction

Sentence segmentation. Answers are first split at line breaks, because they often contain lists and headings, and each line is then segmented with PySBD [115], a rule-based segmenter that handles abbreviations and decimal numbers. Leading list markers such as "3." or "*" are removed, since they are layout rather than content. The same segmenter is used for training, evaluation and inference, so the model always sees sentences cut in the same way.

Sentences without checkable content. Some sentences make no claim about the document: greetings, headings that introduce a list, citation-only lines such as "(Passage 2)" and refusal formulas. They remain in the numbered answer as context but are not decisions, and they carry no label. A short rule detects them, and it applies to 8.2 % of all RAGTruth sentences, mostly list headings. The rule concerns only formatting, such as list markers, citation-only lines and conversational formulas, and it never changes a label; its coverage was checked against the independent segmentation of LLM-AggreFact described in Section 5.5.

Labels. A scored sentence is labelled as contradicted if it overlaps a conflict span, as not mentioned if it overlaps only baseless-information spans, and as supported otherwise. Only 27 of the 14,226 annotated spans (0.19 %) touch no scored sentence, so almost no annotation is lost in the conversion. For LLM-AggreFact the binary label is kept as released. VitaminC labels are mapped directly: *supports* to supported, *refutes* to contradicted and *not enough information* to not mentioned.

Exclusions and splits. Answers flagged by RAGTruth as incorrect refusals or truncated are removed, as are answers without any scored sentence and answers whose document and answer together exceed 2,048 tokens, the length limit imposed by memory on the 16 GB laptop. The official RAGTruth test split becomes test set T1. The training sources are shuffled with a fixed seed: 10 % of them form the development set and the rest the training set, so that answers about the same source never appear in both. Six training answers about one news article that also appears in the LLM-AggreFact test set are removed. From the real part of the VitaminC test split, which is unbalanced, 1,000 pairs per class are sampled with a fixed seed to form T3.

5.4 The resulting dataset

Table 5.2 summarises the five files. The training set contains 13,203 answers and 90,795 labelled sentences. Records are stored per answer, with one label per sentence, so the same data can feed different ways of reading the K decisions of an answer.

RAGTruth is strongly imbalanced: about 88 % of the labelled sentences are supported, and contradictions are only 3 to 4 % (Figure 5.1). This is the natural rate of errors in model answers, and it is kept: the test set reflects what a verifier meets in practice, while the imbalance is handled during training. T3 compensates with a balanced three-way test. Answers have 6.9 scored sentences on average in training and 6.5 in T1, with a maximum of 28 (Figure 5.2). For the analysis by length, T1 is divided into four bands of K: 756 answers with 1–4 sentences,

Table 5.2: The GroundOne dataset. A decision is a labelled sentence; T2 is binary.

Split	Source	Records	Decisions	Supported	Contradicted	Not mentioned
Train	RAGTruth	13,203	90,795	79,898	4,039	6,858
Dev	RAGTruth	1,468	9,996	8,827	410	759
T1	RAGTruth test	2,647	17,284	15,745	584	955
T2	LLM-AggreFact test	29,320	29,320	22,832	6,488 unsupported	
T3	VitaminC test	3,000	3,000	1,000	1,000	1,000

1,288 with 5–8, 497 with 9–13 and 106 with 14 or more. Cells with fewer than 30 sentences of a minority class, such as contradicted sentences in question answering, are reported but marked as indicative.

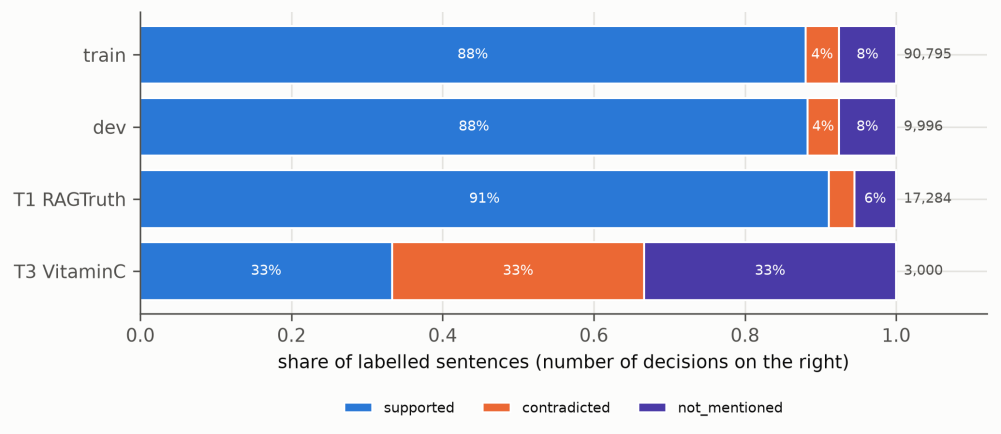


Figure 5.1: Class balance of the labelled sentences in each split. RAGTruth (train, development and T1) is about 88–91 % supported; T3 is sampled to 1,000 pairs per class.

5.5 Reliability of the test sets

The test sets were checked without new annotation, using evidence that is independent of the construction described above (Table 5.3). All test labels come from the human annotators of the published datasets. The two choices that could introduce errors, segmentation and the conversion from spans to sentence labels, were compared with LLM-AggreFact, whose authors converted the same RAGTruth test answers into sentence-level claims independently. The segmenter recovers 96.3 % of their claims exactly, and only 8.5 % of the scored sentences are absent from their

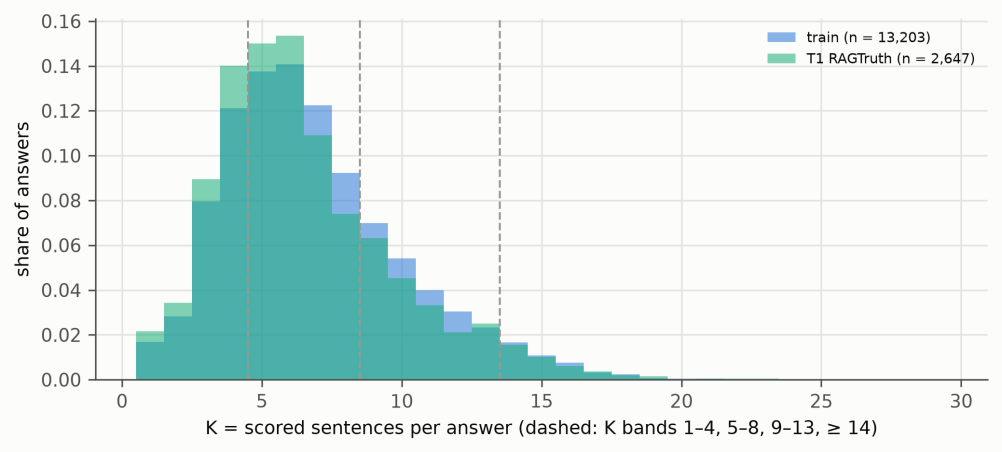


Figure 5.2: Distribution of K, the number of scored sentences per answer, in training and in T1. Dashed lines mark the bands used to report results as a function of K.

set, mostly sentences that their undocumented filter of "non check-worthy" claims removed. On the 15,492 sentences present in both, the binary labels agree in 98.75 % of cases, and all 193 disagreements go in the same direction: the span rule marks a sentence as unsupported where LLM-AggreFact considers it supported. Finally, an encoder baseline evaluated on the unchanged T2 reproduces its published score (74.0 against 73.5), which validates the evaluation code.

Table 5.3: Reliability checks run before the test sets were frozen.

Check	Criterion	Result
Label provenance	every test label comes from human annotators	pass
Segmentation	$\geq 90\,\%$ of LLM-AggreFact claims recovered, $\leq 10\,\%$ of sentences absent from it	96.3 % / 8.5 %
Label derivation	$\geq 97\,\%$ agreement with LLM-AggreFact's binary labels	98.75 %
Reference baseline	MiniCheck-RoBERTa on T2 within 1.5 points of its published score	74.0 vs 73.5
Leakage	no shared source, document or (document, sentence) pair between training and test	pass

These checks validate the binary side of T1. The distinction between contradicted and not mentioned rests on the RAGTruth annotators, whose agreement on the hallucination type is not published. Moreover, a recent re-annotation of the RAGTruth test split found 1.68 times more hallucinated answers than the original

labels [116], which suggests that T1 under-counts the two minority classes. For these reasons the three-way results on T1 are always read together with those on T3. After these checks, and before any model was trained, the three test files were frozen with cryptographic hashes, and the dataset builder refuses to overwrite them with different content; rebuilding the dataset from the raw sources reproduces the frozen files exactly.

5.6 Exploratory analysis

An exploratory analysis of the sources and of the built dataset revealed four properties that shape how the results must be read.

Errors depend on the task and on the model. Data-to-text answers are hallucinated two to four times as often as answers in the other tasks, and answers written by Llama-2 and Mistral contain most of the hallucinations, while GPT-3.5 and GPT-4 answers are almost always faithful (Figure 5.3). Results are therefore reported per task, since a single average would mostly reflect data-to-text.

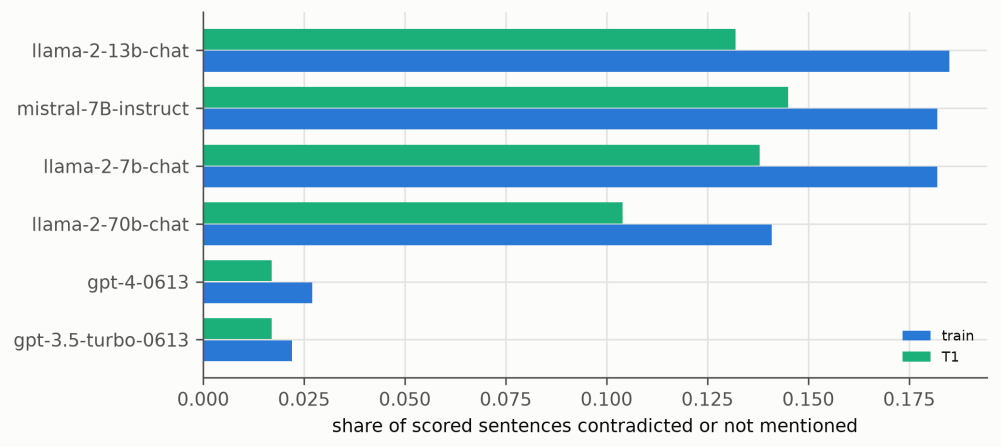


Figure 5.3: Share of contradicted or not-mentioned sentences by the model that wrote the answer, in training and in T1.

The test answers are cleaner than the training answers. In every task, and for every generator, T1 has a lower share of contradicted and not-mentioned sentences than the training set; in question answering, for instance, 0.6 % of T1 sentences are contradicted against 1.6 % in training. Metrics that depend on the base rate, such as precision or F1 on the minority classes, are therefore not

comparable between development and test. The primary metrics are balanced accuracy, macro recall and AUROC, which do not depend on it.

Contradictions are mostly evident. About 85 % of the contradicted training sentences come from spans annotated as *evident* conflicts, such as wrong numbers or names, and only 147 from *subtle* conflicts. The contradicted class is therefore mostly blatant contradictions, and subtle contradictions cannot be evaluated separately.

Surface cues exist and define the baselines. Two simple cues predict the labels without verifying the sentence against the document. The first is word overlap. The share of a sentence's content words that also appear in the document is higher for supported than for contradicted and not-mentioned sentences, in all three test sets (Figure 5.4). On its own it separates supported from unsupported sentences with an AUROC of 0.74 on T1 and 0.70 on T3. The second cue is the wording of the sentence. A classifier that sees only the sentence reaches a three-way balanced accuracy of 0.66 on data-to-text, because some details, such as the number of reviews or the availability of Wi-Fi or valet parking, are often invented in Yelp descriptions. In question answering and summarisation the same classifier stays at chance (0.36 and 0.35), and on T3 as well (0.33), where the contrastive design removes the cue. A verifier is useful only to the extent that it beats these cues, so both are reported as baselines, per task.

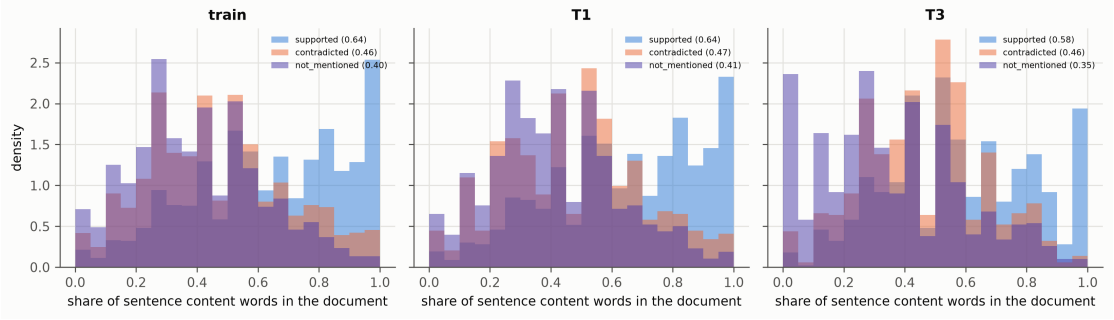


Figure 5.4: Word overlap between a sentence and its document, by class, in training, T1 and T3 (mean in brackets).

Finally, unsupported sentences are not concentrated at the end of an answer. The first sentence is rarely unsupported, the share peaks at positions three to five and stays between 8 and 11 % in later positions (Figure 5.5). Long answers still contain errors near their end, so accuracy at high K matters.

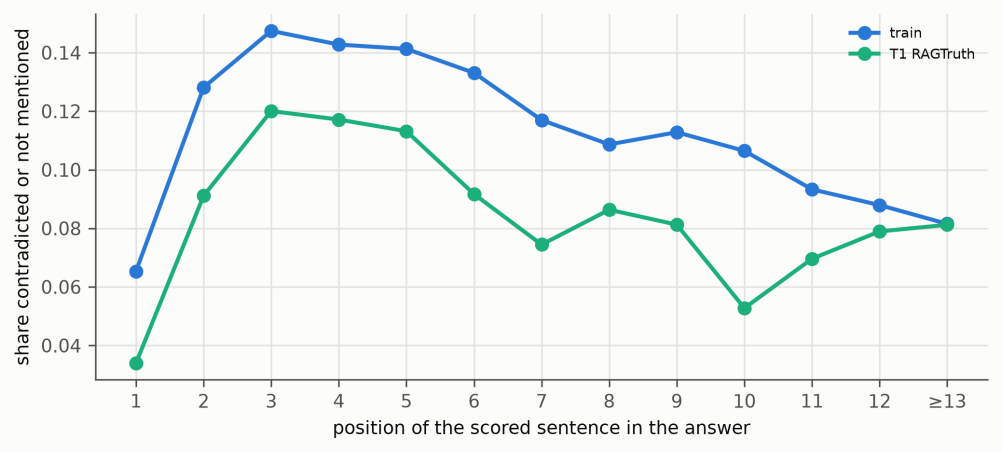


Figure 5.5: Share of contradicted or not-mentioned sentences by position in the answer.

5.7 Leakage control

Splits are made by source, never by sentence, and an automatic check runs on every build. It fails if a source, a document or a (document, sentence) pair of the training or development set appears in any test set, and if the development and training sets share a source. The check passes. It also reports sentence strings that occur in both training and test with different documents: these are 788 template sentences, such as “The business has a rating of 4.0 stars.”, which are expected in data-to-text and do not reveal any test label. Whether the public datasets used here were part of the pre-training data of Qwen2.5 cannot be verified, and this remains a limitation shared by most work on public benchmarks.

Chapter 6

Experimental Setup

This chapter describes the models that were trained, the systems they are compared with, the metrics, the statistical protocol and the way time and energy are measured. All experiments run on one machine, a MacBook Pro with an Apple M4 chip (10-core CPU, 10-core GPU, 120 GB/s of memory bandwidth) and 16 GB of unified memory [45], so that every result reflects what a verifier can do on the device where a local assistant runs. Every baseline, including the encoders and the two general-purpose decision models, also runs on this laptop; no result depends on a remote service.

6.1 Models

Both GroundOne models start from the instruction-tuned Qwen2.5 checkpoints [23] converted to MLX with 8-bit weights, and both are trained with the recipe of Section 4.3. They differ only in size, in the number of training answers and in how micro-batches are formed, which memory dictates.

The 0.5B model is trained on all 13,203 training answers. One epoch of 802 optimiser steps took 5.5 hours of wall-clock time, with an evaluation on the full development set every 150 steps, and training used at most 2.6 GB of memory. The 3B model is about six times slower per token, so it is trained for one night on a random subset of 6,500 answers (about 45,000 decisions): 403 steps in 12.25 hours, with a peak of 5.9 GB. Its development evaluation, every 80 steps, uses a fixed subset of 200 answers, because a full pass would take about an hour. In both cases the checkpoint with the best development score is kept and then calibrated as described in Section 4.4; for the 3B this is step 320.

To measure how much the results depend on the random seed, which sets the initialisation of the adapters and the order of the batches, the 0.5B recipe is repeated with a second seed. Fine-tuning outcomes are known to vary with both factors [97].

One further run with a third seed collapsed early to a constant prediction and was stopped at about 60 % of the epoch; being incomplete, it is excluded from the tables, and it is discussed in Section 7.2. The completed second run is evaluated on all three test sets exactly like the first.

6.2 Baselines

The baselines isolate the sources of any advantage. Each one answers a specific question.

Cues that never read the document as a whole. Two simple predictors give the floor that any verifier must clear. The first scores a sentence by the share of its content words that also occur in the document. The second is a bag-of-words classifier trained on the training sentences alone, without the document. Both are described in Section 5.6.

Untrained readout (B1). The same Qwen2.5 models, without fine-tuning, are asked the three-way question for each sentence and their answer-letter logits are read. Because an untrained model has no reason to treat the `Verdict:` marker as a question, this baseline uses the format with one explicit question per sentence (Section 4.2). It measures what the decision-model mechanism gives before any specialisation.

Generative judges (B0a, B0b). The untrained models are also used in the way that is common in practice: they receive the document, the numbered answer and the list of sentences to check, and they write a verdict for each sentence, with greedy decoding. The *terse* judge writes one line "`<number>: <letter>`" per sentence; the *verbose* judge writes one short sentence of reasoning followed by the verdict. A verdict that cannot be parsed is a format failure; it is counted as an error in accuracy and reported separately.

Equal-output judges (B0d). The terse and verbose judges return less than GroundOne: a letter per sentence and no probability, so they offer no calibration, no adjustable threshold and no choice of policy. Two further judges are therefore asked for exactly GroundOne's content, a label and three probabilities per sentence. The first writes them as a JSON object, with the probabilities expressed in words, which is what a developer would obtain from a prompt alone. The second is the terse judge whose three probabilities are read from the logits at the letter it generates; it obtains the same content at the cost of the terse judge and is the strongest generative opponent. Both judges are calibrated with the same temperature and

bias as GroundOne, fitted on 200 development answers, so that their probabilities receive the same treatment. The JSON prompt was chosen on twelve development answers for format validity only: a numeric example in the prompt was copied verbatim by the models and was replaced by placeholders. Parsing is lenient, in the judges' favour: complete entries of a truncated object are kept, and a missing label is taken from the probabilities. Decoding is not constrained by a grammar [88], so that the protocol stays that of the other judges; this choice affects mainly the format failures of the smallest model.

Readout versus generation (B2). The fine-tuned GroundOne models are run with the same prompt and the same single pass, but the verdict at each marker is the token with the highest probability over the whole vocabulary, which is what greedy generation of one token would produce. The three letters, with or without a leading space, map to the three classes, and any other token is a format failure. This control keeps weights, prompt and cost identical and changes only whether the decision is restricted to the admissible options.

A general-purpose decision model: Rizzo-Flow (B3). This open project [16] implements the decision-model idea with a 4B model fine-tuned on generic typed decisions. It runs in its own server, based on llama.cpp [2], and is queried over HTTP with the document and the numbered answer as shared state and one three-way choice question per sentence. A 4-bit quantised model is used because the 8-bit one does not fit in memory beside the operating system.

A general-purpose decision model of the same size: d1-3B (B3b). Liquid AI's d1-3B [18] is an open decision model of 3.1B parameters, post-trained for typed decisions on a broad mixture of tasks, which reads restricted answer letters with the state shared across questions: the same readout family as GroundOne, at the size of its larger model, but general-purpose. It isolates the value of specialisation against a decision model that was already trained to decide, rather than against an untrained one. It is run zero-shot with its own inference code, in PyTorch on the GPU of the laptop and without network access, with the same state and questions as Rizzo-Flow and at most six questions per call on the same state; on LLM-AggreFact its documents are split exactly as GroundOne's and the maximum over chunks is kept. A practical detail matters for reproducibility: PyTorch keeps compiled kernels for every new input length on the Apple GPU and does not release that memory, so the evaluation runs in worker processes that are restarted when the memory used by the GPU driver exceeds 11 GB, which leaves the outputs unchanged.

Specialised encoders (B4). MiniCheck-RoBERTa [12] and HHEM-2.1-open [13] are the strongest small verifiers available. They score each (document, sentence) pair independently and return one probability of support; long documents are split into chunks of about 400 words and the maximum over chunks is kept, as in MiniCheck. They have no three-way output, so they are compared on the binary decision only. They run with PyTorch on the GPU of the laptop.

Published results. For LLM-AggreFact, the scores of larger systems from the public leaderboard, such as Qwen2.5-7B used zero-shot, are reported as given and not re-run.

The slower systems are evaluated on stratified subsamples of the test sets: RizzoFlow on 300 answers of T1, 600 pairs of T3 and 1,100 items of T2, of which it scored 1,093 (seven exceeded its context); d1-3B on a random subsample of 4,000 items of T2; the untrained 3B on 600 answers of T1; and the generative judges on the 200 answers of the latency protocol below. GroundOne, the untrained 0.5B, the encoders and, on T1 and T3, d1-3B are evaluated on the complete test sets, and the comparisons with a subsampled system are always made on the items that both systems scored.

6.3 Metrics

Because the share of hallucinated sentences differs between the development set and T1 (Section 5.6), the primary metrics do not depend on the base rate. For the three-way decision the main metric is the balanced accuracy, that is the mean recall over the three classes, together with the recall of each class and the share of each predicted class, which reveals a model that collapses onto one answer. For the binary decision the metrics are the balanced accuracy at a threshold of 0.5, as in the LLM-AggreFact leaderboard, and the AUROC of $P(\text{supported})$, which does not depend on any threshold. Calibration is measured with the ECE defined in Section 4.4, the Brier score and the negative log-likelihood, and selective prediction with risk–coverage curves and the area under them (AURC) [96]. All metrics on T1 are also reported per task and per band of K.

6.4 Statistical protocol

Every difference between two systems is accompanied by a 95 % confidence interval from a paired bootstrap with 2,000 resamples [117, 118, 119]. The resampling unit is not the sentence but the group that the leakage check uses: the source document in RAGTruth, the document in LLM-AggreFact and the Wikipedia page in VitaminC. Sentences of the same answer, and answers about the same document, are strongly

correlated, and resampling them independently would produce intervals that are too narrow. On LLM-AggreFact the groups are resampled within each of the eleven datasets, because the macro average gives each dataset the same weight. Both systems of a comparison are evaluated on the same resamples, and the reported p-value is the two-sided share of resamples in which the difference changes sign.

Two limits of this protocol are stated in advance. First, about fifty comparisons are tested and no correction for multiple comparisons is applied; the conclusions therefore rest on intervals that lie far from zero, and a difference with a p-value near 0.05 is not claimed. Second, resampling the data does not capture the variability due to training: the two runs of the 0.5B provide that information separately (Section 7.2).

6.5 Latency, energy and hardware

Latency is measured on 200 answers of T1, 50 from each band of K, so that long answers are as well represented as short ones. GroundOne, the question-per-sentence readout and all generative judges run in process on MLX, one answer at a time, as a verifier would in a local assistant. The first three answers serve as warm-up and are excluded. For the decision-model variants each answer is timed three times and the median is kept; the generative judges are timed once, since greedy decoding is deterministic and each run is much longer. The reported time is the wall-clock time of the model call for one answer, from which the time per decision follows.

Times were also recorded for the encoders, for Rizzo-Flow and for d1-3B, but under conditions that do not allow a fair comparison. The encoders ran with PyTorch and scored each sentence separately, re-reading the document every time, which penalises them; Rizzo-Flow ran in a different inference engine, with 4-bit weights and behind an HTTP interface; d1-3B ran in PyTorch with its own code and with the worker restarts described above. No latency claim is therefore made against them.

Energy is measured with the `powermetrics` tool of macOS, which reports the power of the CPU, the GPU and the Neural Engine every 200 milliseconds. For each system, 100 answers of the latency subset (25 per band of K) are verified once, after 60 seconds of idle measurement with the model already loaded; the energy of a verification is the mean power during the run, minus the idle power, multiplied by the run time and divided by the number of answers. The measurement covers the computing units of the chip but not the memory, the display or the storage, and it was taken overnight with the machine otherwise idle.

Three measurements concern locality directly. The first checks independence from the network: a complete evaluation of T1 is repeated with the libraries in

offline mode and inside a macOS sandbox that denies the process any network access, and its predictions are compared with those of the original run. The second measures the effect of quantisation: the same adapter of the 0.5B model is applied to the original 16-bit (bf16) weights and evaluated on T1 and T3. The third measures co-residence: a single process loads the generator planned for the demonstration application, Qwen2.5-3B-Instruct quantised to 4 bits, and a GroundOne verifier; for 30 T1 documents, ten per task, the generator writes an answer of at most 300 tokens and GroundOne verifies it. The peak memory allocated by MLX and the generation and verification times are recorded. The document is given directly, as in RAGTruth, so the retriever is not included. These measurements also run entirely inside the network-denying sandbox.

Only one model runs at a time, and a memory guard runs every long job and records the free memory and the swap throughout. Results are written once and never overwritten, and each run stores its configuration, so every number in Chapter 7 can be traced to the run that produced it.

Chapter 7

Results

This chapter reports how GroundOne learns and how stable its training is, what it achieves on the three frozen test sets, how it compares with the baselines, and what the decision-model readout contributes by itself. It then examines the quality of the probabilities, the time and energy needed to verify an answer, and the errors that remain. Intervals are 95 % cluster-bootstrap intervals computed as described in Section 6.4.

7.1 Training

Both models learned the task. Before training, a model reading the marker format has no notion of what the marker asks and its verdicts are at chance: the 0.5B answers "contradicted" for 88 % of the development sentences and reaches a three-way balanced accuracy of 0.32. In the completed runs the training loss fell below the value of a model that learned only the class frequencies ($\ln 3 \approx 1.10$) within the first hundred steps, and the development score then rose steadily (Figure 7.1). The 0.5B, trained on all 13,203 answers, reached its best score at step 600 of 802, with a balanced accuracy of 0.633 averaged over the three tasks, after which it stayed flat. The 3B, trained for 12.25 hours on 6,500 answers, reached 0.705 at step 320 of 403 on its development subset. Measured on the full development set, the three-way balanced accuracy is 0.731 for the 0.5B and 0.784 for the 3B, and all three classes were predicted throughout both runs.

7.2 Variability across seeds

The 0.5B recipe was run again with a second seed, which changes the initialisation of the adapters and the order of the batches. When training converges, it converges to the same place (Figure 7.2): the second run reaches a three-way balanced

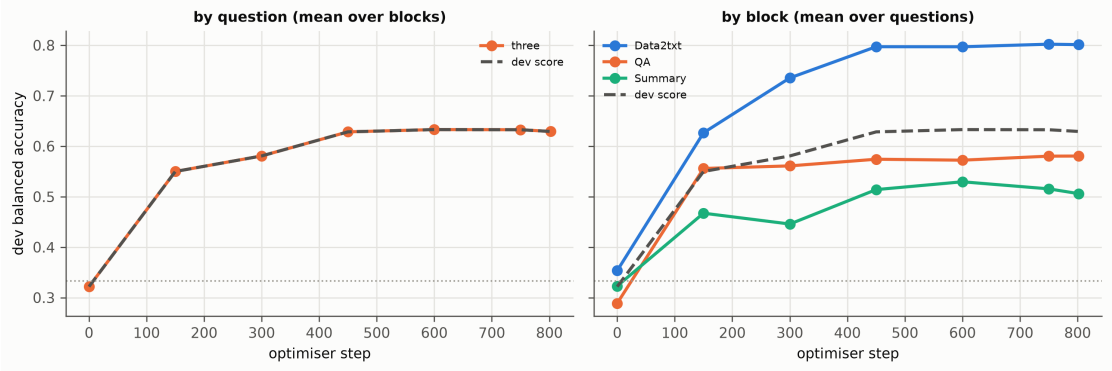


Figure 7.1: Three-way balanced accuracy on the development set during training of the 0.5B model, overall (left) and per task (right). The dotted line marks chance. The 3B shows the same shape on its 200-answer development subset.

accuracy of 0.767 on T1 against 0.769, an AUROC of 0.907 against 0.905, 71.5 against 70.1 on LLM-AggreFact and 0.434 against 0.442 on VitaminC. On T1 the paired difference is +0.001 (−0.008 to +0.011) and is not significant. The variation lies in finer details. The errors are distributed differently between the classes, with the recall of supported and not-mentioned sentences moving by about 0.04 in opposite directions, and the smaller slices move more: the band with fourteen or more decisions by 0.023, the AUROC on VitaminC by 0.025 and single datasets of LLM-AggreFact by up to ten points, as on WiCE (67.9 against 78.0).

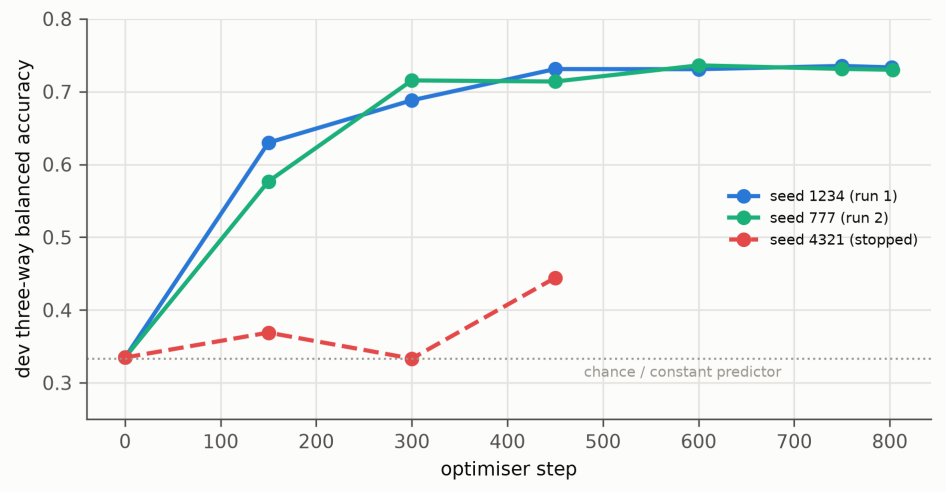


Figure 7.2: Development three-way balanced accuracy of the 0.5B during training with the same recipe and three seeds. Two runs converge to the same level; the third collapsed to a constant predictor and was stopped.

Two consequences follow. First, the bootstrap, which resamples the test data, does not capture this source of variability: on LLM-AggreFact and VitaminC the two runs of the same recipe differ by more than their intervals allow (-0.013 on the macro of T2, -0.025 on the AUROC of T3). Differences between systems that are smaller than the spread between seeds, about 1.5 points on the macro of LLM-AggreFact and 0.025 in AUROC on VitaminC, are therefore not interpreted, and neither are single datasets of LLM-AggreFact unless the gap exceeds about five points.

Second, training does not always converge. A run with a third seed stayed near chance: its development score was 0.369 at step 150 and 0.333 at step 300, where it predicted "supported" for every sentence, and the training loss remained close to the value of a model without information. The run was stopped at step 470 of 801, when its score had climbed to 0.444, so whether it would have recovered by the end of the epoch is unknown; being incomplete, it is excluded from the tables. The recipe has two features that are known to make fine-tuning fragile: gradients are not clipped, and the optimiser runs without bias correction, the default of the framework, a setting that has been identified as a main cause of failed fine-tuning runs [99, 98]. One collapse in three runs does not measure a failure rate, but it shows that the recipe is not fully robust to initialisation.

7.3 Accuracy on the RAG test set (T1)

On the 17,284 sentences of T1 the 0.5B model reaches a three-way balanced accuracy of 0.769 and the 3B model 0.832 (Table 7.1); the difference, $+0.063$ ($+0.052$ to $+0.076$), is the clearest effect of model size. On the binary decision the 0.5B reaches a balanced accuracy of 0.815 and an AUROC of 0.905, and the 3B 0.874 and 0.940. Both are far above the cues that never verify a sentence against its document: word overlap reaches an AUROC of 0.737, and the sentence-only classifier a three-way balanced accuracy of 0.570.

Three observations stand out. First, accuracy does not degrade with the length of the answer: answers with fourteen or more decisions are among the best handled, with 0.723 for the 0.5B and 0.839 for the 3B, and the hardest band is the shortest, which mostly contains question-answering and summarisation answers. This matters because long answers are where a decision model saves the most time; on these answers the gain of the 3B over the 0.5B is $+0.116$ ($+0.050$ to $+0.180$). Second, contradictions are recognised mainly in data-to-text, where RAGTruth concentrates them. In question answering and summarisation the 0.5B separates supported from not-mentioned sentences well but finds almost no contradictions, with a recall of 0.03 and 0.13. The 3B raises both to 0.30, so model size compensates in part for the scarcity of such examples in the training data. Third, question answering and

Table 7.1: GroundOne on T1, by task and by number of decisions per answer (K): three-way balanced accuracy, recall of contradicted sentences, and AUROC of P(supported).

		GroundOne 0.5B			GroundOne 3B		
Subset	Sentences	Bal. acc.	Contr.	AUROC	Bal. acc.	Contr.	AUROC
All	17,284	0.769	0.68	0.905	0.832	0.76	0.940
Question answering	4,876	0.589	0.03	0.916	0.699	0.30	0.948
Summarisation	4,538	0.544	0.13	0.775	0.629	0.30	0.873
Data-to-text	7,870	0.812	0.84	0.929	0.873	0.90	0.955
$K = 1$–4	2,434	0.606	0.18	0.857	0.677	0.25	0.921
$K = 5$–8	8,012	0.758	0.67	0.899	0.829	0.78	0.936
$K = 9$–13	5,201	0.824	0.82	0.926	0.863	0.86	0.945
$K \geq 14$	1,637	0.723	0.50	0.908	0.839	0.68	0.957

summarisation are the clearest evidence that the models read the document: there the sentence-only classifier stays at chance (0.36 and 0.35), while the two models reach 0.59–0.70 and 0.54–0.63.

The same predictions can be read at the level of whole answers, which is how the community Decision Index scores general-purpose decision models on RAGTruth: an answer counts as hallucinated if it contains any unsupported content, and the metric is the F1 score of that class, which a system answering "hallucinated" every time already reaches at 0.52 [22]. Taking the minimum calibrated P(supported) over the sentences of an answer, with a threshold fitted on the development set, GroundOne reaches 0.733 with the 0.5B and 0.784 with the 3B on the 2,647 answers of T1. On the leaderboard no general decision model below one billion parameters exceeds 0.55, Jev reaches 0.765 and the best model, of 27B parameters, 0.803. The comparison is only indicative, because GroundOne was trained on RAGTruth while the leaderboard models are evaluated zero-shot, with a different prompt and on 53 more answers, but it places a 0.5B specialist far above general decision models of its size.

7.4 Comparison with the baselines

Figure 7.3 compares all systems on T1. The specialised encoders reach an AUROC of 0.814 (MiniCheck-RoBERTa) and 0.821 (HHEM), above word overlap but below both GroundOne models: the 0.5B exceeds them by +0.091 (+0.074 to +0.107) and +0.083 (+0.066 to +0.099), the 3B by +0.126 and +0.119. The comparison by task qualifies this result. GroundOne's advantage comes mostly from data-to-text, where it is trained in domain and where the encoders are weakest (AUROC 0.71–0.75),

while on summarisation MiniCheck-RoBERTa is better than the 0.5B model (0.872 against 0.775) and level with the 3B (0.873). The untrained models show how much of the accuracy training provides: the 0.5B is at chance because it never answers "supported", and the 3B, asked the same question, reaches 0.437 against 0.818 after training, measured on the same answers. How much the restricted readout contributes on top of training is measured separately in Section 7.6.

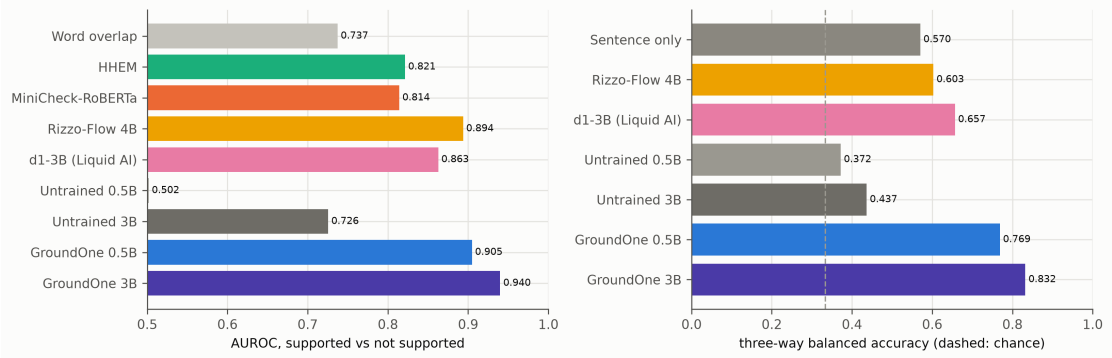


Figure 7.3: T1: AUROC of P(supported) for every system (left), and three-way balanced accuracy for the systems with a three-way output (right). Rizzo-Flow is evaluated on 300 answers and the untrained 3B on 600; all other systems, including d1-3B, on the full test set.

Rizzo-Flow, the larger general-purpose decision model, gives the most instructive comparison. On the same 300 answers of T1, Rizzo-Flow, a 4B model fine-tuned for generic decisions, reaches a three-way balanced accuracy of 0.603, against 0.798 for the 0.5B GroundOne and 0.840 for the 3B; the differences, +0.195 (+0.136 to +0.249) and +0.237 (+0.184 to +0.285), are large. Its AUROC (0.894) is close to that of the 0.5B on these answers (0.916; the difference is not significant) because it ranks sentences well but predicts "supported" for 90 % of them; the 3B ranks them better (+0.056, +0.034 to +0.079). A specialist eight times smaller therefore outperforms the generalist on the decision it was built for, while matching it in the ability to rank sentences.

d1-3B, a general-purpose decision model of the same size as the 3B, gives the sharpest test of specialisation, because it was already trained to make typed decisions and differs from GroundOne mainly in what it was trained on. On the full T1 it reaches a three-way balanced accuracy of 0.657 and an AUROC of 0.863. Both GroundOne models are clearly better: the 0.5B, six times smaller, by +0.112 (+0.092 to +0.133) in three-way accuracy and +0.042 in AUROC, the 3B by +0.175 (+0.157 to +0.195) and +0.077. The gap is widest on long answers, where d1-3B falls to 0.527 with fourteen or more decisions and the 3B GroundOne stays at 0.839 (+0.312, +0.230 to +0.405), and d1-3B recognises only half of the

not-mentioned sentences (recall 0.49). Within the domain of retrieval-augmented answers, specialisation is therefore worth far more than a general training for decisions.

On LLM-AggreFact (T2) the 0.5B model reaches a macro-averaged balanced accuracy of 70.1 over the eleven datasets, against 64.2 for the same model used zero-shot and 72.8 for the zero-shot 7B model on the public leaderboard. It remains below MiniCheck-RoBERTa (74.0) by 3.9 points, a significant difference. On the four datasets closest to retrieval-augmented generation the two are on par (74.4 against 74.6), and on the RAGTruth column GroundOne leads, 82.1 against 76.5, a comparison that favours it because it was trained on RAGTruth training answers. Outside retrieval-augmented generation the 0.5B is weaker than the encoder by seven to nine points on AggreFact-CNN, AggreFact-XSum, FactCheck-GPT and REVEAL, the expected cost of training on RAG data only.

The 3B model, evaluated on the full test set, closes most of this gap. Its macro-averaged balanced accuracy is 74.8, on par with MiniCheck-RoBERTa (74.0; paired difference +0.8 points, interval −0.5 to +2.1) and above the published zero-shot score of the 7B model (72.8, for which no interval can be computed). On the RAG datasets it reaches 76.9, 2.4 points above the encoder (interval +1.4 to +3.3). Its gain over the 0.5B, +4.7 points (+3.4 to +5.9), is broad, with the largest improvements on REVEAL, WiCE, FactCheck-GPT and XSum, so model size improves transfer beyond the training domain. The 3B remains clearly below the encoder only on news summarisation (AggreFact-CNN, 59.3 against 67.1). Rizzo-Flow, on the 1,093 items of its subsample, reaches 61.5; there the 3B GroundOne is 12.0 points better (+7.4 to +16.5). An independent evaluation of the closed decision model Jev on the same test set and with the same metric reports 78.6, and 76.9 for a 27B open model read through its answer letters, with prompts different from those used here, so that no interval can be computed [20]. On a broad grounding benchmark, then, a 3B specialist trained only on RAG answers matches the best small encoder but remains below the closed state of the art and much larger open models. The general-purpose d1-3B confirms the picture at the same size: on a random subsample of 4,000 items it reaches 75.7, against 74.3 for the 3B GroundOne on the same items, a difference that is not significant (−1.3, −4.0 to +1.0), while it beats the 0.5B by 6.5 points. It is weaker on the RAGTruth column (78.9 against 87.3) and clearly stronger on WiCE and on dialogue summarisation.

7.5 Transfer to a different domain (T3)

On VitaminC the picture reverses. On the full test set the 0.5B reaches a three-way balanced accuracy of 0.442 and the 3B 0.527, with AUROCs of 0.722 and 0.817,

and MiniCheck-RoBERTa reaches an AUROC of 0.968. On the 600 pairs evaluated with Rizzo-Flow, the general-purpose model reaches 0.712 and an AUROC of 0.909, against 0.442 and 0.744 for the 0.5B and 0.537 and 0.845 for the 3B. Two facts put these numbers in context. MiniCheck-RoBERTa starts from the AlignScore model, whose training data include VitaminC [11], so T3 belongs to its training distribution, whereas for GroundOne it measures transfer from RAGTruth. Moreover, the gain of the 3B over the 0.5B on T3 comes from model size rather than from the RAG training: the untrained 3B already reaches 0.531 and an AUROC of 0.821. d1-3B, the general-purpose decision model of the same size, makes the contrast explicit: it reaches 0.724 and an AUROC of 0.913 on the full T3, ahead of the 3B GroundOne by 0.197 and 0.096, the reverse of what happens on T1. Specialisation on RAG answers therefore improves the RAG task substantially but does not, by itself, produce a general fact checker. Section 7.12 shows which cases are missed.

7.6 Readout versus generation

The control B2 isolates the readout. With the same fine-tuned weights, the same prompt and the same single pass, the verdict at each marker is either read from the three answer letters, as GroundOne does, or taken as the token that greedy generation would emit (Figure 7.4). For the 0.5B the generated token gives a three-way balanced accuracy of 0.521 against 0.769 for the readout, a difference of +0.248 (+0.228 to +0.269): in a quarter of the markers (24.7 %) the model would not emit a letter at all, but most often a bare space, the beginning of a longer answer. For the 3B the generated token is almost always a letter (1.2 % failures) and agrees with the readout in 92 % of the markers, yet the readout is still better, 0.832 against 0.761 (+0.071, +0.059 to +0.083). On answers with fourteen or more decisions the gap is largest for the small model, +0.346.

The latency of the two options is the same, since both use one pass; what differs is the decision rule. Restricting the choice to the admissible options removes format failures by construction and, since the model is trained only to order the three letters, also gives the more accurate decision (Section 4.2). The smaller the model, the more this restriction matters. The often repeated expectation that reading answer logits and generating a single answer token are equivalent therefore holds for the cost but not for the accuracy of a fine-tuned verifier.

7.7 Calibration and selective prediction

The raw probabilities underestimate support. In the reliability diagram (Figure 7.5) the raw curve lies above the diagonal, because the class weights train the model as if the classes were balanced; the binary ECE on T1 is 0.30 for the 0.5B and 0.18

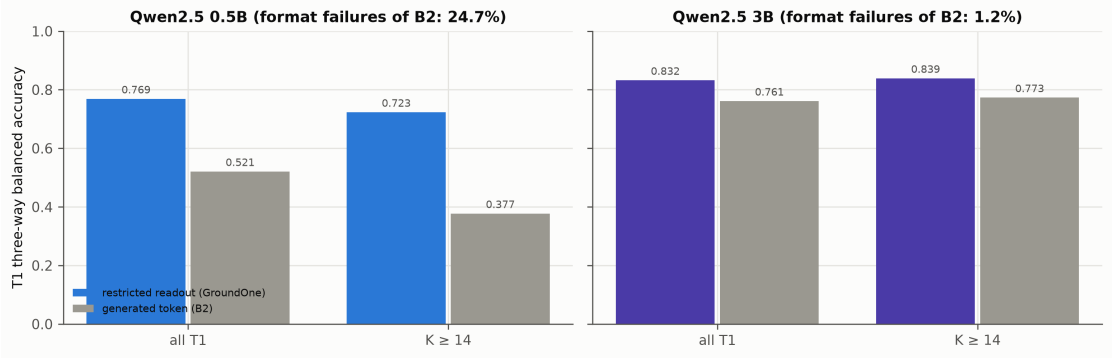


Figure 7.4: Control B2 on the full T1: with the same weights and the same forward pass, the verdict is read from the three answer letters (GroundOne) or taken as the token that greedy generation would emit (format failures counted as errors).

for the 3B. The temperature and bias fitted on the development set bring the ECE on T1 to 0.019 for both models, and to 0.020 for the second run of the 0.5B, and reduce the negative log-likelihood from 0.55 to 0.21 (0.5B) and from 0.36 to 0.17 (3B), while the AUROC is practically unchanged. The map fitted on development data therefore transfers to the test set of the same domain.

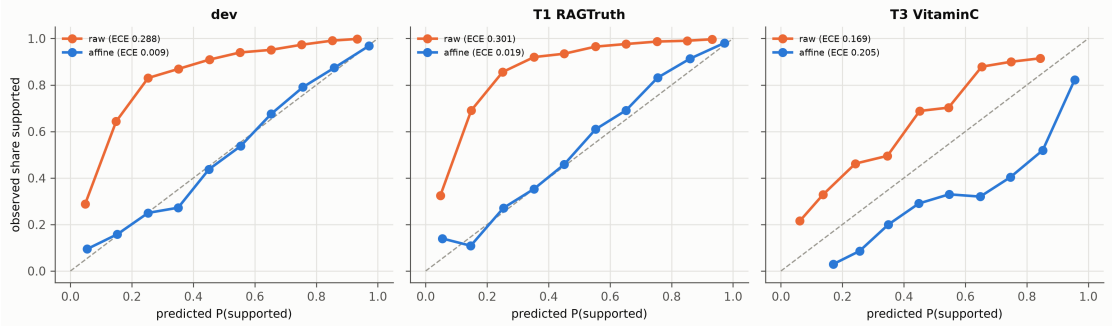


Figure 7.5: Reliability of P(supported) of the 0.5B before (raw) and after calibration on the development set, for the development set, T1 and T3. Points are bins with at least 20 sentences.

Calibration has a consequence that must be kept in mind. The bias restores the true frequency of each class, so the most probable calibrated class becomes "supported" more often: for the 0.5B on T1 the accuracy rises from 0.82 to 0.93, but the three-way balanced accuracy falls from 0.77 to 0.63. Taking the most probable calibrated class is therefore not a good decision rule when the three kinds of error matter equally. The served rule of Section 4.4 resolves this: with the threshold

fitted on development data, $\tau = 0.935$ for the 0.5B, 0.895 for its second run and 0.925 for the 3B, it reaches 0.758, 0.769 and 0.830 on T1, within 0.01 of the raw decision evaluated throughout this chapter, while keeping labels and calibrated probabilities consistent. The thresholds are high because 91 % of the RAGTruth sentences are supported: a sentence is served as supported only when the model is very confident.

Used as confidences, calibrated probabilities make selective prediction effective: deferring the least confident sentences lowers the error rate of the binary decision quickly, with an area under the risk–coverage curve of 0.015 (0.5B) and 0.010 (3B) on T1 (Figure 7.6).

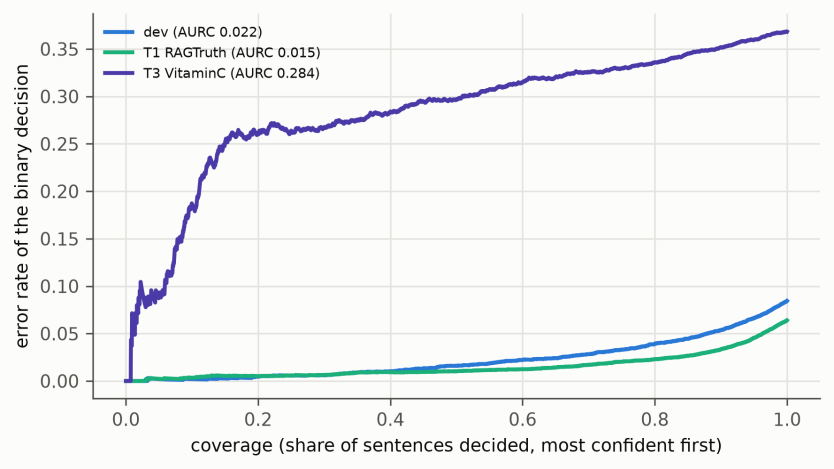


Figure 7.6: Risk–coverage curves of the binary decision of the 0.5B with calibrated probabilities: error rate as the least confident sentences are deferred.

Calibration does not transfer to VitaminC, where the binary ECE rises from 0.17 to 0.21 for the 0.5B and from 0.14 to 0.19 for the 3B, and where the served rule loses 0.02–0.04 with respect to the raw decision. The bias encodes the base rate of RAGTruth, while VitaminC is balanced by construction. A calibration is valid for the distribution it was fitted on, so a deployment needs a small labelled sample of its own data.

7.8 Probabilities at equal output

The equal-output judges answer whether a generative judge of the same size, asked for the same content as GroundOne, can match the quality of its probabilities. It cannot (Figure 7.7). On the 200 answers of the latency subset, with every system calibrated on development data in the same way, the AUROC of P(supported) is 0.94 and 0.97 for GroundOne, 0.48 and 0.68 for the judge whose probabilities are

read from the logits, and 0.47 and 0.58 for the judge that writes them, for the 0.5B and the 3B respectively. The area under the risk–coverage curve is 0.009 and 0.005 for GroundOne against 0.048 to 0.063 for the judges, five to twelve times larger. The accuracy of the decisions follows the same pattern: 0.835 and 0.900 for GroundOne against at most 0.41 for any judge.

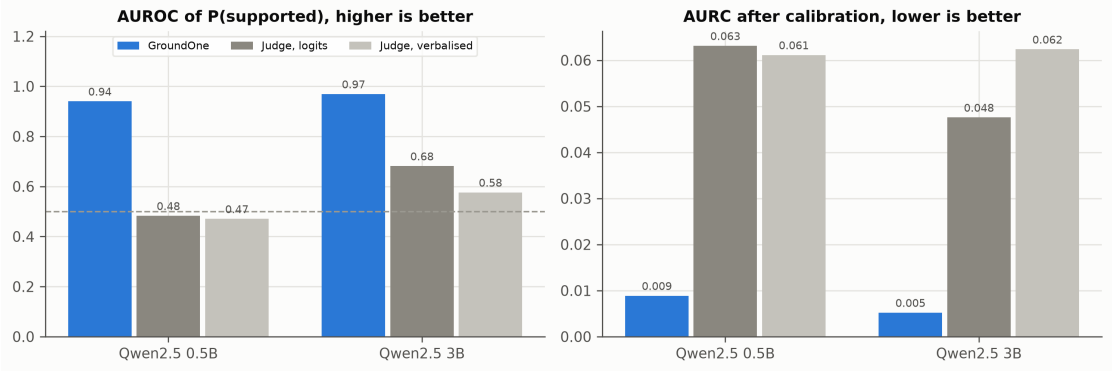


Figure 7.7: Equal-output comparison on the 200 latency answers: generative judges of the same size asked for GroundOne's output, with probabilities read from the logits of the generated letter or written in JSON; all systems calibrated on development data with the same map. Sentences with a valid verdict only.

Two further observations matter for practice. Probabilities written in words carry almost no information: the 3B writes a vector with all the mass on one class for 93 % of the sentences, using only sixteen distinct vectors, and the 0.5B writes uninformative ties such as equal probabilities for all classes; counting the ties without a label as failures, its balanced accuracy falls from 0.234 to 0.081. This agrees with the finding that verbalised confidence of small models is coarse and poorly informative [95, 94]. Second, the ECE alone would hide all of this: after calibration every system, including the judges, reaches a binary ECE between 0.03 and 0.05, because a calibration map can match the average confidence to the base rate even when the probabilities do not separate the classes. Discrimination must be measured separately, with the AUROC and the risk–coverage curve.

Asking for the probabilities also has a cost. The judge that reads them from logits is as fast as the terse judge, since it generates the same letters, but the judge that writes the JSON object needs 17.0 seconds per answer with fourteen or more decisions on the 3B, as much as the verbose judge, for an accuracy no better than that of the terse one.

7.9 Latency as a function of K

Figure 7.8 shows the central measurement of the thesis: the time needed to verify one answer, on the same 200 answers of T1, as the number of decisions grows. Among the systems that run in process on MLX, GroundOne is the fastest at every size and in every band of K, and its advantage grows with K. For answers with fourteen or more decisions, the 0.5B GroundOne needs 0.41 seconds, against 0.80 for the terse judge and 2.51 for the verbose judge of the same size: it is 1.95 times faster than the first (1.82 to 2.14) and 6.16 times faster than the second (5.72 to 6.67). The 3B GroundOne needs 2.57 seconds, against 4.44 and 15.86, which is 1.73 times (1.65 to 1.82) and 6.18 times (5.88 to 6.65) faster. Reading all verdicts in one pass is also faster than asking one question per sentence in a batch, by a factor of 1.4–1.5 at the highest K. Model size is the other lever: the 0.5B GroundOne verifies an answer about six times faster than the 3B (0.33 against 1.99 seconds at the median).

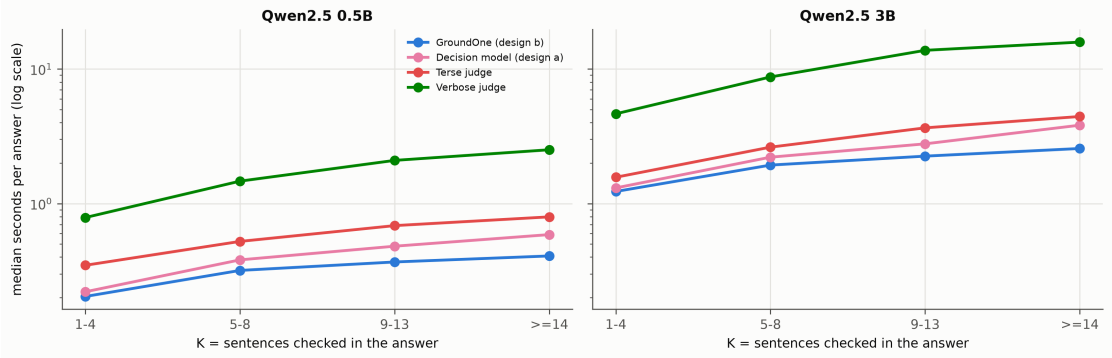


Figure 7.8: Median time per answer as a function of K on the same 200 T1 answers (in-process MLX, one answer at a time, M4 with 16 GB), for GroundOne, the decision model with one question per sentence, and the terse and verbose generative judges. Logarithmic scale.

Speed is not the only difference (Figure 7.9). The untrained generative judges are also much less accurate and do not always respect the output format. The 0.5B verbose judge produces a parsable verdict for only 3 % of the sentences, because it mostly copies the answer back, and the 0.5B terse judge fails for 23 %. The 3B judges respect the format far better (0.1 % and 7.6 % failures) but reach a three-way balanced accuracy of only about 0.40, against 0.90 for the 3B GroundOne on the same answers. The verbose 3B judge writes on average 248 tokens per answer, all of which must be generated before the verdict can be used, while GroundOne generates none.

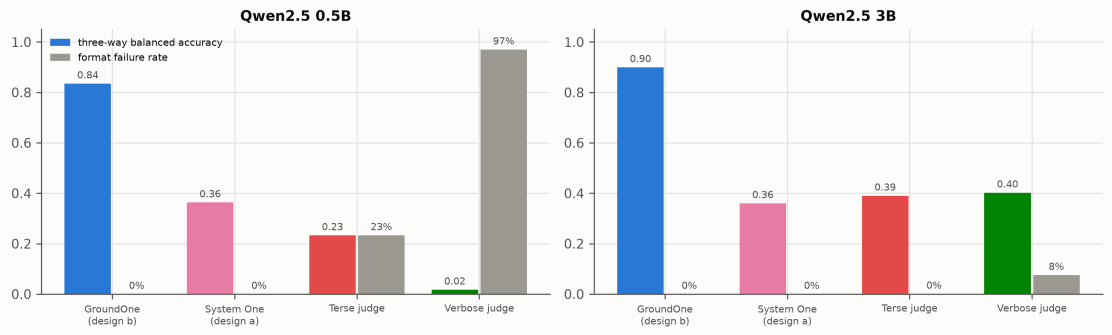


Figure 7.9: On the latency subset: three-way balanced accuracy (format failures counted as errors) and share of sentences whose verdict could not be parsed.

7.10 Energy per verification

The energy measurement adds a qualification that latency alone hides (Figure 7.10). GroundOne uses the least energy at both sizes: 5.5 joules per answer for the 0.5B and 33 for the 3B, against 8.9 and 43 for the terse judge and 22.5 and 112 for the verbose judge. Its advantage, however, is smaller than its advantage in time: 1.6 and 1.3 times less energy than the terse judge, and 4.1 and 3.4 times less than the verbose one. The reason is the power drawn during the computation. GroundOne spends its time in a single long prefill, which keeps the GPU busy and draws about 16 W above idle, while autoregressive decoding is limited by memory bandwidth and draws 11–14 W, but for much longer. This is the behaviour expected from the two phases of transformer inference [42], and it means that a speed-up should never be read as an equal saving of energy [67, 68].

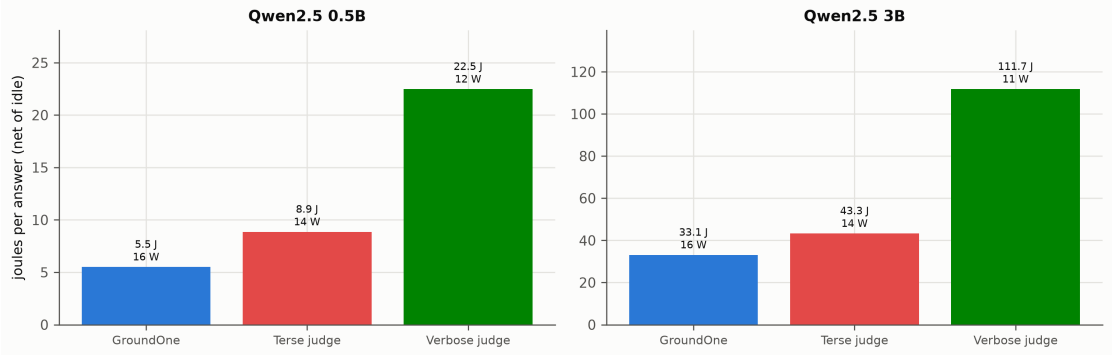


Figure 7.10: Energy per verified answer on 100 latency answers (powermetrics, CPU, GPU and Neural Engine, net of idle power), with the mean net power during the run.

In absolute terms the numbers are small: 5.5 joules is about 0.0015 watt-hours, so the 0.5B GroundOne could verify roughly six hundred answers with the energy of one watt-hour, entirely on the laptop.

7.11 Locality: offline run, quantisation and co-residence

Three measurements complete the picture of on-device cost (Section 6.5). The first concerns independence from the network. The complete evaluation of T1 with the 0.5B model, repeated inside a sandbox that denies the process any network access, produces predictions identical bit for bit to those of the original run on all 17,284 sentences. Loading the model, inference and evaluation therefore need no connection at all. The test is stronger than a disconnected machine, because the process cannot reach the network even when the machine is online.

The second concerns quantisation. When the same adapter of the 0.5B is applied to the original bf16 weights instead of the 8-bit ones, the probabilities change by 0.011 on average on T1 and by 0.007 on T3, and the predicted class agrees on 98.3 % and 98.9 % of the sentences. The metrics do not change: the three-way balanced accuracy is 0.769 with the 8-bit base and 0.768 with the bf16 one on T1, 0.442 in both cases on T3, and the AUROC on T1 is 0.905 against 0.904. The 8-bit base, which halves the memory of the weights, therefore costs no measurable accuracy. The measurement concerns, however, the sensitivity to the precision of the base of an adapter trained on the 8-bit base, not a model trained entirely in bf16.

The third concerns co-residence with the generator. In the same process, the 4-bit Qwen2.5-3B-Instruct generator and GroundOne verify 30 answers generated from T1 documents, with a median of 169 generated tokens and seven scored sentences per answer (Table 7.2). With the 3B verifier the peak memory allocated by MLX is 6.1 GB, with the 0.5B 3.3 GB: both well below the 16 GB of the machine. Verification takes 30 % of the total time per answer with the 3B and 6.7 % with the 0.5B. The small model therefore adds a modest fraction to the generation time, whereas the 3B lengthens it by about a third in exchange for higher accuracy. During the run with the 3B, with the usual desktop applications open, the free memory of the whole system dropped for a few moments to about 1.6 GB: the pair fits comfortably in the process, but on a 16 GB laptop it leaves little headroom for heavy desktop use. The retriever is not included in this measurement.

Table 7.2: Co-residence of generator and verifier in one process: 4-bit Qwen2.5-3B-Instruct as the generator, 30 T1 documents, greedy generation up to 300 tokens. Memory allocated by MLX; median times per answer.

Verifier	Memory after loading	Peak memory	Generation	Verification	Share of verification
GroundOne 3B	5.0 GB	6.1 GB	4.75 s	1.68 s	30 %
GroundOne 0.5B	2.3 GB	3.3 GB	4.65 s	0.28 s	6.7 %

7.12 Error analysis

The weak result on VitaminC and the missed contradictions on T1 have a common cause, which word overlap reveals. On VitaminC, when a supported claim shares few words with its evidence (overlap below 0.4) the 0.5B recognises it in only 2 % of cases, against 64 % when the overlap exceeds 0.7, and it never recognises a contradiction in a claim without digits. The models rely partly on lexical overlap: a paraphrased claim looks "not mentioned", and a small change to a copied sentence, often a number, goes unnoticed.

The same weakness exists on T1, with less weight. Where a sentence shares few or some words with the document, the three-way balanced accuracy of the 0.5B is 0.77, but for sentences that copy more than 70 % of their words from the document it falls to 0.54, with a recall of 0.24 for contradicted and 0.43 for not-mentioned sentences. The 3B reduces this blind spot without removing it: on the same sentences its recall is 0.36 and 0.56, and its balanced accuracy 0.63. Hallucinations hidden in sentences that otherwise copy the source remain the main limitation of both models. VitaminC is not harder in absolute terms: it consists mostly of such cases, paraphrased support and minimal edits, which RAGTruth contains in smaller numbers. Appendix A shows the output of every system on the same answer.

Chapter 8

Discussion, Limitations and Threats to Validity

This chapter interprets the results along the five research questions, then states what the experiments cannot show and which choices could have biased them.

8.1 When a specialist decision model pays off

Specialisation (RQ1). Training on the decision is what makes GroundOne accurate. The untrained models are near chance, and the same models after training reach 0.769 and 0.832 three-way balanced accuracy on the RAGTruth test set, beating the specialised encoders in AUROC and Rizzo-Flow, a general-purpose decision model eight times larger, in three-way accuracy. The advantage is concentrated where the training data are: on LLM-AggreFact the 3B is on par with the best small encoder rather than above it, and below the closed decision model Jev and a 27B open model as measured independently; the 0.5B remains below the encoder, and on VitaminC the trained 3B is no better than the untrained one. The comparison with d1-3B, a general-purpose decision model of the same size, makes the trade-off explicit. d1-3B, already trained to decide but on a broad mixture of tasks, is far behind GroundOne on RAGTruth, especially on long answers, on par with the 3B on LLM-AggreFact and far ahead on VitaminC. A specialist is therefore a strong choice for the domain it was trained on and a weak general fact checker: specialisation buys accuracy and robustness on long answers in its domain at the cost of transfer, while model size, or a broad training mixture, is what improves transfer outside it.

Readout versus generation (RQ2). The decision-model mechanism contributes in two separable ways. The first is cost: one pass over the answer replaces

the generation of K verdicts, so the time saved grows with the length of the answer, reaching a factor of about two against a terse judge and six against a verbose one for answers with fourteen or more sentences. The second, which this thesis did not expect, is accuracy. With identical weights and cost, deciding among the three admissible letters is more accurate than taking the token the model would generate, by 0.25 for the 0.5B and 0.07 for the 3B. A fine-tuned decision model is better understood as a three-way classifier placed on top of a language model than as a generator that stops after one token, and this distinction matters most for small models, which are the ones a laptop can afford. The comparison with generative judges, finally, also measures training: the judges are untrained, so the large differences in accuracy between them and GroundOne should be read as the combined effect of specialisation and readout.

Calibration (RQ3). A temperature and a bias fitted on development data make the probabilities match the observed frequencies on the test set of the same domain, and the calibrated probabilities make selective prediction effective. Two lessons go beyond GroundOne. First, the expected calibration error alone cannot rank systems: after the same calibration, judges whose probabilities barely separate the classes reach the same ECE as GroundOne, and only the AUROC and the risk–coverage curve reveal the difference. Second, decisions and probabilities must be derived consistently: the most probable calibrated class collapses towards the majority class, while a threshold fitted on development data recovers the balanced decision and keeps the label consistent with the probability shown to the user. Calibration is valid only for the distribution it was fitted on, as the results on VitaminC show.

Cost on the device (RQ4). The verifier fits comfortably in the 16 GB laptop: training peaks at 2.6 and 5.9 GB, and verifying a long answer takes 0.41 seconds with the 0.5B and 2.57 with the 3B, at an energy of 5.5 and 33 joules. The 0.5B meets an interactive budget; the 3B is slower than an interactive target for a typical answer and trades time for accuracy. The energy measurements qualify the speed results: a single long prefill draws more power than token-by-token decoding, so the energy saved is smaller than the time saved, and speed should not be read as an equal saving of battery. The verifier also coexists with a generator in the same process: together with a 4-bit 3B generator, the peak memory is 6.1 GB with the 3B and 3.3 GB with the 0.5B. The whole evaluation runs without any network access, with results identical bit for bit, and 8-bit weights cost no accuracy compared with bf16 ones (Section 7.11). The retriever remains to be measured, which will be done in the demonstration application.

The economic cost should also be read with care (Table 8.1). Verifying 10,000 answers with GroundOne takes an amount of energy that, at the European average

household electricity price [120], is worth less than one euro cent with the 0.5B and about three cents with the 3B. Entrusting the same verification to a hosted judge, at the providers' list prices [121, 122, 123], costs instead from about one dollar with the cheapest models to more than two hundred with the top-tier ones. The ratio is thus two to four orders of magnitude, but in absolute terms cheap judges remain inexpensive, and the comparison includes neither the purchase of the hardware nor the quality of hosted judges, which this thesis did not measure. The decisive argument for local verification therefore remains independence: the documents never leave the machine and the service does not depend on an external provider.

Table 8.1: Indicative cost of verifying 10,000 RAG answers. Hosted judges: standard-tier list prices, accessed on 19 October 2026, with about 900 input tokens per answer (median T1 document and answer plus instructions) and the output tokens measured for the 3B judges of this thesis (36 for the terse judge, 248 for the verbose one). GroundOne: measured net chip energy multiplied by the EU average household electricity price (0.29 €/kWh, second half of 2025). Hardware cost and caching or batch discounts are excluded.

System	Input / output price (USD per Mtoken)	Terse judge	Verbose judge
Claude Haiku 5.5, gpt-6-luna	0.10 / 0.50	1.1 USD	2.2 USD
Gemini 3.5 Flash-Lite	0.30 / 2.50	3.6 USD	8.9 USD
Gemini 3.8 Flash	0.75 / 3.75	8.2 USD	16 USD
Claude Sonnet 5.5, gpt-6.1-sol, Gemini 4 Argon	2.00 / 10.00	22 USD	43 USD
Claude Opus 5.5	4.00 / 20.00	43 USD	86 USD
gpt-6-astra	10.00 / 50.00	109 USD	215 USD
GroundOne 0.5B (local)	0.015 kWh, about 0.004 €		
GroundOne 3B (local)	0.092 kWh, about 0.03 €		

Beyond a binary score (RQ5). The three-way output separates contradicted from not-mentioned sentences well in data-to-text, where RAGTruth concentrates its contradictions, and much less in question answering and summarisation, where contradictions are rare in the training data. The 3B improves this considerably, which suggests that more varied contradictions in training, rather than a different architecture, are what the small model lacks. Because the strict and lenient policies and the threshold are derived from the same three probabilities, a user can change the notion of error without retraining, which no binary verifier allows.

8.2 Limitations

One platform and one family of models. Every result comes from one laptop, one runtime and one model family. Latency and energy depend on the hardware and on the implementation, and the ratios between methods, not the absolute numbers, are what should transfer. No latency comparison is made with the encoders or the general-purpose System One, because they ran in different runtimes and under conditions that would make the comparison unfair.

Training stability. Two completed runs of the 0.5B agree on all aggregate metrics, but a third run collapsed early and was stopped. The recipe has no gradient clipping and uses Adam without bias correction, both known causes of unstable fine-tuning [99, 98], so either may explain the collapse; neither was tested. The 3B was trained once, on half of the training answers, and its seed variance is unknown.

Training data. RAGTruth is the only training source. Its contradictions are concentrated in data-to-text, and outside that task they are learned only in part. The models also rely partly on lexical overlap: paraphrased support looks unsupported, and a small change in a sentence that otherwise copies the source often goes unnoticed. This blind spot shrinks with model size but remains the main limitation of both models.

Scope of the decision. Verification is sentence by sentence and only against the retrieved document. A sentence that is true in the world but absent from the document counts as not mentioned, which is the right behaviour for groundedness but not for fact checking. Sentences without checkable content are excluded by a rule, and the segmentation itself can split or merge claims differently from a human reader. The verdict on a sentence depends on the document and on the preceding sentences only.

Contexts made of several documents. When the answer is written from several retrieved passages, GroundOne reads them concatenated as a single document, as in RAGTruth question answering. Three limits remain. The first concerns length: in training, document and answer never exceed 2,048 tokens, so a longer context falls outside the distribution on which the model was measured and must be split into chunks, keeping for each sentence the highest support probability, as on LLM-AggreFact; a fact that requires combining two different chunks may then be missed. The second concerns conflicting sources: when one passage supports a sentence and another contradicts it, the expected verdict is not uniquely defined,

and RAGTruth contains no such cases. The third concerns attribution: GroundOne states whether a sentence is supported, but not by which passage.

Statistical scope. The bootstrap intervals describe the variability of the test data, not that of training, and about fifty comparisons were tested without correction for multiple testing. Conclusions therefore rest on effects far from zero, and differences on LLM-AggreFact and VitaminC smaller than the spread between seeds are not claimed. Several baselines were evaluated on subsamples, which widens their intervals.

8.3 Threats to validity

Labels. The distinction between contradicted and not mentioned rests on the RAGTruth annotators, whose agreement on the type of hallucination is not published, and a recent re-annotation of the RAGTruth test split found considerably more hallucinated answers than the original labels [116]. The test set may therefore under-count the minority classes. For this reason the three-way results on RAGTruth are always read together with those on VitaminC, whose labels have measured agreement. LLM-AggreFact unifies the labels of eleven datasets of different quality, and some of its columns are known to be noisy.

In-domain advantage. GroundOne is trained on RAGTruth, so its results on the RAGTruth test set and on the RAGTruth column of LLM-AggreFact favour it with respect to systems trained elsewhere; conversely, VitaminC lies in the training lineage of MiniCheck-RoBERTa. Each comparison is reported with the relevant caveat, and the claims rest on the pattern across the three test sets rather than on a single one.

Baseline configuration. Generative judges are sensitive to their prompts. Their prompts were fixed before looking at test results, and the prompt of the equal-output judge was chosen on development data for format validity only; parsing was lenient in their favour; decoding was not constrained by a grammar, which mostly affects the format failures of the smallest model. A judge fine-tuned on the same data, or a larger judge, was not evaluated, so the comparison measures what a judge of the same size gives without training. Rizzo-Flow was run with 4-bit weights because the 8-bit model did not fit beside the operating system, which may understate its accuracy. The d1-3B was run zero-shot with the questions used for all decision-model baselines, which may not be the phrasing it works best with, and on a subsample of LLM-AggreFact.

Contamination. The base models were pretrained on large web corpora that may include parts of the public test sets. Because the same base models are used untrained as baselines and reach near-chance accuracy on the three-way task, contamination cannot explain the gains of training, but it cannot be ruled out entirely for the absolute numbers.

One machine. All measurements of time and energy come from one laptop, an Apple M4 with 16 GB, and one runtime, MLX. The cost model of Chapter 2 predicts that the advantage of a decision model holds on every class of local hardware, because it rests on the difference between prefill and decoding, which vendors and independent studies observe everywhere; the lower bounds on decoding computed there for other machines follow from published specifications. The actual speed-ups on a discrete graphics card, a DGX Spark or a phone were not measured: their size depends on the balance between computation and bandwidth of each machine, and the direction of the advantage, not its magnitude, is what the cost model supports.

Chapter 9

Conclusion and Future Work

9.1 Summary of findings

This thesis asked whether a compact language model, specialised in a single decision and used as a decision model, can verify retrieval-augmented answers sentence by sentence on a laptop, and where any advantage comes from. GroundOne, a fine-tuned Qwen2.5 model that reads the verdicts of a whole answer in one forward pass, gives a positive but qualified answer.

Specialisation works where it was aimed. On the RAGTruth test set GroundOne reaches a three-way balanced accuracy of 0.769 with 0.5B parameters and 0.832 with 3B, far above the same models untrained, above the specialised encoders in ranking supported sentences and well above Rizzo-Flow, a general-purpose decision model eight times larger. On LLM-AggreFact the 3B is on par with the best small encoder, and on VitaminC the gain disappears: a specialist is not a general fact checker. d1-3B, a general-purpose decision model of the same size trained to decide on a broad mixture of tasks, confirms the trade-off: GroundOne is far ahead of it on RAGTruth, on par on LLM-AggreFact and far behind on VitaminC.

The decision-model mechanism pays off in two ways. Reading all verdicts in one pass makes verification about two times faster than a terse and six times faster than a verbose generative judge of the same size on long answers, with output that is always valid; and restricting the decision to the admissible answers is, by itself, more accurate than letting the same model generate its verdict. At equal output, generative judges cannot provide probabilities of comparable quality: their probabilities barely separate correct from incorrect decisions, and verbalised ones are almost always all-or-nothing.

The probabilities of GroundOne can be calibrated with a temperature and a bias fitted on development data, and a threshold chosen on the same data turns them into labels that agree with the probabilities and with the evaluated decision. The

expected calibration error alone, however, does not distinguish good probabilities from useless ones, and discrimination must be measured separately.

The whole system runs on a 16 GB laptop. A long answer is verified in 0.41 seconds and 5.5 joules with the 0.5B, and in 2.57 seconds and 33 joules with the 3B. The energy saved with respect to a judge is real but smaller than the time saved, because a single long prefill draws more power than decoding.

The main limitations are the narrow training data, which leave contradictions outside data-to-text and hallucinations hidden in copied sentences only partly learned, and a training recipe that is not fully robust to the random seed.

9.2 Future work

Several directions follow directly from these limits. The training recipe should be made robust, by enabling bias correction and gradient clipping, logging gradient norms and repeating the runs over several seeds, including for the 3B. Training data with more varied contradictions, and with paraphrased support and minimal edits of the kind VitaminC contains, should address both the weak recall outside data-to-text and the blind spot on copied sentences. Training the readout on the letter that the model would naturally write after the marker could align readout and generation, with effects on accuracy that remain to be measured.

On the systems side, the cost model of Chapter 2 should be checked on other hardware: a port of the readout to llama.cpp would allow measuring GroundOne on a discrete graphics card, on a DGX Spark or an AMD Strix Halo machine, and on a phone, whose neural processing units accelerate exactly the prefill on which a decision model relies. The remaining measurement of the locality pillar is the memory of the retriever next to the generator and the verifier, which were already measured together in this thesis. A fair latency comparison with the encoders requires running them in batches, and one with the general-purpose decision models requires comparing both systems through the same HTTP interface. For contexts made of several documents, a natural extension is attribution: reading, besides the verdict, which passage supports or contradicts each sentence, for example with a readout per passage, so as to show users the source of every claim and to handle conflicting sources explicitly. Verification could also run while the answer is being generated, flagging each sentence as soon as it is complete, which the marker design makes natural. Finally, a local assistant built on the served interface would let users choose the policy and the threshold that suit their documents, which is where calibrated probabilities show their value.

Appendix A

Output of every system on one answer

This appendix shows what each system returns for the same answer of T1, so that the numbers of Chapter 7 can be related to concrete outputs. The answer (identifier `rt-7784`) is a data-to-text answer written by a language model from a structured description of a restaurant: a JSON record with its name, address, categories, opening hours (Tuesday to Sunday, 11:00–21:00), attributes, an average rating of 4.5 stars and two customer reviews. It was chosen because every system scored it and because it contains all three classes; it was not selected for the performance of any system. Its six sentences and their reference labels are:

1. *Mesa Verde is a popular business located in Santa Barbara, CA.* — supported.

2. *It offers a variety of services including Garage Door Services, Restaurants, Live/Raw Food, . . . , Vegan, and Salad.* — supported.

3. *The business is open seven days a week from 11 am to 9 pm.* — contradicted (it is closed on Mondays).

4. *Mesa Verde has a rating of 4.5 stars on Google and has received multiple positive reviews.* — not mentioned (the record gives no source for the rating and contains two reviews, one of them negative).

5. *One customer praised the restaurant's fresh, delicious, and inventive cuisine, while another was disappointed by the taste and price of the food.* — supported.

6. *Despite this, many customers have had positive experiences at Mesa Verde and recommend it for its fusion and vegan food options.* — supported.

A.1 Verdicts sentence by sentence

Table A.1 lists the verdict of every system. For GroundOne the label is the served one, derived from the calibrated probabilities with the threshold fitted on development data ($\tau = 0.935$ for the 0.5B and 0.925 for the 3B), and the number is the calibrated P(supported). For the encoders the number is their probability of support and the decision is taken at 0.5. S, C and N stand for supported, contradicted and not mentioned; a dash marks a format failure.

Table A.1: Verdicts on answer `rt-7784`. Errors with respect to the reference are in bold.

System	1	2	3	4	5	6
Reference	S	S	C	N	S	S
GroundOne 0.5B (served, P_{sup})	S 0.96	S 0.98	C 0.50	N 0.05	S 0.98	S 0.97
GroundOne 3B (served, P_{sup})	S 0.96	S 0.99	C 0.58	N 0.21	S 0.99	**N 0.85**
B2 0.5B (generated token)	S	S	C	N	S	–
B2 3B (generated token)	S	S	C	–	S	S
Untrained 0.5B (readout)	**C**	**C**	C	**C**	**C**	**C**
Terse judge 0.5B	S	S	**S**	**S**	S	S
Terse judge 3B	**C**	S	**S**	**S**	**N**	S
Verbose judge 0.5B	–	–	–	–	–	–
Verbose judge 3B	–	–	–	–	–	–
Equal-output judge 3B, logits	**C**	S	**S**	**S**	**N**	S
Equal-output judge 3B, verbalised	S	**C**	**N**	**S**	**C**	**N**
MiniCheck-RoBERTa (P_{sup})	0.78	0.90	0.09	0.03	**0.43**	**0.03**
HHEM-2.1-open (P_{sup})	0.75	0.95	0.01	0.09	0.73	**0.24**
Rizzo-Flow 4B	S	S	**S**	**S**	S	S

The 0.5B GroundOne gets all six sentences right. The contradiction in sentence 3 is caught with a calibrated P(supported) of 0.50: well below the threshold, but far from certain, which is what a calibrated probability should express for a sentence that is correct except for one day of the week. The 3B GroundOne also catches both errors but serves sentence 6, a generalisation over the two reviews, as not mentioned: its P(supported) of 0.85 is below the high threshold that the base rate of RAGTruth imposes, a case where a user who prefers fewer flags would lower the threshold. With the same weights, the generated token (B2) gives the right letter on five of the six sentences, but on the remaining one it produces a space (0.5B) or the word “We” (3B) instead of a verdict. The binary encoders separate the two errors well but also reject sentences 5 or 6, which summarise the reviews in different words; Rizzo-Flow accepts every sentence. The generative judges fail in the ways that Chapter 7 quantifies, as the raw outputs below show.

A.2 Raw outputs

GroundOne 0.5B (served JSON, abridged). One forward pass over 1,193 input tokens, no generated token, 0.47 seconds (2.6 seconds for the 3B):


```
{"design": "b", "threshold": 0.935, "n_decisions": 6,
 "usage": {"input_tokens": 1193, "output_tokens": 0},
 "sentences": [
  {"index": 2, "text": "The business is open seven days a week ...",
   "scored": true, "labels": {"three": "contradicted"},
   "answers": {"three": {"supported": 0.502, "contradicted": 0.462,
                         "not_mentioned": 0.036}},
   "flag_strict": true, "flag_lenient": true},
  {"index": 3, "text": "Mesa Verde has a rating of 4.5 stars ...",
   "scored": true, "labels": {"three": "not_mentioned"},
   "answers": {"three": {"supported": 0.051, "contradicted": 0.003,
                         "not_mentioned": 0.946}},
   "flag_strict": true, "flag_lenient": false}, ...]}
```


Terse judge 3B. One line per sentence, as requested, but with the wrong letters for four sentences:

```
1: B
2: A
3: A
4: A
5: C
6: A
```

Verbose judge 3B (first three verdicts). The reasoning restates the sentence and cites the document as support even for the contradicted and the unsupported sentence, and every verdict is "A". The verdicts are written on a separate line instead of on the same line, as the instruction asked, so the protocol counts them as format failures:

```
1: Mesa Verde is located in Santa Barbara, CA, as stated in the document.
Verdict: A

2: Mesa Verde offers a variety of services including Garage Door Services,
   [...], as mentioned in the categories provided.
Verdict: A

3: The business is open seven days a week from 11 am to 9 pm, as indicated
   by the hours provided.
Verdict: A
```

Verbose judge 0.5B. It copies the six sentences back, numbered, without any verdict.

Equal-output judge 3B, verbalised. The probabilities are all one-hot, and for sentences 3 and 6 the stated label (C) contradicts the stated probabilities, which put all the mass on A:


```
{"1": {"label": "A", "p": [1.0, 0.0, 0.0]},
 "2": {"label": "B", "p": [0.0, 1.0, 0.0]},
 "3": {"label": "C", "p": [1.0, 0.0, 0.0]},
 "4": {"label": "A", "p": [1.0, 0.0, 0.0]},
 "5": {"label": "B", "p": [0.0, 1.0, 0.0]},
 "6": {"label": "C", "p": [1.0, 0.0, 0.0]}}
```


Equal-output judge 0.5B, verbalised. The same vector for every sentence:


```
{"1": {"label": "A", "p": [0.3, 0.3, 0.3]},
 "2": {"label": "A", "p": [0.3, 0.3, 0.3]},
 ...
 "6": {"label": "A", "p": [0.3, 0.3, 0.3]}}
```

Bibliography

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2020 (cit. on pp. 1, 34).

[2] Georgi Gerganov and ggml-org contributors. *llama.cpp: LLM inference in C/C++*. 2023. URL: `https://github.com/ggml-org/llama.cpp` (cit. on pp. 1, 25, 28, 33, 63).

[3] Apple. *MLX: An array framework for Apple silicon*. 2023. URL: `https://github.com/ml-explore/mlx` (cit. on pp. 1, 25, 49).

[4] Tom Gunter, Zirui Wang, Chong Wang, Ruoming Pang, et al. *Apple Intelligence Foundation Language Models*. 2024. arXiv: `2407.21075` (cit. on pp. 1, 12, 16).

[5] Venkat Somala and Luke Emberson. *Frontier AI performance becomes accessible on consumer hardware within a year*. Epoch AI. Aug. 15, 2025. URL: `https://epoch.ai/data-insights/consumer-gpu-model-gap` (visited on 2026-10-11) (cit. on pp. 1, 16).

[6] Stanford Institute for Human-Centered Artificial Intelligence. *The 2025 AI Index Report: Technical Performance*. 2025. URL: `https://hai.stanford.edu/ai-index/2025-ai-index-report/technical-performance` (visited on 2026-10-11) (cit. on pp. 1, 13, 16).

[7] NYU Shanghai RITS. *OpenAI claims a Navier–Stokes proof, amid a dispute over credit*. Sept. 9, 2026. URL: `https://rits.shanghai.nyu.edu/ai/openai-navier-stokes-proof-credit-dispute/` (visited on 2026-10-11) (cit. on pp. 1, 10).

[8] Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, et al. "RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models". In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024 (cit. on pp. 2, 5, 34, 53, 54).

[9] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, et al. "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions". In: *ACM Transactions on Information Systems* (2025). DOI: 10.1145/3703155 (cit. on pp. 2, 34).

[10] Reiner Pope, Sholto Douglas, Aakanksha Chowdhery, Jacob Devlin, et al. "Efficiently Scaling Transformer Inference". In: *Proceedings of Machine Learning and Systems (MLSys)*. 2023 (cit. on pp. 3, 17, 18, 41).

[11] Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. "AlignScore: Evaluating Factual Consistency with a Unified Alignment Function". In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. 2023 (cit. on pp. 3, 35, 73).

[12] Liyan Tang, Philippe Laban, and Greg Durrett. "MiniCheck: Efficient Fact-Checking of LLMs on Grounding Documents". In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2024 (cit. on pp. 3, 35, 53, 54, 64).

[13] Vectara. *HHEM-2.1-Open: Hughes Hallucination Evaluation Model*. 2024. URL: https://huggingface.co/vectara/hallucination_evaluation_model (cit. on pp. 3, 35, 64).

[14] TypeSafe. *Introducing System One Models and Jev*. Blog post. 2026. URL: https://typesafe.ai/blog/introducing-system-one-models-and-jev (cit. on pp. 3, 31, 46).

[15] TheoLeeCJ. *SemIf: open option-logit System One baseline*. 2026. URL: https://github.com/TheoLeeCJ/SemIf (cit. on pp. 3, 5, 32).

[16] Simone Rizzo. *Rizzo Flow: typed decisions from an LLM, without generating a single token*. 2026. URL: https://github.com/Rizzo-AI-Academy/rizzo-flow (cit. on pp. 3, 5, 32, 63).

[17] Hugging Face. *Models tagged decision-model*. URL: https://huggingface.co/models?other=decision-model (visited on 2026-10-10) (cit. on pp. 5, 33).

[18] Liquid AI. *d1-3B: a decision model built on LFM2.5-VL-3B (model card, LiquidAI/d1-3B, revision 051bcc4)*. URL: https://huggingface.co/LiquidAI/d1-3B (visited on 2026-10-10) (cit. on pp. 5, 33, 63).

[19] Cloudflare. *Clef: open decision models (model card, Cloudflare/clef)*. URL: https://huggingface.co/Cloudflare/clef (visited on 2026-10-10) (cit. on pp. 5, 33).

[20] Tobias Deußer, Lorenz Sparrenberg, and Rafet Sifa. *Evaluating and Benchmarking the System One Model Jev*. 2026. arXiv: 2609.37647 (cit. on pp. 5, 32, 46, 72).

[21] Gowthamkumar Nandakishore. *Laya as a Typed Probabilistic Assessor: An Independent Reproduction and a Preregistered Study of Calibration and Selective Escalation*. 2026. arXiv: 2609.33843 (cit. on pp. 5, 32).

[22] multimodalart. *Jev Decision Index: a leaderboard of System One decision models (Hugging Face Space, v0.2.1)*. URL: https://huggingface.co/spaces/multimodalart/jev-decision-index (visited on 2026-10-10) (cit. on pp. 5, 33, 70).

[23] Qwen Team. *Qwen2.5 Technical Report*. 2024. arXiv: 2412.15115 (cit. on pp. 5, 13, 37, 43, 61).

[24] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. "LoRA: Low-Rank Adaptation of Large Language Models". In: *International Conference on Learning Representations (ICLR)*. 2022 (cit. on pp. 5, 37, 43).

[25] Guanqiao Qu, Qiyuan Chen, Wei Wei, Zheng Lin, Xianhao Chen, and Kaibin Huang. "Mobile Edge Intelligence for Large Language Models: A Contemporary Survey". In: *IEEE Communications Surveys & Tutorials* 27.6 (2025), pp. 3820–3860. arXiv: 2407.18921 (cit. on pp. 10, 11).

[26] European Parliament and Council of the European Union. *Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation)*. Official Journal of the European Union, L 119, 4 May 2016, pp. 1–88. 2016. URL: https://eur-lex.europa.eu/eli/reg/2016/679/oj (cit. on p. 10).

[27] Jeremy Kahn. *OpenAI says it cracked a Navier–Stokes math grand challenge; a mathematician raises accusations (Eye on AI)*. Fortune. Sept. 8, 2026. URL: https://fortune.com/2026/09/08/openai-says-it-cracked-navier-stokes-math-grand-challenge-buckmaster-accusation-cheating-intimidation-tao-lament/ (visited on 2026-10-11) (cit. on p. 10).

[28] Larissa G. Capella. *'Why would you ruin your career?': OpenAI claims to have cracked one of math's greatest unsolved problems, but mathematicians allege it played dirty*. Live Science. Sept. 9, 2026. URL: https://livescience.com/physics-mathematics/mathematics/why-would-you-ruin-your-career-openai-claims-to-have-cracked-one-of-maths-greatest-unsolved-problems-but-mathematicians-allege-it-played-dirty (visited on 2026-10-11) (cit. on p. 10).

[29] Apple. *Private Cloud Compute: A new frontier for AI privacy in the cloud*. June 10, 2024. URL: https://security.apple.com/blog/private-cloud-compute/ (visited on 2026-10-11) (cit. on pp. 11, 12).

[30] Senyao Li, Haozhao Wang, Wenchao Xu, Rui Zhang, Song Guo, Jingling Yuan, Xian Zhong, Tianwei Zhang, and Ruixuan Li. *Collaborative Inference and Learning between Edge SLMs and Cloud LLMs: A Survey of Algorithms, Execution, and Open Challenges*. 2025. arXiv: 2507.16731 (cit. on pp. 11, 23).

[31] Open Source Initiative. *The Open Source AI Definition, version 1.0*. Oct. 28, 2024. URL: https://opensource.org/ai/open-source-ai-definition (visited on 2026-10-11) (cit. on p. 13).

[32] Artificial Analysis. *Introducing the Artificial Analysis Openness Index*. Dec. 1, 2025. URL: https://artificialanalysis.ai/articles/announcing-artificial-analysis-openness-index (visited on 2026-10-11) (cit. on p. 13).

[33] Rishi Bommasani, Alice Hau, Kevin Klyman, and Percy Liang. *Foundation models under the EU AI Act*. Stanford Center for Research on Foundation Models. Aug. 1, 2024. URL: https://crfm.stanford.edu/2024/08/01/eu-ai-act.html (visited on 2026-10-11) (cit. on p. 13).

[34] Epoch AI. *Open-weight models lag the closed frontier by about four months on the Epoch Capabilities Index*. May 29, 2026. URL: https://epoch.ai/data-insights/open-closed-eci-gap (visited on 2026-10-11) (cit. on pp. 13, 14).

[35] Artificial Analysis. *Artificial Analysis Intelligence Index*. Leaderboard, version 4.3.2; snapshot of 20 October 2026. URL: https://artificialanalysis.ai/leaderboards/models (visited on 2026-10-20) (cit. on pp. 13, 15).

[36] JustVugg. *colibri: run frontier MoE models on hardware you already own, experts streamed from disk*. URL: https://github.com/JustVugg/colibri (visited on 2026-10-13) (cit. on pp. 14, 25, 27).

[37] Venkat Somala and Luke Emberson. *What's the best model you can run on a single consumer GPU?* Epoch AI. Sept. 9, 2025. URL: https://epochai.substack.com/p/whats-the-best-model-you-can-run (visited on 2026-10-11) (cit. on p. 16).

[38] Zhenyan Lu, Xiang Li, Dongqi Cai, Rongjie Yi, Fangming Liu, Xiwen Zhang, Nicholas D. Lane, and Mengwei Xu. *Small Language Models: Survey, Measurements, and Insights*. 2024. arXiv: 2409.15790 (cit. on p. 16).

[39] Zechun Liu et al. "MobileLLM: Optimizing Sub-billion Parameter Language Models for On-Device Use Cases". In: *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. 235. PMLR. 2024 (cit. on pp. 16, 29).

[40] Google. *Introducing Gemma 3n: the developer guide*. May 20, 2025. URL: https://developers.googleblog.com/en/introducing-gemma-3n/ (visited on 2026-10-11) (cit. on p. 16).

[41] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, et al. "Efficient Memory Management for Large Language Model Serving with PagedAttention". In: *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*. 2023 (cit. on p. 17).

[42] Pratyush Patel, Esha Choukse, Chaojie Zhang, Aashaka Shah, et al. "Splitwise: Efficient Generative LLM Inference Using Phase Splitting". In: *Proceedings of the 51st Annual International Symposium on Computer Architecture (ISCA)*. 2024 (cit. on pp. 17, 22, 29, 41, 78).

[43] Samuel Williams, Andrew Waterman, and David Patterson. "Roofline: An Insightful Visual Performance Model for Multicore Architectures". In: *Communications of the ACM* 52.4 (2009), pp. 65–76. DOI: 10.1145/1498765.1498785 (cit. on pp. 18, 19).

[44] Zhihang Yuan et al. *LLM Inference Unveiled: Survey and Roofline Model Insights*. 2024. arXiv: 2402.16363 (cit. on p. 18).

[45] Apple. *MacBook Pro (14-inch, M4, 2024) – Technical Specifications*. URL: https://support.apple.com/en-us/121552 (visited on 2026-10-10) (cit. on pp. 18, 23, 61).

[46] Keivan Alizadeh, Iman Mirzadeh, Dmitry Belenko, Karen Khatamifard, et al. "LLM in a flash: Efficient Large Language Model Inference with Limited Memory". In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024 (cit. on pp. 19, 27).

[47] Apple. *MacBook Pro – Technical Specifications (M5, M5 Pro and M5 Max)*. URL: https://www.apple.com/macbook-pro/specs/ (visited on 2026-10-11) (cit. on p. 23).

[48] Apple. *Mac mini – Technical Specifications*. URL: https://www.apple.com/mac-mini/specs/ (visited on 2026-10-11) (cit. on pp. 21, 23).

[49] Apple. *Mac Studio (2025) – Technical Specifications*. URL: https://support.apple.com/en-us/122211 (visited on 2026-10-11) (cit. on pp. 22, 23).

[50] NVIDIA. *DGX Spark User Guide: Hardware Overview*. Sept. 10, 2026. URL: https://docs.nvidia.com/dgx/dgx-spark/hardware.html (visited on 2026-10-11) (cit. on pp. 22, 23).

[51] Mark Hachman. *Framework Desktop review*. PCWorld. Aug. 7, 2025. URL: https://www.pcworld.com/article/2866400/framework-desktop-review.html (visited on 2026-10-11) (cit. on pp. 21, 23).

[52] NVIDIA. *Compare GeForce Graphics Cards: specifications*. URL: https://www.nvidia.com/en-us/geforce/graphics-cards/compare/ (visited on 2026-10-11) (cit. on p. 23).

[53] EXO Labs. *NVIDIA DGX Spark and Mac Studio: disaggregated prefill and decode*. 2025. URL: https://blog.exolabs.net/nvidia-dgx-spark/ (visited on 2026-10-11) (cit. on p. 22).

[54] Zhenliang Xue, Yixin Song, Zeyu Mi, Xinrui Zheng, Yubin Xia, and Haibo Chen. *PowerInfer-2: Fast Large Language Model Inference on a Smartphone*. 2024. arXiv: 2406.06282 (cit. on p. 23).

[55] Daliang Xu, Hao Zhang, Liming Yang, Ruiqi Liu, Gang Huang, Mengwei Xu, and Xuanzhe Liu. "Fast On-device LLM Inference with NPUs". In: *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '25)*. ACM, 2025. DOI: 10.1145/3669940.3707239 (cit. on pp. 23, 24, 29).

[56] Apple Machine Learning Research. *Exploring LLMs with MLX and the Neural Accelerators in the M5 GPU*. Nov. 19, 2025. URL: https://machinelearning.apple.com/research/exploring-llms-mlx-m5 (visited on 2026-10-11) (cit. on p. 24).

[57] Subhransu Das, Jiaming Cheng, Arnav Kumar, Sadia Afrose, Mingzhe Han, Michael Silagy, Shreya Palande, Brijesh Soni, and Rajiv Ramnath. *Large Models for Small Devices: Edge AI Deployment*. 2026. arXiv: 2608.15693 (cit. on p. 24).

[58] Apple. *mlx-lm: LLMs with MLX*. 2024. URL: https://github.com/ml-explore/mlx-lm (cit. on pp. 25, 28, 49).

[59] Varun Rajesh, Om Jodhpurkar, Pooja Anbuselvan, Mantinder Singh, Ashok Jallepali, Shantanu Godbole, Pradeep Kumar Sharma, and Hritvik Shrivastava. *Production-Grade Local LLM Inference on Apple Silicon: A Comparative Study of MLX, MLC-LLM, Ollama, llama.cpp, and PyTorch MPS*. 2025. arXiv: 2511.05502 (cit. on p. 25).

[60] Stefanos Laskaridis, Kleomenis Katevas, Lorenzo Minto, and Hamed Haddadi. "MELTing Point: Mobile Evaluation of Language Transformers". In: *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (MobiCom)*. 2024. arXiv: 2403.12844 (cit. on pp. 25, 29).

[61] Salvatore Sanfilippo. *DwarfStar (ds4): DeepSeek 4 Flash and PRO local inference engine for Metal, CUDA and ROCm*. URL: https://github.com/antirez/ds4 (visited on 2026-10-13) (cit. on p. 25).

[62] Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. "LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 35. 2022 (cit. on pp. 28, 29).

[63] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. "GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers". In: *International Conference on Learning Representations (ICLR)*. 2023 (cit. on pp. 28, 29).

[64] Ji Lin et al. "AWQ: Activation-aware Weight Quantization for On-Device LLM Compression and Acceleration". In: *Proceedings of Machine Learning and Systems (MLSys)*. 2024 (cit. on p. 28).

[65] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. "QLoRA: Efficient Finetuning of Quantized LLMs". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023 (cit. on pp. 28, 37, 43).

[66] Yaniv Leviathan, Matan Kalman, and Yossi Matias. "Fast Inference from Transformers via Speculative Decoding". In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. Vol. 202. PMLR. 2023 (cit. on p. 29).

[67] Siddharth Samsi, Dan Zhao, Joseph McDonald, Baolin Li, et al. "From Words to Watts: Benchmarking the Energy Costs of Large Language Model Inference". In: *IEEE High Performance Extreme Computing Conference (HPEC)*. 2023 (cit. on pp. 29, 78).

[68] Alexandra Sasha Luccioni, Yacine Jernite, and Emma Strubell. "Power Hungry Processing: Watts Driving the Cost of AI Deployment?" In: *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. 2024 (cit. on pp. 29, 78).

[69] Jacky Kwok, Hangoo Kang, Tarun Suresh, Jon Saad-Falcon, Marco Pavone, Christopher Ré, and Azalia Mirhoseini. *Contrastive Language Models: A System One Model for Fast and Generalizable Decision-Making*. Blog post, not peer-reviewed. 2026. URL: `https://contrastive-lm.notion.site` (cit. on p. 32).

[70] Convai Innovations. *Laya: a System One decision model (model card, convaiinnovations/laya)*. URL: `https://huggingface.co/convaiinnovations/laya` (visited on 2026-10-10) (cit. on p. 32).

[71] Seyedarmin Azizi, Erfan Baghaei Potraghloo, and Massoud Pedram. *Labels Override Definitions in Jev-Style Typed Decision Models*. 2026. arXiv: `2610.02586` (cit. on pp. 32, 33).

[72] Cho Yin Yong. *DecideBench (dataset card, choyiny/decidebench)*. URL: https://huggingface.co/datasets/choyiny/decidebench (visited on 2026-10-10) (cit. on p. 32).

[73] Perplexity. *pplx-decider-v1.1-27b (model card)*. URL: https://huggingface.co/perplexity-ai/pplx-decider-v1.1-27b (visited on 2026-10-10) (cit. on p. 33).

[74] Ruoqi Guo, Yi Liu, Gelei Deng, Yuekang Li, et al. *Just Ask Jev: Reinforcement Learning for Calibrated Decisions as a Zero-Shot Detector of AI Alignment Failures*. 2026. arXiv: 2609.29429 (cit. on pp. 33, 46).

[75] Yubo Li, Yidi Miao, Ramayya Krishnan, and Rema Padman. *JEV-as-a-Judge: Accept When Confident, Escalate When Unsure*. 2026. arXiv: 2609.26550 (cit. on pp. 33, 46).

[76] Xinpeng Wang, Bolei Ma, Chengzhi Hu, Leon Weber-Genzel, et al. ""My Answer is C": First-Token Probabilities Do Not Match Text Answers in Instruction-Tuned Language Models". In: *Findings of the Association for Computational Linguistics: ACL 2024*. 2024. arXiv: 2402.14499 (cit. on pp. 33, 43).

[77] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. "Large Language Models Are Not Robust Multiple Choice Selectors". In: *International Conference on Learning Representations (ICLR)*. 2024 (cit. on pp. 33, 43).

[78] Adina Williams, Nikita Nangia, and Samuel R. Bowman. "A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference". In: *Proceedings of NAACL-HLT 2018*. 2018 (cit. on p. 34).

[79] Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. "Adversarial NLI: A New Benchmark for Natural Language Understanding". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2020 (cit. on pp. 34, 54).

[80] Tal Schuster, Adam Fisch, and Regina Barzilay. "Get Your Vitamin C! Robust Fact Verification with Contrastive Evidence". In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, 2021, pp. 624–643. DOI: 10.18653/v1/2021.naacl-main.52. URL: https://aclanthology.org/2021.naacl-main.52 (cit. on pp. 34, 53, 54).

[81] Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. "SummaC: Re-Visiting NLI-based Models for Inconsistency Detection in Summarization". In: *Transactions of the Association for Computational Linguistics* 10 (2022), pp. 163–177 (cit. on p. 34).

[82] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, et al. "FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation". In: *Proceedings of EMNLP 2023*. 2023 (cit. on p. 34).

[83] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena". In: *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. 2023 (cit. on p. 35).

[84] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. "RAGAs: Automated Evaluation of Retrieval Augmented Generation". In: *Proceedings of EACL 2024: System Demonstrations*. 2024 (cit. on p. 35).

[85] Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. "ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems". In: *Proceedings of NAACL 2024*. 2024 (cit. on p. 35).

[86] Selvan Sunitha Ravi, Bartosz Mielczarek, Anand Kannappan, Douwe Kiela, and Rebecca Qian. *Lynx: An Open Source Hallucination Evaluation Model*. 2024. arXiv: 2407.08488 (cit. on p. 35).

[87] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. "Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection". In: *International Conference on Learning Representations (ICLR)*. 2024 (cit. on p. 35).

[88] Brandon T. Willard and Rémi Louf. *Efficient Guided Generation for Large Language Models*. 2023. arXiv: 2307.09702 (cit. on pp. 35, 63).

[89] Liyan Tang, Philippe Laban, and Greg Durrett. *LLM-AggreFact Leaderboard*. 2024. URL: https://llm-aggrefact.github.io/ (visited on 2026-10-05) (cit. on pp. 35, 53, 54).

[90] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. "On Calibration of Modern Neural Networks". In: *International Conference on Machine Learning (ICML)*. 2017 (cit. on pp. 35, 45, 47).

[91] Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. "Obtaining Well Calibrated Probabilities Using Bayesian Binning". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2015 (cit. on pp. 35, 47).

[92] Tony Z. Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. "Calibrate Before Use: Improving Few-Shot Performance of Language Models". In: *International Conference on Machine Learning (ICML)*. 2021 (cit. on pp. 35, 45).

[93] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, et al. *Language Models (Mostly) Know What They Know*. 2022. arXiv: 2207.05221 (cit. on p. 35).

[94] Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, et al. "Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback". In: *Proceedings of EMNLP 2023*. 2023 (cit. on pp. 36, 76).

[95] Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, et al. "Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs". In: *International Conference on Learning Representations (ICLR)*. 2024 (cit. on pp. 36, 76).

[96] Yonatan Geifman and Ran El-Yaniv. "Selective Classification for Deep Neural Networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2017 (cit. on pp. 36, 64).

[97] Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. *Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping*. 2020. arXiv: 2002.06305 (cit. on pp. 37, 61).

[98] Marius Mosbach, Maksym Andriushchenko, and Dietrich Klakow. "On the Stability of Fine-tuning BERT: Misconceptions, Explanations, and Strong Baselines". In: *International Conference on Learning Representations (ICLR)*. 2021 (cit. on pp. 37, 69, 84).

[99] Razvan Pascanu, Tomas Mikolov, and Yoshua Bengio. "On the Difficulty of Training Recurrent Neural Networks". In: *International Conference on Machine Learning (ICML)*. 2013 (cit. on pp. 38, 69, 84).

[100] Pritam Deka. *From Retrieval to Typed Decisions: Calibrated System One Models from Biomedical Sentence Encoders*. 2026. arXiv: 2610.02486 (cit. on p. 46).

[101] Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. *MS MARCO: A Human Generated MAchine Reading COmprehension Dataset*. CoCo Workshop at NIPS 2016. 2016. arXiv: 1611.09268 (cit. on p. 53).

[102] Karl Moritz Hermann, Tomáš Kočiský, Edward Grefenstette, Lasse Espeholt, Will Kay, Mustafa Suleyman, and Phil Blunsom. "Teaching Machines to Read and Comprehend". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2015 (cit. on p. 53).

[103] Abigail See, Peter J. Liu, and Christopher D. Manning. "Get To The Point: Summarization with Pointer-Generator Networks". In: *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2017 (cit. on p. 53).

[104] Yelp. *Yelp Open Dataset*. URL: https://business.yelp.com/data/resources/open-dataset/ (visited on 2026-10-05) (cit. on p. 53).

[105] Liyan Tang, Tanya Goyal, Alex Fabbri, Philippe Laban, Jiacheng Xu, Semih Yavuz, Wojciech Kryściński, Justin Rousseau, and Greg Durrett. "Understanding Factual Errors in Summarization: Errors, Summarizers, Datasets, Error Detectors". In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. 2023 (cit. on p. 54).

[106] Liyan Tang, Igor Shalyminov, Amy Wing-mei Wong, et al. "TofuEval: Evaluating Hallucinations of LLMs on Topic-Focused Dialogue Summarization". In: *Proceedings of NAACL 2024*. 2024 (cit. on p. 54).

[107] Nelson F. Liu, Tianyi Zhang, and Percy Liang. "Evaluating Verifiability in Generative Search Engines". In: *Findings of EMNLP 2023*. 2023 (cit. on p. 54).

[108] Chaitanya Malaviya, Subin Lee, Sihao Chen, Elizabeth Sieber, Mark Yatskar, and Dan Roth. "ExpertQA: Expert-Curated Questions and Attributed Answers". In: *Proceedings of NAACL 2024*. 2024 (cit. on p. 54).

[109] Hung-Ting Chen, Fangyuan Xu, Shane Arora, and Eunsol Choi. "Understanding Retrieval Augmentation for Long-Form Question Answering". In: *Conference on Language Modeling (COLM)*. 2024 (cit. on p. 54).

[110] Ryo Kamoi, Tanya Goyal, Juan Diego Rodriguez, and Greg Durrett. "WiCE: Real-World Entailment for Claims in Wikipedia". In: *Proceedings of EMNLP 2023*. 2023 (cit. on p. 54).

[111] Alon Jacovi, Yonatan Bitton, Bernd Bohnet, Jonathan Herzig, Or Honovich, Michael Tseng, Michael Collins, Roee Aharoni, and Mor Geva. "A Chain-of-Thought Is as Strong as Its Weakest Link: A Benchmark for Verifiers of Reasoning Chains". In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024 (cit. on p. 54).

[112] Yuxia Wang, Revanth Gangi Reddy, Zain Muhammad Mujahid, et al. "Factcheck-Bench: Fine-Grained Evaluation Benchmark for Automatic Fact-checkers". In: *Findings of EMNLP 2024*. 2024 (cit. on p. 54).

[113] Liyan Tang. *LLM-AggreFact dataset card (lytang/LLM-AggreFact)*. URL: https://huggingface.co/datasets/lytang/LLM-AggreFact (visited on 2026-10-05) (cit. on p. 54).

[114] Tal Schuster. *VitaminC dataset card (tals/vitaminc)*. URL: https://huggingface.co/datasets/tals/vitaminc (visited on 2026-10-06) (cit. on p. 54).

[115] Nipun Sadvilkar and Mark Neumann. "PySBD: Pragmatic Sentence Boundary Disambiguation". In: *Proceedings of the Second Workshop for NLP Open Source Software (NLP-OSS)*. 2020, pp. 110–114. DOI: 10.18653/v1/2020.nlposs-1.15. URL: https://aclanthology.org/2020.nlposs-1.15 (cit. on p. 54).

[116] Boxi Yu, Yuzhong Zhang, Liting Lin, Lionel Briand, and Emir Muñoz. *Retromorphic Testing with Hierarchical Verification for Hallucination Detection in RAG*. 2026. arXiv: 2603.27752. URL: https://arxiv.org/abs/2603.27752 (cit. on pp. 58, 85).

[117] Philipp Koehn. "Statistical Significance Tests for Machine Translation Evaluation". In: *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2004, pp. 388–395. URL: https://aclanthology.org/W04-3250 (cit. on p. 64).

[118] Taylor Berg-Kirkpatrick, David Burkett, and Dan Klein. "An Empirical Investigation of Statistical Significance in NLP". In: *Proceedings of EMNLP-CoNLL 2012*. 2012, pp. 995–1005. URL: https://aclanthology.org/D12-1091 (cit. on p. 64).

[119] Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. "The Hitchhiker's Guide to Testing Statistical Significance in Natural Language Processing". In: *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2018, pp. 1383–1392. URL: https://aclanthology.org/P18-1128 (cit. on p. 64).

[120] Eurostat. *Electricity price statistics*. Statistics Explained; household prices, second half of 2025. URL: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Electricity_price_statistics (visited on 2026-10-19) (cit. on p. 83).

[121] Anthropic. *Claude API pricing*. URL: https://platform.claude.com/docs/en/about-claude/pricing (visited on 2026-10-19) (cit. on p. 83).

[122] OpenAI. *API pricing*. URL: https://developers.openai.com/api/docs/pricing (visited on 2026-10-19) (cit. on p. 83).

[123] Google. *Gemini API pricing*. URL: https://ai.google.dev/gemini-api/docs/pricing (visited on 2026-10-19) (cit. on p. 83).